SEMESTER 3

ROBOTICS AND ARTIFICIAL INTELLIGENCE

SEMESTER S3

MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL SCIENCE – 3

(Common to B & C Groups)

Course Code	**GYMAT301**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	**Course Type**	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. **(Text 1: Relevant topics from sections 11.7, 11.8, 11.9)**	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w= z^2$, $w=e^z$,$w= \frac{1}{z}$, $w=sinz$. **(Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)**	9
3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof), Cauchy Integral formula (without proof).	9

	(Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	
4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. **(Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)**	**9**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
1. 2 Questions from each module. 2. Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	1. Each question carries 9 marks. 2. Two questions will be given from each module, out of which 1 question should be answered. 3. Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	**K3**
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	**K3**
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	**K3**
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10^{th} edition, 2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1st edition, 2011

SEMESTER:S3

MICROCONTROLLERS AND EMBEDDED SYSTEMS

Course Code	**PCRAT302**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs 30 Mins
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1.To introduce ARM microcontrollers.
2.To introduce the design of embedded electronic systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**ARM Cortex-M Microcontroller:** Introduction to the architecture, Programmer's model, Behavior of the application program status register (APSR), Memory system, Exceptions and interrupts, System control block (SCB), Debug, Reset and reset sequence.	**11**
2	**ARM Cortex-M Instruction Set:** Understanding the assembly language syntax, Use of a suffix in instructions, Unified assembly language (UAL), Instruction set, Barrel shifter, Accessing special instructions and special registers in programming.	**11**
3	**Embedded System Components:** Embedded Systems vs. General Computing Systems, Classification of Embedded Systems, Major Application Areas of Embedded Systems, Purpose of Embedded Systems, Core of the Embedded System, Memory, Sensors and Actuators, Communication Interface, Embedded Firmware, Other System Components.	**11**
4	**Embedded System Design Concepts:** Characteristics of an Embedded System, Quality Attributes of Embedded Systems, Application-Specific Embedded System, Domain Specific Examples of Embedded System, Fundamental Issues in Hardware Software Co-Design, Computational Models in Embedded Design, Embedded Firmware Design Approaches, Embedded Firmware Development Languages.	**11**

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
1.2 Questions from each module. 2.Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	1.Each question carries 9 marks. 2.Two questions will be given from each module, out of which 1 question should be answered. 3.Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain ARM Cortex-M3 microcontroller architecture	**K1**
CO2	Use ARM Cortex-M3 instruction set	**K3**
CO3	Describe embedded system components	**K1**
CO4	Explain embedded system design concepts	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping od Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	2							1
CO2	3	2	2	1	2							1
CO3	3	2	2	1	1							1
CO4	3	2	2	1	1							1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The Definitive Guide to ARM Cortex-M3 and Cortex-M4 Processors	Joseph Yiu	Newnes, (Elsevier)	3rd Edition, 2014.
2	Introduction to Embedded Systems	Shibu K V	Tata McGraw Hill Education Private Limited	2nd Edition

Video Links (NPTEL, SWAYAM…)	
Sl No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc22_cs93/preview
2	https://onlinecourses.nptel.ac.in/noc22_cs93/preview

SEMESTER:S3

ROBOTIC MANIPULATORS AND WORK CELL DESIGN

Course Code	**PCRAT303**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs 30 Mins
Prerequisites (if any)	PCRAT205	**Course Type**	Theory

Course Objectives:

1. Students will be able to apply mathematical modelling to kinematics, dynamics, control and trajectory planning of robotic manipulators
2. Students will be able to apply various concepts to design automated work cells that incorporate robotic manipulators

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Components and Structure of Robots:** Symbolic Representation, Degrees of Freedom and Workspace, Classification of Robots - Power Source, Application Area, Method of Control, Geometry, Common Kinematic Arrangements (Structure and Workspace) - Articulated Configuration (RRR), Spherical Configuration (RRP), SCARA Configuration (RRP), Cylindrical Configuration (RPP), Cartesian configuration (PPP), Parallel Manipulator, Spherical wrist **End Effectors:** Mechanical, Magnetic and Vacuum Grippers, Active and Passive Grippers **Kinematics of Robots:** Rotations - Fundamental and Composite Rotations, Homogeneous coordinates, Translations and Rotations, Composite homogeneous transformations, Screw transformations, Inverse of Homogeneous Transformation Matrices, Other representations of Rotations: Definitions of RPY, Euler Angles and Quaternions	**11**
2	**Forward Kinematics:** Definition, Kinematic Chains, DenavitHartenberg Representation, Assigning the coordinate frames, Algorithm based on DH Convention for Forward Kinematics, Forward Kinematics Examples - up to 3-DoF (Higher degrees can be given as assignment) **Inverse Kinematics:** Definition, The General Inverse Kinematics	**11**

	Problem, General properties of solutions to the inverse kinematics problem, Theory of Kinematic Decoupling, Inverse Kinematics Example - up to 2-DoF (Higher degrees can be given as assignment)	
3	**Velocity Kinematics:** Derivation of the Jacobian – Angular Velocity, Linear Velocity, combining linear and angular Jacobians, Velocity Jacobian Examples given forward kinematic matrices - 2 link planar manipulator, Singularities – Definition and importance of identifying singularities **Dynamic Analysis:**Force-mass-acceleration and torque-inertia-angular-acceleration relationships, Lagrangian Mechanics – Overview, Dynamic Analysis Examples – 1 DOF spring cart system **Robot Control:** The control problem, Single axis PID control - Control Law, Block Diagram, Derivation of Closed Loop Transfer Function, advantages and disadvantages	**11**
4	**Trajectory Planning:** Definitions of Path planning and Trajectory Planning, Joint space versus Cartesian Space trajectory planning, Cubic Polynomial based trajectory planning, Point to Point and Continuous Path Planning **Robot selection considerations for an application**: Number of axes, work volume, capacity and speed, stroke and reach, Repeatability, Precision and Accuracy, Operating environment **Robotic Work Cell Design:** Robot Cell Layouts - Robot-centred Work cell, In-Line Robot Cell, Mobile Robot Cells, Multiple Robots and Machine Interference, Other Considerations in Work cell Design, Work cell Control - Sequence control, Operator interface, Safety monitoring, Interlocks, Error Detection and Recovery, The Work cell Controller	**11**

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
1. 2 Questions from each module. 2. Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	1. Each question carries 9 marks. 2. Two questions will be given from each module, out of which 1 question should be answered. 3. Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe various robotic manipulator configurations and end effectors	**K2**
CO2	Formulate forward and inverse kinematics for robotic manipulators	**K3**
CO3	Analyse robotic manipulator configurations using velocity kinematics and dynamic analysis	**K**
CO4	Plan trajectories and control schemes for robotic manipulators	**K3**
CO5	Design robotic work cells for various applications	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping od Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3		3							3
CO2	3	3	3		3							3
CO3	3	3	3		3							3
CO4	3	3	3		3							3
CO5	3	3	3		3	2					3	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robot Dynamics and Control	Mark W. Spong, Seth Hutchinson, and M. Vidyasagar	Wiley	2nd edition, 2020
2	Fundamentals of robotics – Analysis and control	Robert. J. Schilling	Prentice Hall of India	1996
3	Introduction to Robotics - Analysis, Control, Applications	Saeed Benjamin Niku	John Wiley & Sons	2nd edition, 2010
4	Industrial Robotics – Technology, Programming and Applications	Mikell P. Groover, Mitchel Weiss, Roger N. Nagel, Nicholas G. Odrey, Ashish Dutta	Tata McGraw Hill	2nd edition, 2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Robotics	S K Saha	McGraw-Hill	2008
2	Robotics and Control	R K Mittal and I J Nagrath	Tata McGraw Hill	2003
3	Introduction to Robotics – Mechanics and Control	John J. Craig	Pearson	4th edition, 2017
4	Robotics-Fundamental concepts and analysis	AshitavaGhosal	Oxford University Press	2006

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/112/105/112105249/
2	https://archive.nptel.ac.in/courses/107/106/107106090/
3	https://archive.nptel.ac.in/courses/112/107/112107289/
4	https://archive.nptel.ac.in/courses/112/105/112105249/

SEMESTER: S3

ANALOG AND DIGITAL ELECTRONICS

Course Code	**PBRAT304**	**CIE Marks**	60
Teaching Hours/Week (L: T:P: R)	**3:0:0:1**	**ESE Marks**	40
Credits	**4**	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	**None/ (Course code)**	**Course Type**	Theory

Course Objectives:

1. This course aims to develop the skill to analyse and design different types of analog circuits using discrete electronic components and also to impart the basic knowledge of logic circuits and enable students to apply it to design a digital system.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**BJT and FET Amplifiers:** BJT Amplifiers, RC Coupled amplifier: Need of different components and design. Multistage amplifiers: Effect of cascading on gain and bandwidth. **NMOSFET amplifiers:** MOSFET circuits at DC, MOSFET as an amplifier, Biasing of discrete MOSFET amplifier, Small signal analysis of CS configuration using small signal hybrid-pi model, Derivation of small signal voltage and current gain, input and output impedance.	**11**
2	**Operational Amplifiers:** Op-Amp Basics, Block diagram of an Op Amp, Op Amp parameters -Slew Rate, Band Width, CMRR, Gain, Offset Voltage and Offset Current, differential and common mode operation of Op Amps, Inverting & Non Inverting Amplifier. **Oscillators:** Condition for oscillations, RC phase shift and wein bridge oscillators. **555 timer**: Functional block diagram, Astable and Monostable operation.	**11**
3	**Boolean Algebra:** Binary logic functions, Boolean laws, truth tables, associative and distributive properties, De Morgans theorems,	**11**

	realization of switching functions using logic gates. **Combinational Logic**: Canonical logic forms, sum of product & product of sums, Karnaugh maps, two and three and four variable Karnaugh maps, simplification of expressions.	
4	**Combinational and Sequential Logic**: Analysis & design of Combinational Logic: Introduction to combinational circuits, code conversions, decoder, encoder, priority encoder, binary adder, subtractor. **Sequential Logic:** Sequential circuits, flip-flops : JK,D & T, clocked and edge triggered flipflop, block diagram, timing diagram and working of asynchronous and synchronous four bit counters using JK FF. Registers , serial in serial out shift registers.	**11**

Course Assessment Method
(CIE: 60 marks,ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**30**	**12.5**	**12.5**	**60**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
1.2 Questions from each module. 2.Total of 8 Questions, each carrying 2 marks **(8x2 =16marks)**	1.Each question carries 6 marks. 2.Two questions will be given from each module, out of which 1 question should be answered. 3.Each question can have a maximum of 2 sub divisions. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Design and analyse basic amplifiers using BJT and MOSFET	**K3**
CO2	Familiarize the fundamental principles of operational amplifiers and design multivibrators using IC 555.	**K3**
CO3	Familiarize digital logic and Boolean algebra.	**K2**
CO4	Design and Analyse sequential and combinational circuits.	**K4**
CO5	Apply the concepts learnt in the course to develop a project and prepare a brief report.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									2
CO2	3	2	2									2
CO3	3	2	2									2
CO4	3	2	1									2
CO5	3	3	3		3				3	3	2	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Electronic Devices and Circuit Theory	Robert Boylestad and L Nashelsky	11/e Pearson	2015
2	Digital Design	Mano M.M, Ciletti M.D	Pearson India	4th Edition. 2006
3	Linear Integrated Circuits	Roy D. C. and S. B. Jain	3/e, New Age International	2010
4	Microelectronic Circuits	Sedra A. S. and K. C. Smith	6/e, Oxford University Press	2013

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Electronic Circuits, Analysis and Design	Neamen D	3/e, Tata Mc Graw Hill	2007.
2	Modern Digital Electronics	R.P. Jain	Tata McGraw Hill	4th edition, 2009
3	Digital Circuits and Systems	D.V.Hall	Tata Mc Graw Hill	1989

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	**5**
2	Contribution in Progress Presentations and Question Answer Sessions	**4**
3	Involvement in the project work and Team Work	**3**
4	Execution and Implementation	**10**
5	Final Presentations	**5**
6	Project Quality, Innovation and Creativity	**3**
Total		**30**

Project Assessment and Evaluation criteria (30 Marks)

1. **Project Planning and Proposal (5 Marks)**

    - Clarity and feasibility of the project plan
    - Research and background understanding
    - Defined objectives and methodology

2. **Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**

    - Individual contribution to the presentation
    - Effectiveness in answering questions and handling feedback

3. **Involvement in the Project Work and Team Work (3 Marks)**

    - Active participation and individual contribution
    - Teamwork and collaboration

4. **Execution and Implementation (10 Marks)**

    - Adherence to the project timeline and milestones
    - Application of theoretical knowledge and problem-solving
    - Final Result

5. **Final Presentation (5 Marks)**

    - Quality and clarity of the overall presentation
    - Individual contribution to the presentation
    - Effectiveness in answering questions

6. **Project Quality, Innovation, and Creativity (3 Marks)**

    - Overall quality and technical excellence of the project
    - Innovation and originality in the project

Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	**GNEST305**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to AI and Machine Learning**: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11
2	**Mathematical Foundations of AI and Data science**: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	11
3	**Applied Probability and Statistics for AI and Data Science**: Basics of probability-random variables and statistical measures - rules in probability- Bayes theorem and its applications- statistical estimation-Maximum	11

	Likelihood Estimator (MLE) - statistical summaries- Correlation analysis- linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	
4	**Basics of Data Science**: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	**11**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
1. 2 Questions from each module. 2. Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	1. Each question carries 9 marks. 2. Two questions will be given from each module, out of which 1 question should be answered. 3. Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	**K3**
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	**K3**
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	**K3**
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6^{th} edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	AurélienGéron	O'Reilly Media, Inc.	2^{nd} edition,2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1^{st} edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9^{th} edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1^{st} edition, 2016

Reference Books				
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2^{nd} edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1^{st} edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1^{st} edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1^{st} edition, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1^{st} edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.math.arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	**6**
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return -– Payback – Accounting Rate of Return.	**6**

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	**15**	**12.5**	**12.5**	**50**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	**50**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	**K2**
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	**K3**
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	**K2**
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7^{TH} Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: UCHUT346

Course Name: Economics for Engineers

Max. Marks: 50		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 8 marks

Module 1

9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

		Module 2		
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$.. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
		Module 3		
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNPfc from GDPmp.	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDPmp and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
		Module 4		
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

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SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	**UCHUT347**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	**Syllabus Description**	**Contact Hours**
1	**Fundamentals of ethics** - Personal vs. professional ethics, Civic Virtue, Respect for others, **Profession and Professionalism,** Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, **Technology and digital revolution**-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, **High technologies: connecting people and places**-accessibility and social impacts, **Managing conflict**, Collective bargaining, **Confidentiality**, Role of confidentiality in moral integrity, **Codes of Ethics**. **Basic concepts in Gender Studies** - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, **Gender disparity and discrimination in education**, employment and everyday life, History of women in Science & Technology,	**6**

	Gendered technologies & innovations**, Ethical values and practices in connection with gender** - equity, diversity & gender justice, **Gender policy and women/transgender empowerment initiatives.**	
2	**Introduction to Environmental Ethics:** Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). **Sustainable Engineering Principles:** Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. **Ecosystems and Biodiversity:** Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. **Landscape and Urban Ecology:** Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	**6**
3	**Hydrology and Water Management:** Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. **Zero Waste Concepts and Practices:** Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. **Circular Economy and Degrowth:** Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. **Mobility and Sustainable Transportation:** Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	**6**
4	**Renewable Energy and Sustainable Technologies:** Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. **Climate Change and Engineering Solutions:** Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. **Environmental Policies and Regulations:** Overview of key environmental policies and regulations (national and international), Role of engineers in policy	**6**

	implementation and compliance, Ethical considerations in environmental policy-making. **Case Studies and Future Directions:** Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	

Course Assessment Method

(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12

3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
		Total Marks		**50**

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis**: Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts**: Ability to apply course concepts to real-world problems and local contexts.
- **Creativity**: Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills**: Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	**K3**
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	**K4**
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	**K5**
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	**K4**
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessmen	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?

- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).

  Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER:S3

ANALOG AND DIGITAL ELECTRONICS LAB

Course Code	**PCRAL307**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	PBRAT304	**Course Type**	Lab

Course Objectives:

1. Familiarize students with the Analog Circuits Design through the implementation of basic Analog Circuits using discrete components.
2. Help the learners to get familiarized with Digital Logic Design through the implementation of logic circuits using ICs of basic logic gates & flip flops.

Expt. No.	Experiments
	Part A: Analog Circuits Lab (Minimum 5 experiments are to be done)
1	RC coupled CE amplifier - frequency response characteristics
2	MOSFET amplifier (CS) - frequency response characteristics
3	Measurement of Op-Amp parameters.
4	Inverting & Non Inverting Amplifier
5	RC phase shift oscillator using OpAmp
6	Wien bridge oscillator using OpAmp
7	AstableMultivibrator using IC555.
8	MonostableMultivibrator using IC555.
	Part B: Digital Lab (Minimum 6 experiments are to be done, 3 combinatorial and 3 sequential circuits)
1	Realization of functions using basic and universal gates (SOP and POS forms).
2	Design and realization of half /full adder and subtractor using basic gates and universal gates
3	Design and implementation of code converters using logic gates(i) BCD to excess-3 code and vice versa (ii) Binary to gray and vice-versa
4	Design and implementation of multiplexer and de-multiplexer using logic gates
5	Construction and verification of 4 bit ripple counter.

6	Design and implementation of Mod-N ripple counters
7	Synchronous Counter: Realization of Mod-N counters.
8	Design and implementation of any of SISO/SIPO/PISO shift registers
9	Design and implementation of Ring counter and Johnson Counter

It is compulsory to conduct a minimum of 5 experiments from Part A and a minimum of 6 experiments from Part B.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

1. ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
2. ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Design analog circuits using BJT/FETs and evaluate their performance characteristics	**K3**
CO2	Design and implement basic circuits using IC (OPAMP and 555 timers).	**K3**
CO3	Design and implement combinational logic circuits using logic gates.	**K3**
CO4	Design and implement sequential logic circuits using integrated circuits.	**K3**
CO5	Function effectively as an individual and in a team to accomplish the given task.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	3					3		2	
CO2	3	3	3	3					3			
CO3	3	3	3	3					3			
CO4	3	3	3	3								
CO5	2	2	2	2					3	3	3	

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Electronic Devices and Circuit Theory	Robert Boylestad and L Nashelsky	11/e Pearson	2015
2	Linear Integrated Circuits	Roy D. C. and S. B. Jain	3/e, New Age International	2010
3	Digital Design	Mano M.M, Ciletti M.D	Pearson India	4th Edition. 2006

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**

   - Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
   - Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. **Conduct of Experiments (7 Marks)**

   - Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
   - Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
   - Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

   - Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
   - Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

   - Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

   - Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
   - Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
   - Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
   - Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

3. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

4. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER:S3

MICROCONTROLLERS AND EMBEDDED SYSTEMS LAB

Course Code	**PCRAL308**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs 30 Mins
Prerequisites (if any)	PCRAT302	**Course Type**	Lab

Course Objectives:

1. To familiarize the students with Embedded C/Assembly Language Programming of modern microcontrollers.
2. Impart the skills for interfacing the microcontroller with the help of Embedded C/Assembly Language Programming.

Expt. No.	Experiments
1	Interface a simple Switch and display its status through LED.
2	Display the Hex digits 0 to F on a 7-segment LED interface, with an appropriate delay in between.
3	Demonstrate the use of an external interrupt to toggle an LED On/Off.
4	Display "Hello World" message using Internal UART.
5	Using the Internal PWM module of ARM controller generate PWM and vary its duty cycle.
6	Interface a simple Switch and display its status through LED using mbed OS.
7	With PWM control the brightness of LED using mbed OS.
8	Measure Ambient temperature with a sensor and SPI ADC IC using mbed OS.
9	Interface and Control a DC Motor.
10	Interface a Stepper motor and rotate it in clockwise and anti-clockwise direction.

Conduct **ANY EIGHT** of the above experiments on an ARM Cortex-M3/M4 evaluation board using ARM assembly/Embedded 'C' &KeiluVision tool/compiler or other similar tools.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

5. ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
6. ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Implement communication via UART to microcontroller	**K3**
CO2	Demonstrate interfacing and control of DC and stepper motor	**K3**
CO3	Operate the PWM module of ARM controller	**K3**
CO4	Demonstrate the use of an external interrupt to toggle an LED On/Off	**K3**
CO5	Execute a 7-segment LED interface	**K3**
CO6	Demonstrate interfacing a switch and display its status	**K3**
CO7	Implement thermometer using sensor and SPI ADC IC	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	1	3					2	1	1
CO2	3	2	3	1	3					2	1	1
CO3	3	2	3	1	3					2	1	1
CO4	3	2	3	1	3					2	1	1
CO5	3	2	3	1	3					2	1	1
CO6	3	2	3	1	3					2	1	1
CO7	3	2	3	1	3					2	1	1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The Definitive Guide to ARM Cortex-M3 and Cortex-M4 Processors	Joseph Yiu	Newnes, (Elsevier)	3rd Edition, 2014.
2	Introduction to Embedded Systems	Shibu K V	Tata McGraw Hill Education Private Limited	2nd Edition

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**
   - Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
   - Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.
2. **Conduct of Experiments (7 Marks)**
   - Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
   - Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.

   Teamwork: Collaboration and participation in group experiments.
3. **Lab Reports and Record Keeping (6 Marks)**

   Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.

- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

ROBOTICS AND ARTIFICIAL INTELLIGENCE

SEMESTER S4

MATHEMATICS FOR ELECTRICAL SCIENCE – 4

Course Code	**GBMAT401**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs 30 Mins
Prerequisites (if any)	Basic calculus	**Course Type**	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To expose the students to the basics of random processes essential for their subsequent study of analog and digital communication

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. **[Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]**	**9**
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. **[Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]**	**9**

3	about a Population Mean (for large sample), t Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. **[Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]**	**9**
4	Random process concept, classification of process, Methods of Description of Random process, Special classes, Average Values of Random Process, Stationarity- SSS, WSS, Autocorrelation functions and its properties, Ergodicity, Mean-Ergodic Process, Mean-Ergodic Theorem, Correlation Ergodic Process, Distribution Ergodic Process. **[Text 2: Relevant topics from Chapter 6]**	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	**K3**
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	**K3**
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using *z*-tests and the one-sample *t*-test.	**K3**
CO4	Analyze random processes by classifying them, describing their properties, utilizing autocorrelation functions, and understanding their applications in areas like signal processing and communication systems.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9th edition, 2016
2	Probability, Statistics and Random Processes	T Veerarajan	The McGraw-Hill	3rd edition, 2008

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6th edition, 2020
3	Probability and Random Processes	Palaniammal, S.	PHI Learning Private Limited	3rd edition, 2015
4	Introduction to Probability	David F. Anderson, Timo, Benedek	Cambridge	1st edition, 2017

Video Links (NPTEL, SWAYAM…)	
Sl. No.	**Link ID**
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
4	https://archive.nptel.ac.in/courses/117/105/117105085/

SEMESTER: S4

SIGNAL PROCESSING FOR ROBOTICS

Course Code	PCRAT402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To understand and analyse various signal processing techniques for Robotics
2. To analyse continuous and discrete time systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Elementary signals, Continuous time and Discrete time signals and systems, Signal classifications, operations, linearity, stability and causality of systems, LTI systems-definition and examples Convolution, Correlation and Orthogonality of signals	11
2	Frequency domain representation-Continuous time Fourier series, Continuous time Fourier transform, Laplace transform, Inverse Laplace transform, Differential equation representation of continuous time systems,Transfer function, Frequency response	11
3	Sampling, Aliasing, Z transform, ROC, Inverse Z transform, Difference equation representation of discrete time systems, Frequency domain representation of discrete time signals-Discrete time Fourier transform (DTFT), Analysis of discrete time LTI systems using the above transforms	11
4	Discrete Fourier Transform (DFT), computational complexity, circular convolution, long convolution, Fast Fourier Transform (FFT)-Decimation in Time algorithm.	11

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Perform basic operations on continuous time discrete time signals and systems	**K2**
CO2	Analyze continuous time systems in the frequency domain by finding the frequency response	**K4**
CO3	Analyze discrete time systems using tools like Z-transform and DTFT	**K4**
CO4	Efficiently compute the DFT using fast algorithms (FFT)	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2									2
CO2	3	3	3									2
CO3	3	3	3									2
CO4	3	3	3									2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Discrete-Time Signal Processing	Alan V Oppenheim, Ronald W. Schafer	Pearson	3rd, 2010
2	Signals & Systems	Simon Haykin	PHI	2nd , 2003

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Signals & Systems	Alan V. Oppenheim and Alan Willsky,	PHI	2nd, 2009
2	Signals & Systems	Anand Kumar,	PHI	3rd, 2013
3	Signals & Systems	P Ramakrishna Rao, Shankar Prakriya	Mc Graw Hill	2013
4	Signals & Systems-Continuous and Discrete,	Rodger E. Ziemer	Pearson	4th 2013
5	Digital Signal Processing	Proakis J. G. and Manolakis D. G	Pearson Education	2007
6	Digital Signal Processing-a practical approach	Ifeachor E.C. and Jervis B. W	Pearson education	2009
7	Schaums outline: Digital Signal Processing	Monson H Hayes	McGraw HillProfessional	1999
8	Digital Signal Processing	Salivahanan S	McGraw Hill	2019

<table>
<tr><th colspan="2">Video Links (NPTEL, SWAYAM…)</th></tr>
<tr><th>Module No.</th><th>Link ID</th></tr>
<tr><td>1</td><td rowspan="4">NPTEL videos on Signals and Systems, by Prof. Aditya K Jagannatham, IIT Kanpur (https://archive.nptel.ac.in/courses/108/104/108104100/)
NPTEL videos on DSP, by Prof. S. C. Dutta Roy, IIT Delhi (https://nptel.ac.in/courses/117102060)
NPTEL videos on Real TIme DSP, by Prof. Rathna G. N., IISc, Bangalore (https://archive.nptel.ac.in/courses/108/108/108108185/)</td></tr>
<tr><td>2</td></tr>
<tr><td>3</td></tr>
<tr><td>4</td></tr>
</table>

SEMESTER: S4

KINEMATICS OF MACHINERY

Course Code	**PCRAT403**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	**Course Type**	Theory

Course Objectives:

1. To understand the basic components and layout of linkages in the assembly of a system/machine
2. Provide insight into analysis of various mechanisms for position, velocity and acceleration
3. Understand the basics of synthesis of mechanisms
4. Provide insight into the concepts of toothed gearing and kinematics of gear trains.
5. Provide an introduction to spatial mechanisms

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to kinematics and mechanisms - various mechanisms, kinematic diagrams, degree of freedom- Grashof's criterion, inversions, mechanical advantage, transmission angle. Position Posture & Displacement Analysis, Loop closure equation, velocity and acceleration analysis - graphical and analytical methods – complex number methods - computer oriented methods. Instant centre -Kennedy's theorem	**12**
2	Kinematic synthesis (planar mechanisms) - type, number and dimensional synthesis precision points. Graphical synthesis for motion - path and prescribed timing - function generator. 2 position and 3	**10**

	position synthesis – overlay Method. Freudenstein's equation.	
3	Gears – Classification- terminology of spur gears – law of gearing-tooth profiles- involute spur gears- contact ratio - interference - backlash - gear standardization – interchangeability. Classification of gears- internal gears - theory and details of bevel, helical and worm gearing. Gear trains - simple and compound gear trains - planetary gear trains	**11**
4	Introduction to Spatial Mechanisms-Mobility, spatial posture analysis- Loop closure equation, Classification of solutions of the vector tetrahedron equation, velocity and acceleration of RRRR, RSSR configurations.	**11**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of kinematics, various planar mechanisms	**K2**
CO2	Perform analysis a of mechanisms for its displacement, velocity and acceleration	**K4**
CO3	Solve the problem on gear drives, including selection depending on requirement	**K3**
CO4	Synthesize four bar mechanisms for body guidance and function generation	**K2**
CO5	Understand the basics of spatial mechanisms	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1			1		1				1
CO2	2	1	1			1		1				1
CO3	2	1	1			1		1				1
CO4	2	1	1			1		1				1
CO5	2	1	1			1		1				1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Theory of Machines and Mechanisms	Ballaney P. L	Khanna Publishers	2005
2	Theory of Machines	S. S. Rattan	Tata McGraw Hill	2009

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Theory of Machines and Mechanisms,	J. E. Shigley, J. J. Uicker	McGraw Hill	2010
2	Kinematics and Dynamics of Machinery,	Norton	Tata McGraw Hill	2009

SEMESTER: S4

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Course Code	**PBRAT404**	**CIE Marks**	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	**ESE Marks**	40
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	**Course Type**	Theory

Course Objectives:

1. Introduce fundamental artificial intelligence and machine learning concepts for robotics through project-based learning.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of artificial intelligent systems, different types, intelligent agents, problem solving by search. Basics of machine learning, different types, examples, features, feature vector, training set, target vector, test set, feature extraction, over-fitting, curse of dimensionality. Data transformation: scaling, normalization, encoding.	**11**
2	Linear discriminant-based algorithm: perceptron, gradient descent method, perceptron algorithm, support vector machines, separable classes, non-separable classes, multiclass case.	**11**
3	Introduction to robot localization problem, Probabilistic techniques for robot pose estimation, Monte Carlo Localization (MCL), Bayes Rule, Theorem of Total Probability.	**11**
4	Introduction to motion planning problem, Search Algorithms - Breadth-First Search (BFS), A* Search, Dynamic Programming.	**11**

Course Assessment Method

(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks **(8x2 =16marks)**	• Each question carries 6 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 2 sub divisions. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Apply the basics of artificial intelligence, machine learning and different types	**K3**
CO2	Employ discriminant-based algorithms	**K3**
CO3	Use statistical methods for localisation of robots	**K3**
CO4	Operate motion planning techniques	**K3**
CO5	Apply the concepts learnt in the course to develop a project and prepare a brief report.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	2	3			1	2	2	1	1
CO2	3	1	2	2	3			1	2	2	1	1
CO3	3	1	2	2	3			1	2	2	1	1
CO4	3	1	2	2	3			1	2	2	1	1
CO5	3	3	3		3				3	3	2	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Pattern Recognition and Machine Learning	Bishop, C. M	Springer	2006
2	Pattern Recognition	Theodoridis, S. and Koutroumbas, K.	Academic Press	2003
3	Principles of Robot Motion Theory, Algorithms, and Implementation	Choset H. et al.	MIT Press	2005

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Mastering Machine Learning with scikit-learn	Gavin Hackeling	Packt	2014
2	Deep Learning with Pytorch	Eli Stevens et al.	Manning	2020
3	Mobile Robot Localization and Map Building	Jose A. et al.	Springer Science	2000

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc24_cs81/preview
2	https://onlinecourses.nptel.ac.in/noc24_cs81/preview
3	https://onlinecourses.nptel.ac.in/noc24_cs81/preview https://nptel.ac.in/courses/107106090
4	https://onlinecourses.nptel.ac.in/noc24_cs81/preview https://nptel.ac.in/courses/107106090

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	**5**
2	Contribution in Progress Presentations and Question Answer Sessions	**4**
3	Involvement in the project work and Team Work	**3**
4	Execution and Implementation	**10**
5	Final Presentations	**5**
6	Project Quality, Innovation and Creativity	**3**
Total		**30**

Project Assessment and Evaluation criteria (30 Marks)

1. **Project Planning and Proposal (5 Marks)**

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. **Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. **Involvement in the Project Work and Team Work (3 Marks)**

- Active participation and individual contribution
- Teamwork and collaboration

4. **Execution and Implementation (10 Marks)**

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. **Final Presentation (5 Marks)**

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. **Project Quality, Innovation, and Creativity (3 Marks)**

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER: S4

SOFT COMPUTING TECHNIQUES

(Common to RU Branches/ B-Group/Groups)

Course Code	**PERAT411**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	**Course Type**	Theory

Course Objectives:

1. Understand Soft Computing concepts, technologies, and applications
2. Understand applications Fuzzy and ANN
3. Develop Neuro-Fuzzy and Neuro-Fuzzy-GA expert system.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to soft computing and Artificial Neural Networks:** Introduction to Soft Computing, Difference between Soft and Hard computing, Artificial neural networks Vs Biological neural networks, ANN architecture, Basic building block of an artificial neuron, Connections, Learning, Activation functions, Introduction to Early ANN architectures (basics only) - McCulloch & Pitts model	8
2	**ANN-Supervised Learning:** Neuron as a simple computing element, The perceptron, Backpropagation networks: architecture, multilayer perceptron, backpropagation learning-input layer, accelerated learning in multilayer perceptron, The Hopfield network, Bidirectional associative memories (BAM), Radial Basis Function Neural Network	9

3	**ANN: Unsupervised Learning:** Hebbian Learning, Generalized Hebbian learning algorithm, Competitive learning, Self- Organizing Computational Maps: Kohonen Network. **Fuzzy Logic**: Crisp & fuzzy sets, fuzzy relations, Min-Max Composition- features of membership functions, fuzzy conditional statements, fuzzy rules, fuzzy algorithm. Fuzzy logic controller. Defuzzification Methods: Lambda cut method.	**10**
4	Genetic algorithms: basic concepts, encoding, reproduction-Roulette wheel, tournament, rank, and steady state selections, cross over, mutation, fitness function, Convergence of GA, Applications of GA case studies.	**9**

Course Assessment Method (CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe soft computing techniques and the basic models of ANN	**K2**
CO2	Develop the concepts of supervised and unsupervised learning of Neural Networks	**K2**
CO3	Illustrate the operations, model and applications of Fuzzy logic	**K3**
CO4	Understand the concepts of Genetic Algorithm	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									3
CO2	3	2	1	2								3
CO3	3	2	1	2								3
CO4	3	2	1									3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Neural Networks, Fuzzy Logic, and Genetic Algorithms: Synthesis and Applications	R. Rajasekaran and G. A Vijayalakshmi Pai	1. Prentice Hall of India	June 2013
2	Principles of Soft Computing	S.N Sivandam and S.N Deepa	Wiley Publications.	2nd edition Wiley
3	Genetic Algorithms in Search, Optimisation, and Machine Learning	D. E. Goldberg	Addison-Wesley	
4	Fuzzy Logic with Engineering Applications	Timothy J Ross,	McGraw Hill	1997
5	Introduction to Evolutionary Computing.	Eiben A. E. and Smith J. E.	Springer,Natural Computing Series	Second Edition,2007

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Soft Computing & Intelligent Systems: Theory & Applications	N. K. Sinha and M. M. Gupta	Academic Press /Elseivier	Elsevier. 2009
2	Neural Network- A Comprehensive Foundation	Simon Haykin	Prentice Hall International, Inc.	
3	Computational Intelligence: Concepts to Implementation	R. Eberhart and Y. Shi	Morgan Kaufman/Elsevier,	2007
4	Neural Network and Fuzzy Systems	Bart Kosko	Neural Network and Fuzzy Systems	

n

Video Links (NPTEL, SWAYAM…)	
Sl No.	**Link ID**
1	Introduction To Soft Computing By Prof. Debasis Samanta, IIT Kharagpur (swayam course)

SEMESTER: S4

PLC & DISTRIBUTED CONTROL SYSTEMS

(Common to RU Branch)

Course Code	**PERAT412**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	**Course Type**	Theory

Course Objectives:

1. Understand the fundamental concepts of Programmable Logic Controllers
2. Provide insight into the Programming of PLC
3. Understand the basics of Distributed control system

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Programmable logic controllers: Architecture of PLC -Types of PLC –PLC modules, Input and Output modules –Digital and Analog Input/Output- examples of Digital and Analog Inputs/Outputs in PLC-AC- DC power supplies in PLC- isolators- PLC Configuration -Scan cycle -Capabilities of PLC-Selection criteria for PLC –PLC Communication with PC and software- PLC Wiring-Installation of PLC and its modules. PLC programming languages: Ladder Logic, Functional Block Diagram FBD -Sequential Flow Chart SFC - Structured Text - Instruction List	9
2	PROGRAMMING OF PLC: – Ladder Logic Programming – Realization of simple logic circuits programming on-off inputs/ outputs. Auxiliary commands and functions: PLC Basic Functions: Register basics- timer functions- counter functions. Program control instructions- math instructions- sequencers- PLC based traffic light system, stepper motor & servo motor control using PLC, Analog sensor interfacing with PLC APPLICATIONS OF PLC: Case studies of manufacturing automation and process automation.	9

3	PLC programming tools as per IEC 61131-Developing programs using Sequential Function Chart and FBD Networking PLC: usingRS232, RS485, Protocols-ethernet- Modbus, CANOpen, OPC-Open Platform Communication- OPC UA Industrial Automation: General Block diagrams of DDC- Supervisory control- Data Acquisition system.	**9**
4	Distributed Control System- DCS - Architectures, Comparison, Local control unit, Process interfacing issues, Communication facilities- Distributed Control System Basics: DCS introduction- Various function Blocks- DCS components/block diagram- DCS Architecture of different makes- comparison of these architectures with automation pyramid- DCS specification- latest trend and developments in DCS- DCS support to Enterprise Resources Planning (ERP) - performance criteria for DCS and other automation tools.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Learn PLC hardware and its general architecture.	**K2**
CO2	Design logic circuits to perform industrial control functions of medium complexity and realise the same using ladder logic.	**K3**
CO3	Apply the industrial protocols used with PLC in real applications.	**K3**
CO4	Familiarize the DCS architecture and its functions.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										3
CO2	3	3	2									3
CO3	3	3	2									3
CO4	3	3										3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Programmable Logic Controllers	Frank D Petruzella	McGraw Hill	6th Edition, 2022
2	Distributed Control Systems	Michael P. Lukas	Van Nostrand Reinhold Co.,Canada	1st Edition, 1986

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Programmable Controllers- An Engineers's Guide	E.A. Parr, Newnes,	Butterworth-Heinemann	2nd Edition, 1999
2	Programmable controllers, Hardware, Software & Applications	George L. Batten Jr	Mc GrawHill	2nd Edition, 1994.
3	Programmable logic controllers	W. Bolton	Butterworth-Heinemann	6th Edition, 2015
4	Programmable Logic Controllers: Programming Methods and Applications	John R Hackworth and Fredrick D Hackworth Jr	Pearson Education	2006
5	Programmable Logic Controllers - Principles and Applications	John. W. Webb Ronald A Reis	Prentice Hall Inc	4thEdition, 1998

SEMESTER: S4

ENGINEERING OPTIMIZATION

Course Code	PERAT413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Mins
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Enable the learner to model engineering minima/maxima problems as optimization problems
2. Enable the learner to deploy various constrained and unconstrained optimization algorithms to obtain the minima/maxima of engineering problems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Engineering application of Optimization – Statement of an Optimization problem–Classification, Review of basic calculus concepts –Stationary points; Functions of single and two variables; Convexity and concavity of functions – Definition of Global and Local optima – Optimality criteria, Linear programming methods for optimum design – Standard form of linear programming (LP) problem; Canonical form of LP problem; Simplex Method, Duality, Application of LPP models in engineering	9
2	Optimization algorithms for solving unconstrained nonlinear optimization problems – Search based techniques: Direct search: Fibonacci and golden section search, Hookes and Jeeves , Gradient based method:, Newton's method	9
3	Optimization algorithms for solving constrained optimization problems– direct methods – penalty function methods, barrier method -Optimization of function of multiple variables subject to equality constraints; Lagrangian function– Inequality constrained techniques-KKT conditions-steepest descent method	9
4	Modern methods of Optimization– Metaheuristic techniques: Genetic Algorithms – Simulated Annealing – Particle Swarm optimization –Ant colony optimization–:Engineering Applications of unconstrained and constrained optimization	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Formulate an optimization problem to optimize an engineering application using the principles of basic calculus.	**K2**
CO2	Apply the Simplex method to solve a linear programming problem	**K3**
CO3	Solve the unconstrained optimization problems using gradient based method.	**K3**
CO4	Apply the various optimization techniques to solve a constrained optimization problem	**K3**
CO5	Use metaheuristic algorithms to solve constrained and unconstrained	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									2
CO2	3	3	3									2
CO3	3	2	3									2
CO4	3	2	3									2
CO5	3	2	3									2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Engineering Optimization, Theory and Practice	S.S RAO	New Age International Publishers	4^{th} Edition ,2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Optimization Techniques and Applications with Examples	Xin-She Yang	John Wiley & Sons	2018
2	Optimization for Engineering Design Algorithms and Examples	Deb K	Prentice Hall India	2000
3	Introduction to Optimization Design	Arora J	Elsevier Academic Press, New Delhi	2004
4	Linear Programming	Hardley G	Narosa Book Distributors Private Ltd	2002
5	Genetic Algorithms and engineering optimization	Mitsuo Gen, Runwei Cheng	John Wiley & Sons	2002
6	An introduction to optimization	Edwin KP Chong, Stanislaw, H Hak	John Wiley & Sons	Fourth Edition, 2013

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	NPTEL https://www.youtube.com/watch?v=a2QgdDk4Xjw
2	NPTEL https://www.youtube.com/watch?v=dPQKltPBLfc
3	NPTEL https://www.youtube.com/watch?v=qY-gKL7GxYk
4	NPTEL https://www.youtube.com/watch?v=Z_8MpZeMdD4 https://www.youtube.com/watch?v=FKBgCpJlX48

SEMESTER: S4

COMPUTER ORGANIZATION AND ARCHITECTURE

Course Code	PERAT414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To introduce principles of computer organization and basic architectural concepts.
2. To introduce the programming of a simple digital computer and simple register transfer language to specify various computer operations.
3. To understand computer arithmetic, instruction set design, microprogrammed control unit, pipelining and vector processing, memory organization and I/O systems, and multiprocessors

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Basic Structure of computers** – functional units - basic operational concepts - bus structures. Memory locations and addresses - memory operations, Instructions and instruction sequencing, addressing modes. Basic processing unit – fundamental concepts – instruction cycle – execution of a complete instruction - single bus and multiple bus organization	9
2	**Register transfer logic**: inter register transfer – arithmetic, logic, and shift micro-operations. **Processor logic design:** - processor organization – Arithmetic logic unit - design of arithmetic circuit - design of logic circuit - Design of arithmetic logic unit - status register – design of shifter - processor unit – design of accumulator.	9
3	**Arithmetic algorithms:** Algorithms for multiplication and division (restoring method) of binary numbers. Array multiplier, Booth's multiplication algorithm. **Pipelining:** Basic principles, classification of pipeline processors, instruction and arithmetic pipelines (Design examples not required), hazard detection and resolution.	9

4	**Control Logic Design:** Control organization – Hardwired control-microprogram control –control of processor unit - Microprogram sequencer, micro programmed CPU organization - horizontal and vertical micro instructions. **I/O organization:** accessing of I/O devices – interrupts, interrupt hardware - Direct memory access.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Recognize and express the relevance of basic components, I/O organization, and pipelining schemes in a digital computer	**K2**
CO2	Explain the types of memory systems and mapping functions used in memory systems	**K2**
CO3	Demonstrate the control signals required for the execution of a given instruction	**K3**
CO4	Illustrate the design of the Arithmetic Logic Unit and explain the usage of registers in it	**K3**
CO5	Explain the implementation aspects of arithmetic algorithms in a digital computer	**K3**
CO6	Develop the control logic for a given arithmetic problem	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	1								3
CO2	3	1	1	1								3
CO3	3	1	1	1								3
CO4	3	2	1	1								3
CO5	3	2	1	1								3
CO6	3	2	1	1								3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Computer Organization , 5/e,	Hamacher C., Z. Vranesic and S. Zaky,	McGraw Hill,	2011
2	Digital Logic & Computer Design,	Mano M. M.,	PHI,	2004
3	Computer architecture and parallel processing	KaiHwang, Faye Alye Briggs,	McGrawHill,	1984

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Digital Logic & Computer Design, 3/e,	Mano M. M.,	Pearson Education	2013.
2	Computer Organization and Design, 5/e,	Patterson D.A. and J. L. Hennessy,	Morgan Kaufmann Publishers	2013.
3	Computer Organization and Architecture: Designing for Performance,	William Stallings,	Pearson, 9/e	2013.
4	Computer Organization and Design, 2/e,	Chaudhuri P.,	Prentice Hall	2008
5	Computer Organization and Architecture,	Rajaraman V. and T. Radhakrishnan,	Prentice Hall	2011

SEMESTER: S4

PRINCIPLES OF COMMUNICATION TECHNIQUES

Course Code	PERAT416	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To introduce the principles of analog and digital communication techniques
2. To introduce the principles of computer communication

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Elements of a communication system, Need for modulation. Amplitude modulation, DSB-SC and SSB systems, AM Transmitter and receiver. Frequency and phase modulation, Narrow and wideband FM and their spectra, FM transmitter and receiver.	9
2	Geometric representation of signals, Signal space, Gram-Schmitt procedure for orthogonisation. Digital modulation schemes: BASK, BFSK, BPSK, QPSK, QAM (analysis not required), Signal constellation of digital modulation schemes, Transmission rate.	9
3	Introduction to computer communication: OSI Layering, TCP/IP layering. Switching: Circuit switching and packet switching Data link layer: Flow control (stop & wait and sliding window), Media access control: Aloha, Slotted Aloha, CSMA, CSMA/CD.	9
4	Network layer: Logical addressing, IPv4 & IPV6, Classes of IP addresses, Subnetting. Routing schemes: Distance vector routing, Link state routing. Transport layer: TCP, UDP, Congestion control.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

<table>
<tr><th>Part A</th><th>Part B</th><th>Total</th></tr>
<tr><td>2 Questions from each module.
Total of 8 Questions, each carrying 3 marks
(8x3 =24marks)</td><td>Each question carries 9 marks.
Two questions will be given from each module, out of which 1 question should be answered.
Each question can have a maximum of 3 sub divisions.
(4x9 = 36 marks)</td><td>60</td></tr>
</table>

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the principles of AM and FM systems.	**K1**
CO2	Use orthogonisation methods to obtain signal constellation of digital communication schemes.	**K3**
CO3	Describe the low-level functionalities for computer communication	**K1**
CO4	Explain the higher-level functionalities for computer communication	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	2							1
CO2	3	2	2	1	2							1
CO3	3	2	2	1	1							1
CO4	3	2	2	1	1							1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Communication Systems	Simon Haykin	Wiley	4th Ed
2	Digital Communications: Fundamentals and Applications",	Bernad Sklar	Pearson	2nd Ed
3	Computer Networks- A Systems Approach	Larry L. Peterson, Bruce S. Davie	Morgan Kaufmann	4th Ed.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Computer Networks	S. Tanenbaum, D. J. Wetherall	Prentice Hall	5th Ed
2	Modern Digital And Analog Communication Systems	B.P. Lathi, Zhi Ding, et al	Oxford University Press	4th Ed.

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	http://www.digimat.in/nptel/courses/video/117102059/117102059.html
2	https://archive.nptel.ac.in/courses/108/104/108104091/
3	https://archive.nptel.ac.in/courses/106/105/106105183/
4	https://onlinecourses.swayam2.ac.in/cec21_cs04/preview

SEMESTER: S4

ROBOT SAFETY AND STANDARDS

Course Code	**PERAT417**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	**Course Type**	Theory

Course Objectives:

1. Understand basic safety guidelines for robotic applications
2. Identify key elements of the robot safety standard
3. Apply provisions of the standard in daily operations

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Robot Safety:** Safety Basics and Features, Need for Safety and the Role of Engineers Robot Safety: Problems and Hazards, Roles of Robot Manufacturers and Users in Robot Safety, Safety Considerations in Robot Design, Installation and Operations, Robot Safeguard Approaches, Robot-Related Safety Problems Causing Weak Points in Planning, Design, and Operation, Workplace Safety	**11**
2	**Robot System Reliability:** Reliability Basics and Configurations, Classifications of Robot Failures and their Causes and Corrective Measures, Robot Effectiveness Dictating Factors, Robot-Related Reliability Measures, Robot Reliability Analysis, Reliability Analysis of Hydraulic and Electric Robots.	**11**

3	**Roboethics:** Roboethics and Levels of Robomorality, Ethics Fundamental Elements and Theories, Top-Down and Bottom Up Roboethics Approach, Ethics in Human-Robot Symbiosis, Robot Rights, Socialized Roboethics, Ethical Issues of Socialized Robots, Case Studies, Additional Roboethics Issues	**11**
4	**Robot Standards:** The Concept of Standard and Standardization, Characteristics and Benefits of Standardization, Standardization Bodies, Standard-Setting, Robot Standards: Electrical Interfaces on Robots for Industrial Environments- End Effector, Manipulating Industrial Robots — Mechanical Interfaces, Safety Requirements for Robotics in Industrial Environment, Safety Design for Industrial Robot Systems, Connectors for Electronic Equipment. Safety of Machinery, Performance Criteria and Related Test Methods for Service Robots	**11**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand basic safety guidelines for robotic applications	**K2**
CO2	Understand robot system reliability	**K2**
CO3	Understand roboethics	**K2**
CO4	Apply provisions of the standard in daily operations	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1			2		2				2
CO2	2		1			2		2				2
CO3	2		1			2		2				2
CO4	2		1			2		2				2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robot System Reliability and Safety: A Modern Approach	B. S. Dhillon	CRC Press, Taylor & Francis Group	2015
2	Roboethics: A Navigating Overview	Spyros G, Taffetas	1st Edition, Springer	2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Human-Robot Interaction: Safety, Standardization, and Bench marking	Paolo Barattini, Federico Vicentini	CRC Press	2019
2	Robot Ethics: The Ethical and Social Implications of Robotics	Abney, Keith, Bekey, George A, Lin, Patrick	The MIT Press	2012
3	SO/TC 299, Robotics	ISO	ISO	2018
4	Robot Framework Test Automation	SumitBisht	PACKT Publishing	2013

COURSE NAME – DIGITAL IMAGE PROCESSING (Common to ------Branches/------Group/Groups)			
Course Code	PERAT 415	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:1	**ESE Marks**	60
Credits	5	**Exam Hours**	2 Hrs 30 mts
Prerequisites (if any)	None/ **(Course code)**	**Course Type**	Theory

Course objectives:

- To introduce the fundamental concepts of Digital Image Processing and study the various transforms required for image processing.
- To study spatial and frequency domain image enhancement and image restoration methods.
- To understand image compression and segmentation techniques.

	Syllabus	**Contact Hours**
Module-I	Digital Image Fundamentals: Image representation, Types of images, Elements of DIP system, Basic relationship between pixels, Distance Measures, Simple image formation model. Brightness, contrast, hue, saturation, Mach band effect. Colour image fundamentals-RGB, CMY, HIS models, 2D sampling and quantization.	**9 hrs**
Module-II	2D Image transforms: DFT, Properties, Walsh transform, Hadamard transform, Haar transform, DCT, KL transform and Singular Value Decomposition. Image Compression: Image compression model, Lossy, lossless compression, Concept of transform coding, JPEG Image compression standard.	**9 hrs**
Module- III	Image Enhancement: Spatial domain methods: Basic Gray Level Transformations, Histogram Processing, Enhancement Using Arithmetic/Logic Operations, Basics of Spatial Filtering, Smoothing spatial Filters, Sharpening spatial Filters. Frequency domain methods: low pass filtering, high pass filtering, homomorphic filtering.	**9 hrs**
Module- IV	Image Restoration: Degradation model, Inverse filtering- removal of blur caused by uniform linear motion, Minimum Mean Square Error (Wiener) Filtering. Image segmentation: Region based approach, clustering , Segmentation based on thresholding, edge based segmentation,	**9 hrs**

	Hough Transform.	

Course Assessment Method (CIE: -40 Marks, ESE: 60 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	*Internal Exam*	*Assignment/ Micro Project*	*Total*
5	15	20	40

Assignments/Micro Projects: 20 Marks

Students should analyze real world image processing problems and implement using any available software.

The following topics may be identified for assignment/ mini project.

1. Illustration of different colour image models and its application.

2. Implementation of image transforms and compression algorithms

3. Examine different spatial and frequency domain filtering techniques on real world example images.

4. Implement image restoration techniques, adjust parameters, and evaluate results qualitatively and quantitatively

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. **(4x9 = 36 marks)**	60

Course Outcomes (COs)

At the end of the course the student will be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Compare different colour model representations of image processing system	**K4**
CO2	Analyse the various concepts and mathematical transforms and compression schemes necessary for image processing	**K4**
CO3	Illustrate the various schemes of image filtering	**K5**
CO4	Determine the techniques for restoration of images	**K5**

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

Course Articulation Matrix (Mapping of course outcomes with program outcomes)**:**

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									2
CO2	3	3	3									2
CO3	3	3	3									2
CO4	3	3	3									2
CO5	3	3	3									2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Textbooks

SL No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Image Processing	Gonzalez Rafel C	Pearson Education	2009
2	Digital Image Processing	S Jayaraman, S Esakkirajan, T Veerakumar	Tata Mc Graw Hill	2015
3				
4				

Reference Books

1	Digital Image Processing	Kenneth R Castleman	Pearson Education	2/e,2003
2	Fundamentals of digital image processing	Anil K Jain	PHI	1988
3	Digital Image Processing	Pratt William K	John Wiley	4/e,2007
4				

Video Links (NPTEL, SWAYAM etc):

Module - I	https://nptel.ac.in/courses/117105079 https://nptel.ac.in/courses/117104069
Module - II	same as above
Module - III	same as above
Module - IV	same as above

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	**UCHUT346**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	**6**
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	**6**

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	**6**
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return -– Payback – Accounting Rate of Return.	**6**

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	**50**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	**K2**
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	**K3**
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	**K2**
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7^{TH} Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: UCHUT346

Course Name: Economics for Engineers

Max. Marks: 50 | Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
		PART B		
		Answer any one full question from each module. Each question carries 8 marks		
		Module 1		
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$.. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNPfc from GDPmp.	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDPmp and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)
*****				

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	**UCHUT347**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Fundamentals of ethics -** Personal vs. professional ethics, Civic Virtue, Respect for others, **Profession and Professionalism,** Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, **Technology and digital revolution**-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, **High technologies: connecting people and places**-accessibility and social impacts, **Managing conflict**, Collective bargaining, **Confidentiality**, Role of confidentiality in moral integrity, **Codes of Ethics**. **Basic concepts in Gender Studies** - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, **Gender disparity and discrimination in education**,	6

	employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations**, Ethical values and practices in connection with gender** - equity, diversity & gender justice, **Gender policy and women/transgender empowerment initiatives.**	
2	**Introduction to Environmental Ethics:** Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). **Sustainable Engineering Principles:** Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. **Ecosystems and Biodiversity:** Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. **Landscape and Urban Ecology:** Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	**6**
3	**Hydrology and Water Management:** Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. **Zero Waste Concepts and Practices:** Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. **Circular Economy and Degrowth:** Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. **Mobility and Sustainable Transportation:** Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	**6**
4	**Renewable Energy and Sustainable Technologies:** Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. **Climate Change and Engineering Solutions:** Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to	**6**

	mitigate, adapt and build resilience to climate change. **Environmental Policies and Regulations:** Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. **Case Studies and Future Directions:** Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	

Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of	G	12

		sustainable development* - Module II, Module III & Module IV		
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
		Total Marks		**50**

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis**: Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts**: Ability to apply course concepts to real-world problems and local contexts.
- **Creativity**: Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills**: Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	**K3**
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	**K4**
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	**K5**
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	**K4**
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessmen	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-

based product) and present findings on its sustainability.

- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on

local communities and ecosystems.

- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).

  Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER: S4

COMPUTER AIDED MODELLING AND SIMULATION OF MECHANICAL SYSTEMS

Course Code	**PCRAL407**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Lab

Course Objectives:

1. To introduce modern CAD packages for drafting and modelling of engineering components
2. To create a digital mock-up of engineering components

Expt. No.	**Experiments**
1	Modelling of machine parts, and brackets using 2D drawings
2	Assembling of machine parts a)Universal joint b) Flange coupling
3	Generation of manufacturing drawings from 3D models/assembly
4	Model based dimensioning of parts
5	Modelling of thin walled parts using sheet metal workbench
6	Modeling of surfaces of the given geometry like helmet, mouse, arms of robots etc
7	Rapid prototyping of simple CAD models
8	Creation of joints at assembly points and animate their relative motion
9	Kinematic simulation of four bar mechanisms
10	Kinematic simulation of slider crank mechanisms
11	Stress Analysis of a cantilever beam
12	Plane Stress Analysis of thin sheets

Course Assessment Method

(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Apply the knowledge of engineering drawings to interpret 2D drawings and model them using software**.**	**K3**
CO2	Prepare standard assembly models and drawings of machine components using part drawings	**K1**
CO3	Prototype models using rapid prototyping	**K3**
CO4	Analyse simple mechanism for its kinematics	**K4**
CO5	Analyse simple structures for stress	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3		2		3					3		
CO2	3		2		3					3		
CO3	3		2		3					3		
CO4	3		2		3					3		
CO5	3		2		3					3		

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Latest manuals of CAD and solid modelling software packages like CATIAV5, UG-NX, Creo, Inventor etc		Respective OEM of software	
2	CAD/CAM: Theory & Practice	Ibrahim Zeid, R Sivasubrahmanian	Tata McGraw Hill Education Private Limited, Delhi	
3	Machine Drawing	N. D. Bhatt and V.M. Pancha	Charotar Publishing House.	

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. **Conduct of Experiments (7 Marks)**

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

   - Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

   - Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.

   - Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
   - Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
   - Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

   - Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

   - Accuracy of Results: Precision and correctness of the obtained results.
   - Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER: S4

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING LAB

Course Code	**PCRAL408**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Lab

Course Objectives:

1. To put into practice different types of regression and classification techniques as a foundation for solving the real-world problems.
2. To understand the effectiveness of clustering and classification techniques via simulation**.**

Expt. No.	**Experiments**
1	Generate a random array representing a set of observations and apply min-max scaling, standardizing and normalizing to it (see 4.1, 4.2 and 4.3 [1])
2	Train and evaluate linear regression model using any available housing dataset (see 13.1 and 11.8 [1])
3	Train and evaluate a polynomial regression model (see 13.3 and 11.8 [1])
4	Reduce the variance of the regression model using regularization (see 13.4 [1])
5	Train and evaluate a naïve-Bayes classifier given continuous, discrete and count features (see 18.1, 18.2 [1])
6	Use k-means clustering to group observations and evaluate the performance (see 19.1 and 11.9 [1])
7	Train, evaluate and visualize a decision tree classifier (see 14.1, 14.3 [1])
8	Train and evaluate classification model using a forest of randomized decision trees (see 14.4 [1])
9	Train and evaluate a boosted model using AdaBoost Classifier (see 14.10 [1])
10	Reduce features using the Principal Components and Matrix factorization (see 9.1 and 9.4[1])

11	Implement robot path plan using BFS algorithm
12	Implement robot path plan using A* search algorithm

Conduct any 10 experiments from above.

Course Assessment Method

(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand sklearn features, functions and built-in datasets to train and evaluate ML models	**K2**
CO2	Implement different clustering schemes using typical ML techniques	**K3**
CO3	Implement different classification schemes using typical ML techniques	**K3**
CO4	Implement robot path planning algorithms	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			2							2
CO2	3	2			2							2
CO3	3	2			2							2
CO4	3	2			2							2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Machine Learning with Python Cookbook-Practical Solutions from Preprocessing to Deep Learning	Chris Albon	O'Reilly Media, Inc.,	1st Ed., 2018
2	Hands-on machine learning with Scikit-Learn, Kerals& TensorFlow-concepts, Tools and Techniques to Build Intelligent Systems	Aurélien Géron,	O Reilly Media Inc.	2nd Ed., 2019

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1				
2				
3				
4				

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**

   - Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
   - Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. **Conduct of Experiments (7 Marks)**

   - Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
   - Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
   - Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

   - Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
   - Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

   - Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

   - Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
   - Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
   - Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
   - Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

   - Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

   - Accuracy of Results: Precision and correctness of the obtained results.
   - Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

   - Ability to explain the experiment, procedure results and answer related questions
   - Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

   - Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

ROBOTICS AND ARTIFICIAL INTELLIGENCE

SEMESTER S5

CONTROL SYSTEMS

Course Code	**PCRAT501**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3 :1: 0: 0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. This course aims the study of analysis and design of control systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**System modelling - Transfer function approach:** Introduction to control systems – Classification of control systems. Principles of automatic control. Feedback control systems – Practical examples – Transfer function – Transfer function of electrical, mechanical and electromechanical systems – Block diagram – Signal flow graph – Mason's gain formula.	**11**
2	**Time domain analysis:** Standard test signals - Response of systems to standard test signals – Step response of second order systems in detail – Time domain specifications – delay time, rise time, peak time, maximum percentage overshoot and settling time. Steady state response – Steady state error- Static & Dynamic error coefficients.	**11**
3	**Stability of linear systems in time domain:** Asymptotic and BIBO stability, Routh-Hurwitz criterion of stability. Root locus - Construction of root locus – Effect of addition of poles and zeros on root locus.	**11**
4	**Controller Design:** Basic Control actions and Controller characteristics: Classification of	**11**

	Controllers, Two position control, proportional, integral and derivative controllers- Principle of working, advantages and disadvantages. Transfer Function of PI, PD and PID controllers.	

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze the control systems by transfer function approach.	**K4**
CO2	Get an adequate knowledge in the time response of systems & steady state error analysis	**K2**
CO3	Learn the concept of stability of control systems and methods of stability analysis.	**K3**
CO4	Design of basic control actions like proportional, integral & derivative controllers.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	3	2										
CO3	2	2										
CO4	3	2										

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Automatic Control Systems	S. Hassan Saeed	KATSON Books	Third Edition
2	Control System Engineering	Norman S Nise	McGraw Hill	Sixth Edition

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Modern Control Engineering	Katsuhiko Ogata	Pearson Education	
2	Control Systems	M. Gopal	McGraw Hill Education	2012
3	Automatic Control Systems.	B.C. Kuo	Pearson Education	
4	Modern Control Systems.	Richard C Dorf and Robert H. Bishop	Pearson Education	

SEMESTER S5

ADVANCED TOPICS IN ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Course Code	PCRAT502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBRAT404	Course Type	Theory

Course Objectives:

1. To introduce advanced topics in artificial intelligence and machine learning.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Convolutional Neural Network (CNN):** Convolutional operation, filters, padding, and stride, activation functions - ReLU, Sigmoid, Tanh, Pooling layers - average pooling, max pooling, Fully connected layers, Training CNN - Loss functions, Optimizers, Backpropagation, Regularization Techniques - Overfitting and underfitting, Dropout, Data augmentation, Advanced CNN Architectures – ResNets, VGG networks, Inception network.	**11**
2	**Sequential Neural Network:** Recurrent Neural Networks, Long Short-Term Memory (LSTM) Networks, Gated Recurrent Units (GRUs), Backpropagation Through Time (BPTT), Gradient Clipping, Attention mechanism.	**11**
3	**Robot Control with AI:** Proportional-Integral-Derivative (PID) control, Tuning PID gains for optimal performance, model predictive control, feedback linearization control.	**11**
4	**Simultaneous Localization and Mapping (SLAM):** Introduction to SLAM, Sensor Modalities for SLAM, Probabilistic Framework for SLAM, SLAM Algorithms - Kalman Filter, Extended Kalman Filter (EKF), Particle Filter, Mapping algorithms: Occupancy grids,	**11**

	landmark-based maps, SLAM for dynamic environments, Loop closure detection.	

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe Convolutional Neural Network.	**K2**
CO2	Explain Sequential Neural Network.	**K2**
CO3	Discuss robot control with AI.	**K2**
CO4	Explain SLAM techniques.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	1	2	-	-	-	-	-	-	1
CO2	3	1	2	1	2	-	-	-	-	-	-	1
CO3	3	1	2	1	2	-	-	-	-	-	-	1
CO4	3	1	2	1	2	-	-	-	-	-	-	1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Pattern Recognition and Machine Learning	Bishop, C. M	Springer	2006
2	Pattern Recognition	Theodoridis, S. and Koutroumbas, K.	Academic Press	2003
3	Principles of Robot Motion Theory, Algorithms, and Implementation	Choset H. et al.	MIT Press	2005

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc24_cs81/preview
2	https://onlinecourses.nptel.ac.in/noc24_cs81/preview
3	https://onlinecourses.nptel.ac.in/noc24_cs81/preview https://nptel.ac.in/courses/107106090
4	https://onlinecourses.nptel.ac.in/noc24_cs81/preview https://nptel.ac.in/courses/107106090

SEMESTER S5

ELECTRIC DRIVES AND CONTROL

Course Code	**PCRAT503**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of electric drives, their components, and their applications, particularly in robotics.
2. To introduce students to various special electrical machines, their construction, working principles, and their roles in modern automation systems.
3. To impart knowledge about power electronics components such as SCRs, controlled rectifiers, DC-DC converters, and inverters, including their operation, control, and protection.
4. To explore the control and operation of DC and AC motor drives, including closed-loop control techniques and advanced methods for speed and torque control

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Electric Drives:** –Block diagram. Advantages, Uses in Robotic Applications. **Introduction to Special Electrical Machines-** DC servo motor, AC servo motors, stepper motor- variable reluctance - permanent magnet- hybrid, BLDC, PMSM- Construction, working and principle of operation	**9**
2	**SCR:** Structure, working, V-I and switching characteristics- Turn on methods of SCR- Gate triggering circuits, R and RC triggering circuits, line synchronised triggering – natural and forced commutation (concept only). Protection of SCR. Requirements of isolation and synchronization	**9**

	in gate drive circuits- Opto and pulse transformer-based isolation. **Controlled Rectifiers** –Single phase fully controlled bridge rectifier with R, RL and RLE loads (continuous & discontinuous conduction) – output voltage equation. Three-phase fully controlled converter with RLE load	
3	**DC-DC converters** – Step down and step up choppers – control methods- two-quadrant & four-quadrant chopper. **DC Motor drives-** Solid state speed control of DC motors- Armature control and field control, Single phase fully controlled converter drives (Rectifier and inverter mode). **Chopper-controlled DC drives-** Regenerative braking control- Four quadrant chopper drives. Closed loop speed and torque control. **Inverters** — 1-phase full-bridge voltage source inverters inverter with R & RL loads- 3-phase bridge inverter with R load – 120° & 180° conduction mode. Voltage control in inverters– Pulse Width Modulation – single pulse width, multiple pulse width & sine PWM-elimination of harmonics- Variable Voltage Variable Frequency Drive (Block diagram only).	**9**
4	Control of Servomotors-Components of typical servo system-with DC and brushless DC servo motor, Feedback system -Sizing of servomotors Position control of Stepper motor- Drive circuit – modes of excitation- open loop and closed loop control of Stepper Motor- applications Permanent Magnet Motor drives- Speed control of BLDC motor- converter circuits, modes of operation - applications. Speed control of PMSM-Self-control- Sensorless Control Digital Control of Electric Drives: Digital Controllers; Microcontroller and DSP-Based Control	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe the construction, working principles, and applications of special electrical machines such as DC servo motors, AC servo motors, stepper motors, BLDC, and PMSM.	**K2**
CO2	Analyze the operation, characteristics, and control methods of power electronic components including SCRs, controlled rectifiers, DC-DC converters, and inverters.	**K4**
CO3	Apply knowledge of DC and AC motor drives to design and implement speed and torque control systems using various control methods.	**K3**
CO4	Design and simulate drive circuits for different types of motors, including closed-loop control systems, using digital controllers such as microcontrollers and DSPs.	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	-
CO2	3	3	2	2	1	-	-	-	-	-	-	-
CO3	3	2	3	2	2	-	-	-	-	-	-	-
CO4	3	2	3	3	2	1	-	-	-	-	-	-

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Power electronics converters applications and design	Ned Mohan, Tore m Undeland, William P Robbins,	John Wiley and Sons.	Third Edition, 2022
2	Power semiconductor control drives	Dubey G. K.	Prentice Hall, Englewood Cliffs, New Jersey,	Third Edition
3	Special Electrical Machines	E. G. Janardhanan	PHI Learning Private Limited.	2014
4	Electrical machines	Nagarath.I.J& Kothari .D.P	Tata McGraw-Hill.	1998

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electric Drives: Concepts and Applications	Vedam Subrahmanyam	McGraw-Hill Education	2nd Edition 2001
2	Fundamentals of Electric Drives	Gopal K. Dubey	Narosa Publishing House	2nd Edition, 2002
3	Electric Motor Drives: Modeling, Analysis, and Control	R.Krishnan	Prentice Hall	2st Edition 2001
4	Control of Electric Machine Drive Systems	Seung-Ki Sul	Wiley-IEEE Press	1st Edition, 2011

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	Fundamentals of Electric Drives by Prof. Shyama Prasad Das, IIT Kanpur: https://onlinecourses.nptel.ac.in/noc22_ee94 NPTEL :: Electrical Engineering - NOC: Fundamentals of Electric Drives
2	Power Electronics by Prof. L. Umanand, IISc Bangalore https://onlinecourses.nptel.ac.in/noc21_ee01/preview
3	Fundamentals of Electric Drives by Prof. Shyama Prasad Das, IIT Kanpur: https://onlinecourses.nptel.ac.in/noc22_ee94 NPTEL :: Electrical Engineering - NOC: Fundamentals of Electric Drives
4	Control of Electrical Drives by Prof. K. R. Rajagopal, IIT Kharagpur

SEMESTER S5

ROBOT OPERATING SYSTEM

Course Code	PBRAT504	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Introduce the Robot Operating System (ROS)
2. Introduce the applications of ROS

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**ROS Fundamentals, Topics:** The ROS Graph, roscore, catkin, Workspaces, and ROS Packages, rosrun, Names, Namespaces, and Remapping, roslaunch, tf: Coordinate Transforms - Poses, Positions, and Orientations, tf, Publishing to a Topic, Subscribing to a Topic, Latched Topics, Defining Message Types, Mixing Publishers and Subscribers	9
2	**Services and Actions:** Defining a Service, Implementing a Service, Using a Service, Defining an Action, Implementing a Basic Action Server, Using an Action, Implementing a More Sophisticated Action Server, Using the More Sophisticated Action	9
3	**Robots and Simulators:** Subsystems - Actuation: Mobile Platform, Actuation: Manipulator Arm, Sensors, Computation Complete Robots – PR2, Fetch, Robonaut 2, TurtleBot, Simulators – Stage, Gazebo. Other Simulators	9

4	**Building Maps of the World and Navigating About the World:** Maps in ROS, Recording Data with rosbag, Building Maps, Starting a Map Server and Looking at a Map, Localizing the Robot in a Map, Using the ROS Navigation Stack, Navigating in Code	**9**

Suggestion on Project Topics

Projects Related to applications of Robot Operating System can be given as micro projects and evaluation can be done by the faculty

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	**30**	**12.5**	**12.5**	**60**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks **(8x2 =16 marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 2 sub divisions. • Each question carries 6 marks. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Apply ROS fundamentals and topics	**K3**
CO2	Apply ROS services and actions	**K3**
CO3	Describe robots and simulators	**K2**
CO4	Apply ROS to build maps of the World and navigate	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	1	3	-	-	1	2	2	1	1
CO2	3	1	2	1	3	-	-	1	2	2	1	1
CO3	3	1	2	1	3	-	-	1	2	2	1	1
CO4	3	1	2	1	3	-	-	1	2	2	1	1

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Programming Robots with ROS: A Practical Introduction to the Robot Operating System	Morgan Quigley, Brian Gerkey, and William D. Smart	O'Reilly	2015
2	A Gentle Introduction to ROS	Jason M. O'Kane		2016

Video Links (NPTEL, SWAYAM…)

Module No.	Link ID
1	'Robotics' Video Lectures from IIT Bombay by Prof. P. Seshu, Prof. P.S. Gandhi, Prof. K. Kurien Issac, Prof. B. Seth, Prof. C. Amarnath - Mechanical Engineering NPTEL Video Lectures (nptelvideos.com)

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		**30**

1. **Project Planning and Proposal (5 Marks)**
    - Clarity and feasibility of the project plan
    - Research and background understanding
    - Defined objectives and methodology
2. **Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**
    - Individual contribution to the presentation
    - Effectiveness in answering questions and handling feedback
3. **Involvement in the Project Work and Team Work (3 Marks)**
    - Active participation and individual contribution
    - Teamwork and collaboration
4. **Execution and Implementation (10 Marks)**
    - Adherence to the project timeline and milestones
    - Application of theoretical knowledge and problem-solving
    - Final Result
5. **Final Presentation (5 Marks)**
    - Quality and clarity of the overall presentation
    - Individual contribution to the presentation
    - Effectiveness in answering questions
6. **Project Quality, Innovation, and Creativity (3 Marks)**
    - Overall quality and technical excellence of the project
    - Innovation and originality in the project
    - Creativity in solutions and approaches

SEMESTER S5

DESIGN OF TRANSMISSION ELEMENTS

Course Code	**PERAT521**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To learn fundamentals of designing transmission elements such as, belts and Gear
2. To study design principles and criteria for clutches
3. To learn about design criteria for bearings

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Belt and Chain Drives: Flat Belt Drive: Belt Construction, Design of flat belt drive with pulley and Selection of Belts from catalogues. V-Belt Drive: Design of V belt drive Selection of V-Belts from catalogues, Timing belt drives selection procedure.	9
2	Gear Design: Design of straight tooth spur & helical gears based on strength and wear considerations — Bevel and Worm gears: Specifications and design of bevel and worm gears	9
3	Design of plate clutches –axial clutches-cone clutches-internal expanding rim clutches- Band and Block brakes — external shoe brakes — Internal expanding shoe brake.	9
4	Sliding contact bearing: - lubrication, lubricants, viscosity, journal bearings, hydrodynamic theory, Petroff's equation, bearing characteristic number, Sommerfeld number, Heat generated in bearings, Heat dissipated by	9

	bearings, Design procedure of Journal bearings. Ball and roller bearings: - Types, bearing life, static and dynamic load capacity, Stribeck's Equation, selection of bearings, selection of taper roller bearings, Design procedure of Ball and roller bearings, Needle bearings.	

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Ability to Select and Design Belt and Chain Drive Systems	**K3, K4**
CO2	Able to get Proficiency in Designing various types Gears	**K3, K4**
CO3	To study design clutches mechanism	**K3, K4**
CO4	Able to demonstrate competence in designing different types of bearings	**K3, K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3		1								
CO2	3	3		1								
CO3	3	3		1								
CO4	3	3		1								

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	A Textbook of Machine Design	R S Khurmi and J K Guptha	S Chand	25 ,2020
2	Design of Transmission element	T J Prabhu	T J Prabhu	7,2015
3	Mechanical Engineering Design	J. E. Shigley,	McGraw Hill,2003	10,214
4	Machine Design	Jalaludeen,	Anuradha Publications,	2,2016
5	Design of Machine elements, , 2016	V.B.Bhandari,	McGraw Hill	4,2006

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fundamentals of Machine Component Design	Juvinall R.C &Marshek K.M.,	, John Wiley,	**6,** 2011
2	Design of Machine Elements	Spottes, M.F	Prentice Hall	3,1994.
3	Machine Design	RajendraKarwa,	Laxmi Publications (P) LTD,	3,2006
4	Mechanical Design of Machines,	Siegel, Maleev& Hartman	International Book Company	6,2008

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/112/106/112106137/
2	https://archive.nptel.ac.in/courses/112/106/112106137/
3	https://archive.nptel.ac.in/courses/112/106/112106137/
4	https://archive.nptel.ac.in/courses/112/106/112106137/

Design Data Books (permitted for reference in the university examination)

1. Mahadevan, K., and K. Balaveera Reddy, Design Data Handbook, Mechanical Engineers in SI and Metric Units. CBS Publishers & Distributors, New Delhi, 2018.

2. NarayanaIyengar B.R &Lingaiah K, Machine Design Data Handbook, Tata McGraw Hill/Suma Publications, 1984

3. PSG Design Data, DPV Printers, Coimbatore, 2012

SEMESTER S5

ADVANCED MECHANISM DESIGN

Course Code	PERAT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Ability to analyse and design complex mechanisms involving multiple degrees of freedom and constraints.
2. Mastery of advanced synthesis methods for designing mechanisms that meet specified kinematic requirements.
3. To integrate mechanism design principles with robotics and automation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction – review of fundamentals of kinematics - analysis and synthesis – terminology, definitions and assumptions – planar, spherical and spatial mechanisms‟ mobility – classification of mechanisms – kinematic Inversion – Grashoff`s law Position and displacement – complex algebra solutions of planar vector equations – coupler curve generation velocity – analytical methods - vector method – complex algebra methods – Freudenstein‟s theorem	9
2	Planar complex mechanisms - kinematic analysis - low degree complexity and high degree complexity, Hall and Ault`s auxiliary point method – Goodman‟s indirect method for low degree of complexity mechanisms Acceleration – analytical methods – Chase solution - Instant centre of acceleration. Euler-Savory equation - Bobillier construction	9

3	Synthesis of mechanisms: Type, number and dimensional synthesis – function generation – two position synthesis of slider crank and crankrocker mechanisms with optimum transmission angle – three position synthesis – structural error – Chebychev spacing - Cognate linkages – Robert-Chebychev theorem – Block"s method of synthesis, Freudenstein"s equation	**9**
4	Kinematic analysis of spatial revolute-Spherical-Spherical-Revolute mechanism – Denavit- Hartenberg parameters – forward and inverse kinematics of robotic manipulators	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To compute mobility and motion parameters	**K3, K4**
CO2	Ability to analyse low degree and high degree planar complex mechanisms	**K3, K4**
CO3	To design two- , and three- position synthesis; apply Chebychev spacing; describe cognate linkages	**K3, K4**
CO4	To analyse RSSR mechanism and apply D–H notation for direct and inverse kinematics	**K3, K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3		1								
CO2	3	3		1								
CO3	3	3		1								
CO4	3	3		1								
CO5												

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Theory of Mechanism and Machines	Amitabha Ghosh, Asok Kumar Mallik	East-West Press	3,1998
2	Advanced Mechanism Design: Analysis and Synthesis Vol. I	Sandor and Arthur G. Erdman	PHI	2,1985
3	Advanced Mechanism Design: Analysis and Synthesis Vol.II	Sandor and Arthur G Erdman	PHI	2,1985

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Theory of Machines and Mechanisms	Joseph E. Shigley ,John Joseph Uicker	McGraw Hill	2,1995

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/112/104/112104121/
2	https://archive.nptel.ac.in/courses/112/104/112104121/
3	https://archive.nptel.ac.in/courses/112/104/112104121/
4	https://archive.nptel.ac.in/courses/112/104/112104121/

SEMESTER S5

ROBOT DYNAMICS AND CONTROL

Course Code	**PERAT523**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To learn and understand generalized co-ordinates, Jacobian matrix Mass Distribution and other fundamental equations
2. To understand Lagrangean and Hamiltonian mechanics
3. To understand linearity's nonlinearities in control system

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	ROBOT FORCE MODELS: Generalized co-ordinates - Generalized Forces - Equation of Motions – Static Forces in Manipulators - Jacobian matrix - Jacobians in The Force Domain - Cartesian Transformation of Velocities and Static Forces - Acceleration of A Rigid Body - Mass Distribution –Non-rigid Body Effects - Newton's Equation - Euler's Equation – Langrage Equation	9
2	ROBOT DYNAMICS General Expressions for Kinetic and Potential Energy - Kinetic Energy for an n-Link Robot - Potential Energy for an n-Link Robot - Equations of Motion - Lagrangean Multiplier - Langrage's Equation - Hamilton Equation - Hamilton vector Field- Euler - Langrage Equation – State Vector and Equation Formulation	9

3	ROBOT CONTROL SYSTEM The manipulator control problem, Linear second-order model of manipulator. Functions of controller and power amplifier. Joint actuators-stepper motor, servo motor. Control Schemes: PID control scheme – Position and force control schemes. Robotic sensors and its classification, Internal sensors – Position, velocity, acceleration and force information, External Sensors – Contact sensors-Limit switches, piezoelectric, pressure pads, Non-contact sensors – Range sensors, Vision sensor- robotic vision system, Description of components of vision system.	**9**
4	CONTROL OF MANIPULATORS Linear Time Varying and Linearization – Input and Output Stability - Background: The Frobenius Theorem - Single-Input Systems. Introduction to nonlinear system – time varying systems - multi-input, multi-output control systems - the control problem for manipulators - practical considerations - current industrial-robot control systems - Lyapunov stability analysis – Cartesian - based control systems - adaptive control - Limit Cycle - Describing Function	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	To understand the dynamics and control of robotics and manipulators.	**K2, K3**
CO2	To model and analyse the dynamics of robotic systems using advanced mathematical techniques such as Lagrangian and Hamiltonian mechanics.	**K3, K4**
CO3	To analyse, design, and implement control systems with sensors for robotic manipulators.	**K3, K4**
CO4	To study linear time-varying systems, nonlinear control methods for robotic manipulators.	**K3, K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3		1								
CO2	3	3		1								
CO3	3	3		1								
CO4	3	3		1								

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robot modelling and control"	Mark W. Spong, Seth Hutchinson, M. Vidyasagar.,"	Wiley	2,2020
2	Introduction to Robotics – Mechanics and control	John J. Craig,	Prentice hall	3,2022
3	Industrial Robotics Technology, Programming and Applications",.	Groover. M.P., Weis. M., Nagel. R.N. and Odrey.N.G. "	McGraw Hill	2,2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robotics Control, Sensing, Vision and Intelligence	K.S.Fu, Gonzalez, R.C. and Lee, C.S.G.	McGraw Hill	**2,** 1987
2	Introduction to Robotics: Analysis, Control, Applications	Saeed B. Niku,	John Wiley & sons	2,2020
3	Robotics Engineering – An Integrated Approach	Klafter. R.D., Chmielewski, T.A. and Negin. M.	Prentice-Hall of India Pvt. Ltd	3,2006

Video Links (NPTEL, SWAYAM…)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/107/112107289/
2	https://archive.nptel.ac.in/courses/112/107/112107289/
3	https://archive.nptel.ac.in/courses/112/107/112107289/
4	https://archive.nptel.ac.in/courses/112/107/112107289/

SEMESTER S5

ADDITIVE MANUFACTURING AND 3D PRINTING

Course Code	PERAT524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Understand the working principle and process parameters of Additive Manufacturing processes.
2. Apply the knowledge in Modelling of AM components
3. Select the suitable material and process for fabricating a given product.
4. Understand the different post processing methods for AM.
5. Explore the applications of AM processes in various fields.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Additive manufacturing: Importance of Additive Manufacturing- Role of additive manufacturing in product design and development-Basic principle of additive manufacturing- Procedure of product development in additive manufacturing. -Classification of additive manufacturing processes, Materials used in additive manufacturing- Benefits & Challenges in Additive Manufacturing.	9
2	CAD Data Processing for AM: - CAD model development; Overview on Data requirements, Data formats (STL, SLC, CLI, RPI, LEAF, IGES, HP/GL, CT, STEP), Overview on slicing methods; Part orientation and support generation; Design of support structure for AM; Outline on AM software's-Introduction to slicing software's: Cura.	9

3	Liquid, Solid & Powder Based AM Technologies: Stereolithography; FDM; LOM; SLS, Direct Metal Laser Sintering; Working Principles, products, materials, merits, drawbacks and applications Direct Energy Deposition AM Process: Laser Engineered Net Shaping (LENS), Direct Metal Deposition (DMD), Electron Beam based metal deposition: working principles, products, benefits and drawbacks, applications.	**9**
4	Post Processing of AM Parts: Overview on support material removal, Surface quality and aesthetic improvement. Conventional Tooling Vs. Rapid Tooling, Classification of Rapid Tooling, Direct and Indirect Tooling Methods Applications of Additive Manufacturing: –Biomedical-Manufacturing–Aerospace- Automotive-Electronics.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals and process parameters of AM processes.	**K1, K2**
CO2	Design and develop a product for AM process	**K2, K3**
CO3	Discuss various additive manufacturing processes and explore the suitability of AM process for various applications.	**K1, K2**
CO4	Apply Additive Manufacturing techniques for obtaining solutions	**K1, K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			2							
CO2	3	2	3		3							
CO3	3	2			2							
CO4	3	2	2		2							

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Rapid prototyping: Principles and applications	Chua C.K., Leong K.F., and Lim C.S.,	World Scientific Publishers	3rd Edition, 2010
2	Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing	Ian Gibson, David W Rosen, Brent Stucker	Springer	2nd Edition, 2015
3	3D Printing and Additive Manufacturing: Principles and Applications.	Chee Kai Chua, Kah Fai Leong	, World Scientific Publishers	Fourth Edition of Rapid Prototyping, 2014

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Rapid Manufacturing: The Technologies and Applications of Rapid Prototyping and Rapid Tooling.	D.T. Pham, S.S. Dimov	Springer	2001
2	Rapid Prototyping: Principles and Applications in Manufacturing.	Rafiq Noorani	John Wiley & Sons	2006

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/112/103/112103306/
2	https://archive.nptel.ac.in/courses/112/103/112103306/
3	https://archive.nptel.ac.in/courses/112/103/112103306/
4	https://archive.nptel.ac.in/courses/112/103/112103306/

SEMESTER S5

INDUSTRIAL HYDRAULICS

Course Code	**PERAT526**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	**Course Type**	Theory

Course Objectives:

1. To know about the basic elements of closed loop hydraulic system
2. To do the sizing and system design of typical hydraulic systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Hydraulic Power Generators – Selection and specification of pumps, pump characteristics. Linear and Rotary Actuators –election, specification and characteristics.	8
2	Control and regulation elements: Pressure - direction and flow control valves -relief valves, non-return and safety valves - actuation systems.	8
3	Hydraulic circuits: Reciprocation, quick return, sequencing, synchronizing circuits - accumulator circuits- industrial circuits - press circuits – hydraulic milling machine - grinding, planning, copying, - forklift, earth mover circuits	10
4	Circuits: sequential circuits - cascade methods - mapping methods – step counter method, compound circuit design - combination circuit design. Selection and sizing: selection of components – design calculations – application -fault finding of circuits - use of microprocessors for sequencing -PLC, Low cost automation – Robotic circuits	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the various components in industrial hydraulic system	**K2**
CO2	Understand the various hydraulic circuits used in industries	**K2**
CO3	To design simple hydraulic circuits	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	3							
CO2	3	3		2	3							
CO3	3	3	3	2	2							

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Hydraulic and Pneumatics	Andrew Parr	Jaico Publishing House,	1999

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fluid Power with Applications	Antony Espossito	Prentice Hall	1980
2	Pneumatic and Hydraulic Systems	Bolton. W.	Butterworth – Heinemann,	1997
3	Hydraulic and Pneumatic Controls: Understanding made Easy	K. Shanmuga Sundaram,	Chand & Co Book publishers,	2006 (Reprint 2009)

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	NPTEL :: Mechanical Engineering - Fundamentals of Industrial Oil Hydraulics and Pneumatics
2	Industrial Hydraulics and Automation - Course (nptel.ac.in)

SEMESTER S5

DYNAMICS OF MACHINERY

Course Code	PERAT527	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Impart knowledge on dynamic analysis and balancing of mechanisms
2. Familiarise with operational characteristics of a mechanical drive and brake systems.
3. Familiarize with mathematical modelling and analysis of mechanical vibration systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Static force Analysis:** Static equilibrium, Equilibrium of two and three force members, Members with two forces and torque, Free body diagrams, Principle of superposition. Static force analysis of four bar mechanism and Slider-crank mechanism without friction. (numerical problems) **Dynamic force Analysis:** D'Alembert's principle, Inertia force, Inertia torque. (theory only)	9
2	**Balancing of rotating masses**: Static and dynamic balancing, balancing of single rotating mass by balancing masses in same plane and in different planes. Balancing of several rotating masses by balancing masses in same plane and in different planes. Analytical and/or graphical methods. (numerical problems) **Bearings:** Sliding contact bearings - types, Lubrication in bush bearings, thick & thin film lubrication, Coefficient of friction in boundary & hydrodynamic regions, hydrostatic & hydrodynamic bearings; Rolling contact bearings - conformity, types of ball and roller bearings. (theory only)	9

3	Belt drives: Angular Velocity Ratio, Effect of Slip, Law of Belting, Length of Open Belt, Length of Cross Belt, Angle of Arc of Contact, Ratio of Belt Tensions, Power Transmitted, Centrifugal Tension, Condition for Maximum Power Transmission, Initial Belt Tension, Effect of Initial Tension on Power Transmission, Belt Creep. (numerical problems) **Brakes:** Mechanical brakes- block brake with short shoe, analysis of forces acting on drum, Pivoted block brake with long shoe, Band brakes, differential band brake. (numerical problems)	**9**
4	Free vibration: Free vibration of single degree undamped systems, Classical method, Energy method, equivalent systems, Damped free vibration-Viscous damping-underdamping, critical damping, overdamping; Coulomb damping, logarithmic decrement. (numerical problems)	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Perform static force analysis on simple mechanisms	**K3**
CO2	Analyse rotating masses for its unbalance	**K4**
CO3	Analyse belt drives and calculate power, torque, and speed	**K3**
CO4	Determine natural frequency of undamped and damped single degree freedom systems	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								3
CO2	3	3	3	3								3
CO3	3	3	3	3								3
CO4	3	3	3	3								3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Theory of Machines	S. S. Rattan	Tata Mc Graw Hill	2009
2	Theory of Machines: Kinematics and Dynamics	Sadhu Singh	Pearson Education India	2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Digital Logic & Computer Design, 3/e,	Mano M. M.,	Pearson Education	2013.
2	Computer Organization and Design, 5/e,	Patterson D.A. and J. L. Hennessy,	Morgan Kaufmann Publishers	2013.
3	Computer Organization and Architecture: Designing for Performance,	William Stallings,	Pearson, 9/e	2013.
4	Computer Organization and Design, 2/e,	Chaudhuri P.,	Prentice Hall	2008
5	Computer Organization and Architecture,	Rajaraman V. and T. Radhakrishnan,	Prentice Hall	2011

SEMESTER S5

ADVANCED MANUFACTURING PROCESSES

Course Code	**PERAT528**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Manufacturing Processes	**Course Type**	Theory

Course Objectives:

1. Learn how to program CNC machines
2. To introduce machining principles and processes in the manufacturing of precision components and products using non-conventional technologies.
3. To offer a basic understanding of machining capabilities, limitations, and productivity of advanced manufacturing techniques.
4. Understand the working principle and process parameters of Additive Manufacturing processes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	CNC: systems – Principle of operation, components of CNC system, coordinate systems, classification of CNC systems, point-to-point and contouring systems, incremental and absolute programming methods, open loop and closed loop systems, feedback devices. Manual part programming - Computer aided part programming - APT. Programming exercises in turning, drilling and milling operations.	**9**
2	Introduction to Advanced Non-conventional Machining Processes: ultrasonic machining (USM), Abrasive jet machining (AJM), Water jet machining (WJM), Abrasive water jet machining (AWJM), Electrochemical machining (ECM), Electro discharge machining (EDM), Electron beam machining	**9**

	(EBM), Laser beam machining (LBM), Plasma arc machining (PAM), Ion beam machining (IBM).	
3	Introduction to principles of micro and nano fabrication techniques- standard micro machining flow chart. Magnetorheological Finishing (MRF) processes, Magneto-rheological Abrasive Flow Finishing (MRAFF) processes- Principle, equipment and applications. Magnetic float polishing (MFP), Elastic Emission machining (EEM), Ion Beam Machining (IBM)- Nano fabrication using CVD, PVD and LIGA process.	**9**
4	Introduction to Additive manufacturing: Importance of Additive Manufacturing- Role of additive manufacturing in product design and development-Basic principle of additive manufacturing- Benefits & Challenges in Additive Manufacturing- Principle and process-stereo-lithography, selective laser sintering, fused deposition modelling.	**8**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To be conversant with the CNC programming.	**K2, K3**
CO2	Explain various Non-traditional material removal process.	**K1, K2**
CO3	Analyse the processes and evaluate the role of each process of micro machining.	**K1, K2**
CO4	Explain the processes used in additive manufacturing.	**K1, K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			2							
CO2	3	2	3		3							
CO3	3	2			2							
CO4	3	2	2		2							

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Computer control of manufacturing systems	Yoram Koren	TMH	2017
2	Advanced Machining Processes	Jain V.K.	Narosa publishers	2014
3	Introduction to Micromachining	Jain V.K.	Narosa publishers	2014
4	Rapid prototyping: Principles and applications	Chua C.K., Leong K.F., and Lim C.S.,	World Scientific Publishers	3rd Edition, 2010

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	CAD/CAM: Theory & Practice	Ibrahim Zeid, R Sivasubrahmanian	Tata McGraw Hill Education Private Limited	2^{nd} Edition 2009
2	Computer-Aided Design and Manufacturing	M.P. Groover, E.M. Zimmers, Jr.	Prentice Hall of India	1987

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/112/105/112105211/
2	https://archive.nptel.ac.in/courses/112/103/112103202/
3	https://archive.nptel.ac.in/courses/117/108/102108078/
4	https://archive.nptel.ac.in/courses/112/103/112103306/

SEMESTER S5

STRUCTURAL ANALYSIS OF ROBOTIC PARTS

Course Code	**PERAT525**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Students will be able to understand basic concepts of stress, strain and their relations based on linear elasticity.
2. Students will be able to calculate buckling load on columns.
3. Students will be able to develop shear-moment diagrams of a beam and find the maximum moment/shear and their locations.
4. Students will be able to calculate bending stresses on beams.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Simple Stresses & Strains:** Types of stresses & strains, Hooke's law, Stress–strain diagram for mild steel, Lateral strain, Poisson's ratio & volumetric strain, Elastic moduli & the relationship between them, Principles of superposition, Total elongation of tapering bars of circular and rectangular cross sections. Elongation due to self–weight, Thermal stresses.	**9**
2	**Shear Force and Bending Moment:** Definition of beam, Types of beams loadings and supports, Concept of shear force and bending moment, Sign convention, Shear Force and Bending Moment diagrams for cantilever, simply supported and overhanging beams subjected to point loads, uniformly distributed loads, point couples and combination of these loads, Point of inflexion and contraflexure. **Theories of Failure:** Various theories of failure, Maximum Principal Stress	**9**

	Theory, Maximum Principal Strain Theory, Strain Energy and Shear Strain Energy Theory (Von Mises Theory).	
3	**Compound Stresses:** Stress components on inclined planes, General two-dimensional stress system, Principal planes and stresses and Mohr's circle of stresses. **Torsion of circular shafts:** Theory of pure torsion, Derivation of Torsion equations, Torsional rigidity and polar modulus, Assumptions made in the theory of pure torsion, Power transmitted by shaft of solid and hollow circular sections, Combined bending, torsion and end thrust.	**9**
4	**Flexural Stresses:** Theory of simple bending, Assumptions, Derivation of bending equation, Neutral axis, Determination of bending stresses of rectangular section (Solid and Hollow), circular section (Solid and Hollow), I section, T section, angle section and channel section. **Elastic stability of columns:** Short and long columns, Euler's theory of columns, Assumptions, Effective length, Slenderness ratio, Effect of end conditions, Derivations of Euler's Buckling load for different end conditions, Limitations of Euler's theory, Rankine's formula and problems.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	***Internal Ex***	***Evaluate***	***Analyse***	***Total***
5	**15**	**10**	**10**	**40**

Criteria for Evaluation (Evaluate and Analyse): 20 marks

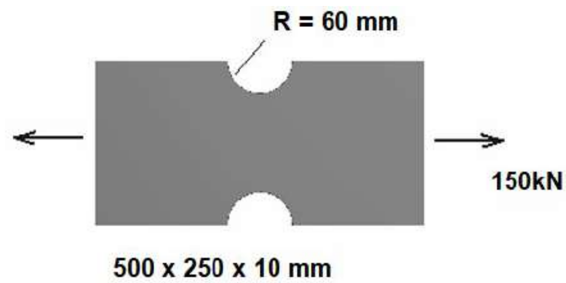
Criteria for Evaluation (Evaluate and Analyse): 20 marks

Evaluation Method: Experiments Using FEM based solvers: (20 marks)

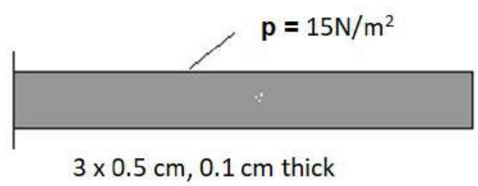
Students can perform specific experiments using tools ANSYS, ABACUS etc.
Each experiment can focus on structural analysis of robotic parts.

Three sample questions for experiments are given below:

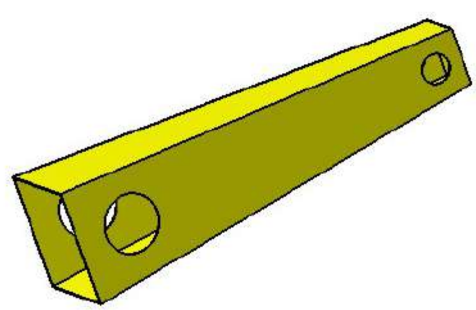
1: Consider a thin structural steel plate as shown below. Compute the maximum normal stress in the horizontal, X direction if 100 kN tensile load is applied to the left and right end surfaces. Compare your result with the value you compute with the help of an FEM based solver



2: Consider the long, thin structural steel beam shown below, fixed at one end, and free at the other. Apply a pressure of $15 N/m^2$ on the upper edge. Compute the vertical deflection at the free end and the maximum bending stress at the top edge at the mid-length of the beam. Look up the theoretical solution in your solid mechanics book and compare it with the help of an FEM based solver.



3: The figure shows the robotic arm made of sheet metal and aluminium. The hollow cross-section is square and measures 80 and 40 mm at the bigger and smaller ends respectively. The thickness of the sheet is 2mm. The arm is fixed at the bigger hole through a bolt and at the smaller hole position, a torque of 10Nm is applied through a shaft. The distance between the center of the holes is 240mm. The arm's length is 300mm and each hole is positioned 30mm from the end. The bigger hole measures 20mm and the smaller hole measures 10mm in diameter. Find the deformation and stress distribution using any FEM package.



End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Analyze the behavior of the solid bodies subjected to axial loading;	**K3**
CO2	Develop shear-moment diagrams of a beam and find the maximum moment/shear and their locations	**K3**
CO3	Determine the stresses in shafts. Determine principal planes and stresses, and apply the results to combined loading case.	**K3**
CO4	Determine the stresses in beams and columns.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								3
CO2	3	3	3	3								3
CO3	3	3	3	3								3
CO4	3	3	3	3								3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Strength of Materials	S S Rattan	McGraw Hill Education India	2nd edition, 2011
2	Mechanics of solids	R. K. Bansal	Laxmi Publications	2004

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Engineering Mechanics of Solids	E. P. Popov, T. A. Balan	Pearson Education	2012
2	Mechanics of Materials	R. C. Hibbeler	Pearson Education	2008

SEMESTER S5

ROBOT OPERATING SYSTEM LAB

Course Code	**PCRAL507**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Lab

Course Objectives:

1. To familiarize the students with how to model and simulate robots in ROS.

Conduct ANY SEVEN of the following experiments on any computing system using ROS.

Expt. No.	Experiments
1	Simulate the turtlebot robot that alternates between driving forward and stopping.
2	Simulate the turtlebot robot that finds the distance to an obstacle in front of the robot.
3	Simulate the turtlebot robot that wanders the environment avoiding obstacles.
4	Simulate the turtlebot robot that moves according to keyboard teleoperation commands.
5	Simulate the turtlebot robot that follows a line.
6	Create a new mobile robot.
7	Simulate the new mobile robot to navigate autonomously.
8	Create a new robotic arm.
9	Simulate a new robotic arm to move as per plan.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	**25**	**20**	**50**

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Use ROS to simulate a wandering robot.	**K3**
CO2	Apply ROS to simulate a robot that can avoid obstacles.	**K3**
CO3	Demonstrate using ROS a simulation of a teleoperated mobile robot.	**K3**
CO4	Operate ROS to simulate a line follower robot.	**K3**
CO5	Implement using ROS a simulation of a new robot that navigates autonomously.	**K3**
CO6	Execute using ROS a simulation of a new robotic arm.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	1	3	-	-	-	2	2	1	1
CO2	3	2	3	1	3	-	-	-	2	2	1	1
CO3	3	2	3	1	3	-	-	-	2	2	1	1
CO4	3	2	3	1	3	-	-	-	2	2	1	1
CO5	3	2	3	1	3	-	-	-	2	2	1	1
CO6	3	2	3	1	3	-	-	-	2	2	1	1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Programming Robots with ROS: A Practical Introduction to the Robot Operating System	Morgan Quigley, Brian Gerkey, and William D. Smart	O'Reilly	2015
2	A Gentle Introduction to ROS	Jason M. O'Kane		2016

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	Robotics' Video Lectures from IIT Bombay by Prof. P. Seshu, Prof. P.S. Gandhi, Prof. K. Kurien Issac, Prof. B. Seth, Prof. C. Amarnath - Mechanical Engineering NPTEL Video Lectures (nptelvideos.com)

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**

   - Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
   - Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. **Conduct of Experiments (7 Marks)**

   - Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
   - Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
   - Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

   - Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
   - Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

   - Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

   - Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
   - Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
   - Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
   - Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

   - Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

   - Accuracy of Results: Precision and correctness of the obtained results.
   - Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

   - Ability to explain the experiment, procedure results and answer related questions
   - Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

   - Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S5

ADVANCED ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING LAB

Course Code	**PCRAL508**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Lab

Course Objectives:

1. To put into practice different advanced artificial intelligence and machine learning algorithms for robot perception and control**.**

Expt. No.	Experiments
1	Image Classification: • Build a CNN to classify images into predefined categories. Experiment with different architectures • Experiment with different architectures (LeNet, AlexNet, VGG, ResNet) and hyperparameters. Evaluate performance using standard metrics.
2	Object Detection: • Train a CNN-based object detection model (e.g., YOLO/Faster R-CNN etc.) on a custom dataset. • Evaluate performance using standard metrics.
3	Image Segmentation: • Implement a CNN-based image segmentation model (e.g., U-Net/DeepLabv3 etc.) for semantic or instance segmentation. Evaluate performance using standard metrics.
4	Text Classification: • Build a Recurrent Neural Network (RNN) or Long Short-Term Memory (LSTM) network for text classification (e.g., sentiment analysis, spam detection). • Experiment with different word embedding techniques (Word2Vec, GloVe).

	Evaluate performance using standard metrics.
5	Time Series Prediction: • Use a Recurrent Neural Network (RNN) or LSTM network to predict future values of a time series (e.g., stock prices, weather data). • Experiment with different window sizes and hyperparameters. Evaluate performance using standard metrics.
6	2D SLAM with Lidar: • Implement a 2D SLAM algorithm (e.g., Gmapping, Hector SLAM) using a simulated or real lidar sensor. Evaluate performance using standard metrics.
7	Visual SLAM: • Implement a visual SLAM algorithm (e.g., ORB-SLAM, DSO) using a simulated or real camera. Evaluate performance using standard metrics.
8	Robot Arm Control using PID: Simulate the control of robot arm using PID controller.

Conduct any 7 experiments from above.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Demonstrate neural network models for robot vision tasks.	**K3**
CO2	Use neural network models for sequential data processing.	**K3**
CO3	Implement machine learning algorithms for robot mapping and localisation.	**K3**
CO4	Use artificial intelligence techniques for robot control.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	2	-	-	-	1	2	-	2
CO2	3	2	1	-	2	-	-	-	1	2	-	2
CO3	3	2	1	-	2	-	-	-	1	2	-	2
CO4	3	2	1	-	2	-	-	-	1	2	-	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Deep Learning with Pytorch	Eli Stevens	Manning	1st Ed., 2020
2	Hands-on machine learning with Scikit-Learn, Kerals& TensorFlow-concepts, Tools and Techniques to Build Intelligent Systems	Aurélien Géron	O Reilly Media Inc.	2nd Ed., 2019

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Machine Learning with Python Cookbook-Practical Solutions from Preprocessing to Deep Learning	Chris Albon	O'Reilly Media, Inc.,	1st Ed., 2018

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	Deep Learning for Computer Vision, https://onlinecourses.nptel.ac.in/noc24_cs89/course
2	Modern Robotics: Mechanics, Planning, and Control Specialization, https://www.coursera.org/specializations/modernrobotics

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

    - Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
    - Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
    - Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
    - Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

    - Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

    - Accuracy of Results: Precision and correctness of the obtained results.
    - Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

    - Ability to explain the experiment, procedure results and answer related questions
    - Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

    - Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

ROBOTICS AND ARTIFICIAL INTELLIGENCE

SEMESTER S6

FIELD ROBOTICS

Course Code	**PCRAT601**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	**Course Type**	Theory

Course Objectives:

1. To give students the ability to apply mathematical modelling to kinematics, localization and path planning of mobile robots
2. To give students the ability to apply various strategies to system level control of mobile robots

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Locomotion:** Introduction, Key issues for locomotion, **Legged Mobile Robots** - Leg configurations and stability, **Wheeled Mobile Robots** - Wheeled locomotion: the design space, Wheel design, Wheel geometry, Stability, Maneuverability, Controllability, **Aerial robots** – advantages of UAVs over conventional manned aircraft, Types of UAVs - Fixed-wing UAV or airplane – characteristics, Rotary-wing UAV - Multi-rotor, Tricopter, Quadcopter or quadrotor, Hexacopter, Y3/Y6, Octocopter **Mobile Robot Kinematics:** Introduction, Kinematic Models and Constraints - Representing robot position, forward kinematic models, Wheel kinematic constraints for Fixed standard wheel, Steered standard wheel and Castor wheel only, Robot kinematic constraints Example of Robot kinematic models and constraints: Differential-drive robot only Mobile Robot Maneuverability - Degree of mobility, Degree of steerability,	**11**

	Robot maneuverability, Mobile Robot Workspace - Degrees of freedom, Holonomic robots	
2	**Mobile Robots Fundamental problems**: Path Planning, Localization, Sensing, Mapping, Simultaneous Localization and Mapping **Mobile Robot Localization:** Introduction, The Challenge of Localization - Noise and Aliasing, Localization-Based Navigation versus Programmed Solutions, Belief Representation - Single-hypothesis belief, Multiplehypothesis belief, Differences between Markov and Kalman Localization (detailed explanation not required) **Representation and Reasoning about Space:** Representing Space - Spatial Decomposition, Geometric Representations, Topological Representations **Representing the Robot:** Configuration Space, Simplification of C-space	**11**
3	**Path Planning for Mobile Robots:** Free-space path versus Semi-free-space path, Non-homotopic paths versus Homotopic paths **Constructing a discrete search space** - Visibility graph, Tangent graph, Retraction Methods **Searching a Discrete State Space** - Graph search - Depth-first Search, Breadth-first Search, Dijkstra's Algorithm, Best first search – A and A* Algorithm **Searching a Continuous State Space** – Potential Fields, Vector field histogram, Bug 1, 2 and Tangent Bug algorithms **Probabilistic path planning** – RPP, PRM, RRT	**11**
4	**System Control:** Horizontal Decomposition, Vertical Decomposition, Hybrid Control Architectures, Middleware, Alternative Control Formalisms, The Human Robot Interface **Off-Board Communication:** Tethered, Untethered – types of untethered communication	**10**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Derive kinematic constraints and models for mobile robots	MISSING
CO2	Describe the techniques used in mobile robot localization	
CO3	Apply various algorithms to path planning for mobile robots	
CO4	Apply various techniques for mobile robotic system control	
CO5	Implement a mobile robot that incorporates functionalities suitable for an application	

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3		3							3
CO2	3	3	3		3							3
CO3	3	3	3		3							3
CO4	3	3	3		3							3
CO5	3	3	3	3	3				3	3	3	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Autonomous Mobile Robots	R Siegwart, IR Nourbakhsh, D Scaramuzza	The MIT Press	2nd edition, 2011
2	Computational Principles of Mobile Robotics	Gregory Dudek, Michael Jenkin	Cambridge University Press	2nd edition, 2010
3	Chapman & Hall_CRC Artificial Intelligence and Robotics Series - A First Course in Aerial Robots and Drones-	Yasmina Bestaoui Sebbane	CRC Press	1st edition, 2022

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Mobile Robot Control	Spyros G. Tzafestas	Elsevier Science Publishing Co Inc	1st edition, 2013
2	Autonomous Mobile Robots and Multi-Robot Systems - Motion-Planning, Communication, and Swarming	Kagan, Shvalb, Ben-Gal	Wiley	1st edition, 2019

Video Links (NPTEL, SWAYAM…)

Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/106/112106298/
2	https://archive.nptel.ac.in/courses/112/106/112106298/
3	https://archive.nptel.ac.in/courses/112/106/112106298/
4	https://archive.nptel.ac.in/courses/112/107/112107289/

SEMESTER S6

ROBOTIC VISION

Course Code	PCRAT602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To introduce image processing and robot vision techniques

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Digital Image Formation and low-level processing:** Fundamentals of Image Formation, Transformation: Orthogonal, Euclidean, Affine, Projective, etc; Fourier Transform, Convolution and Filtering, Image Enhancement, Restoration, Histogram Processing.	9
2	**Depth estimation and Multi-camera views:** Perspective, Binocular Stereopsis: Camera and Epipolar Geometry; Homography, Rectification, DLT, RANSAC, 3-D reconstruction framework; Auto-calibration.	9
3	**Feature Extraction:** Edges - Canny, LOG, DOG; Line detectors (Hough Transform), Corners - Harris and Hessian Affine, Orientation Histogram, SIFT, SURF, HOG, GLOH, Scale-Space Analysis- Image Pyramids and Gaussian derivative filters, Gabor Filters and DWT.	9
4	**Image Segmentation:** Region Growing, Edge Based approaches to segmentation, Graph-Cut, Mean-Shift, MRFs, Texture Segmentation; Object detection.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe digital image formation and low level image processing techniques.	**K2**
CO2	Explain depth estimation and multi-camera views	**K2**
CO3	Execute feature extraction.	**K3**
CO4	Demonstrate image segmentation	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	1	2							1
CO2	3	1	2	1	2							1
CO3	3	1	2	1	2							1
CO4	3	1	2	1	2							1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Computer and Machine Vision -Theory Algorithm and Practicalities	E.R. Davies	Academic Press	2012
2	Computer Vision: Algorithms and Applications,	Richard Szeliski	Springer	2011

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Computer Vision: A Modern Approach	David Forsyth and Jean Ponce	Pearson India	2002

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc23_ee39/preview https://nptel.ac.in/courses/106105216
2	https://onlinecourses.nptel.ac.in/noc23_ee39/preview https://nptel.ac.in/courses/106105216
3	https://onlinecourses.nptel.ac.in/noc23_ee39/preview https://nptel.ac.in/courses/106105216
4	https://onlinecourses.nptel.ac.in/noc23_ee39/preview https://nptel.ac.in/courses/106105216

SEMESTER S6

DIGITAL CONTROL SYSTEMS

Course Code	PERAT631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCRAT 501 (CONTROL SYSTEMS)	Course Type	Theory

Course Objectives:

1. This course aims to analyze and design digital control systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction:** Basic Elements of discrete data control systems, advantages of discrete data control systems, examples. Signal conversion & processing: Digital signals & coding, data conversion & quantization, sample and hold devices. Mathematical modelling of the sampling process. Zero order hold & first order Hold.	9
2	**Discrete time control systems:** Pulse transfer function, Z transform analysis of closed loop and open loop systems- Modified z- transfer function- Steady state error analysis of digital systems- Examples on static error coefficients. Bilinear transformation- mapping from s-plane to z-plane.	9
3	**Analysis of digital control systems:** Stability analysis of linear digital control systems - Routh Hurwitz criteria, Jury's test. Root loci of digital control systems – rules for construction of root locus. Frequency domain analysis - Bode plots- Gain margin and Phase margin.	9
4	**State Space Techniques:** State space representation of discrete time systems- Transfer function from	9

	state space model-various canonical forms, discrete time state transition matrix- conversion of transfer function model to state space model-characteristics equation- solution of discrete state equations.	

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic elements, their functions and interconnections in a digital control system.	**K2**
CO2	Develop the pulse transfer function and steady state error analysis of digital control systems.	**K4**
CO3	Understand frequency domain analysis and analyse stability of linear digital control systems.	**K2**
CO4	Develop state space representation of discrete time systems and find solution of state equation.	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	3	2										
CO3	2	3										
CO4	2	3	2									

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Digital Control and State variable methods	M. Gopal	Tata McGraw Hill	4th Edition
2	Digital control systems	B. C. Kuo	Oxford University Press	2nd Edition
3	Digital Control Engineering-Analysis & Design	M. Sami Fadali & Antonio Visiol	ELSIEVER	2nd Edition
4	Discrete-Time Control Systems	Katsuhiko Ogata	Prentice Hall of India	2nd Edition

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Continuous & Discrete Control Systems	John Dorsey	Tata McGraw Hill	
2	Modern Control Systems	Richard C Dorf and Robert H. Bishop	Pearson Education	

SEMESTER S6

ADAPTIVE CONTROL

Course Code	**PERAT632**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	**Course Type**	Theory

Course Objectives:

1. Understand basic theories of adaptive control
2. Be able to analyse an adaptive control system
3. Differentiate between MRAS and STR

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Adaptive Control – Introduction, block diagram, adaptive systems – 4 approaches with block diagrams, applications of adaptive control, importance of adaptive control, Robust Adaptive control – overview, robust control laws	9
2	Deterministic Self tuning regulators (STR): Pole placement design, Indirect self tuning regulators, Continuous time self tuners, Direct self tuning regulators, disturbances with known characteristics	9
3	Stochastic and Predictive Self tuning regulators: Design of minimum variance and moving average controllers Stochastic self tuning regulators, Unification of direct self tuning regulators. Linear quadratic STR, adaptive predictive control	9
4	Model reference adaptive control (MRAS): The MIT Rule, Determination of adaptation gain, Lyapunov theory, Design of MRAS using Lyapunov theory, BIBO stability, Output feedback, Relations between MRAS and STR	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Familiarize with different types of adaptive control and its importance	**K1,K2**
CO2	Analyze pole placement design and STR	**K2,K3**
CO3	Differentiate between Deterministic Self tuning regulators and Stochastic Self tuning regulators	**K2,K3**
CO4	Discuss the Model Reference Adaptive Control method	**K2,K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2						1	1	1			1
CO2	2		1				2	1	1			1
CO3	2		1				1	1	1			1
CO4	2						1	1	1			1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Adaptive control	Karl.J.Astrom and Bjorn Wittenmark	Dover publications	2^{nd} edition, 2008
2	Adaptive Control: Stability, Convergence and Robustness.	S. Sastry and M. Bodson	Prentice Hall	1989
3	Robust Adaptive Control	P. A. Ioannou and J. Sun	PTR Prentice-Hall	1996

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Stable Adaptive Systems	K. S. Narendra and A. M. Annaswamy	Prentice-Hall	1989
2	Adaptive Control Verlag	I.D. Landau, R. Lozano, and M. M'Saad	Springer	1998

SEMESTER S6

SUPERVISORY CONTROL

Course Code	PERAT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to understand computer controlled systems and supervisory control

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Process Control Loops: Feedback control, feed forward control, Cascade control, Ratio control, Split range control. Computerized control: Basic building blocks of Computer controlled systems, advantages. Programmable Logic Controller: Definition of PLC Architecture of PLC, Input and output modules, Types of inputs and outputs, analog and discrete I/O modules, power supply of PLCs. PLC programming languages, Ladder programming, Relay logic, Timers and counters, simple programs in PLC, math operations, sequencers, program control instructions, analog instructions, case studies- bottle filling system, gate control system	9
2	Direct digital control: block diagram, multiplexers, demultiplexers, ADC, Data Acquisitions systems. SCADA: SCADA Fundamentals: Introduction, Open system: Need and advantages, Building blocks of SCADA systems, Remote terminal unit (RTU), Human-Machine Interface (HMI) subsystem.	9
3	SCADA communication systems, Master Station: Master station software components, Master station hardware components, Server systems in the master station, Small, medium, and large master stations, Global positioning systems (GPS), software architecture of SCADA Distributed Control System : DCS - Architectures, Comparison, Local	9

	control unit, Process interfacing issues, Communication facilities. Various function Blocks	
4	LCU communication Facilities - Communication system requirements – Architectural Issues Operator displays, Operator Interfaces – Engineering Interfaces. Development of Field Control Unit (FCU) diagram for simple control applications. Introduction to HART and Field bus protocol. Interfacing Smart field devices (wired and wireless) with DCS controller. Introduction to Object Linking and Embedding (OLE) for Process Control, Automation in the cloud with case studies.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand basic process control loops	**K2**
CO2	Design and develop ladder-based PLC programs	**K3**
CO3	Illustrate simple computerized process control systems such as DAQ and DDC	**K2**
CO4	Illustrate SCADA systems and its building blocks for industrial automation	**K2**
CO5	Understand Distributed Control System and its applications	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3	2										2
CO3	3	2										2
CO4	3	3										2
CO5	3	3										2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Programmable Logic Controllers	Frank D Petruzella	McGraw Hill	4th Edition, 2011
2	Power System SCADA and Smart Grids	Mini S. Thomas	CRC Press	3rd Edition,2015
3	Distributed Control Systems: Their Evaluation and Design,	Michael P. Lukas	Van Nostrand Reinhold Co.,	1986
4	Distributed computer control for industrial Automation'	D. Popovic and V.P.Bhatkar,'	Marcel Dekker, Inc., Newyork	1990.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Elements of Process Control Applications	Deshpande P.B and Ash R.H	ISA Press, New York,	1995
2	Process Control Instrumentation Technology,	Curtis D. Johnson,	Pearson New International	8th Edition, 2013
3	Computer-based Industrial Control	Krishna Kant	Prentice Hall, New Delhi,	2nd Edition, 2011
4	Principles of Process Control	D. Patranabis , P	Tata McGraw Hill Education	2012

SEMESTER S6

NON-LINEAR CONTROL

Course Code	**PERAT634**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	PCRAT501 Control Systems	**Course Type**	Theory

Course Objectives:

1. To understand the behavior of nonlinear system
2. To familiarize the analysis of nonlinear system with various methods

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Characteristics of non-linear systems - Methods of analysis – Classifications of nonlinearities – Common physical nonlinearities – Saturation or limiter – Deadzone or Threshold – Types of relays - Types of springs – Backlash in Gears – Inherent nonlinearity. Linearisation of Nonlinear system - Autonomous and non autonomous systems-modelling of simple pendulum- mass spring system.	9
2	**Phase plane analysis** Phase Plane Analysis: Singular points- Phase portraits – Analytical method for the construction of Phase trajectories – Graphical method f construction of Phase trajectory – Isoclines Method – Analysis and classification of singular points - Local behaviour of nonlinear systems - Limit cycles on phase plane - Stability- poincare- bendixon theorems.	9
3	**Describing function analysis** Describing Function: Definition - Describing Function Analysis- Describing functions of Typical nonlinearities- Ideal Relay – Relay with deadzone – Simple deadzone – Saturation or Limiter	9

	Closed loop stability using describing function- Stability of limit cycles – Accuracy of describing function analysis - Relative stability from describing from describing function.	
4	**Stability of the system- Lyapunov Stability Theory** Stability of nonlinear systems- Concept - equilibrium points- stability in small and stability in large- Lyapunov's stability definitions – Localisation and stability in the small - Stability by the method of Lyapunov – First methd of Lyapunov- Concept of Lyapunov's stability theorems - Sign definiteness of scalar function Lyapunov stability theorems - Stability of nonlinear system by method of Lyapunov – Krasovskii' theorem- Variable gradient method of generating Lyapunov function	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand nonlinear system behaviour	**K2**
CO2	Analyse nonlinear system using phase plane method	**K3**
CO3	Understand and apply describing function method for analysing nonlinear system	**K3**
CO4	Analyse the stability of nonlinear system using Lyapunov method	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										2
CO2	3	2	2	2								2
CO3	3	2	2	2								2
CO4	3	2	2	2								2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Nonlinear systems	H. K. Khalil	Prentice Hall	3rd Edition, 2002.
2	Applied Nonlinear Control	Jean Jacques Slotine, Weiping Li	Prentice Hall Inc.	1991
3	Nonlinear Systems; Analysis, Stability and Control	Shankar Sastry	Springer	1999
4	Advanced Control System	Nagoor Kani	Rba Publication	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Nonlinear systems analysis	M. Vidyasagar	Society of Industrial and Applied Mathematics	2nd Edition,2002
2	Nonlinear Control Systems, Analysis and Design	H. Marquez	Wiley	2003
3	Nonlinear Control Systems	A. Isidori,	Springer	3rd Edition, 1995
4	Nonlinear Differential Equations and Dynamical Systems	F. Verhulst	Springer	2nd Edition, 1996.

SEMESTER S6

POWER ELECTRONICS

Course Code	PERAT636	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the basic principles of power electronics and the characteristics of power semiconductor devices.
2. To design various power converter circuits including rectifiers, DC-DC converters, inverters, and AC-AC converters.
3. To learn the application of power electronics in controlling electric drives and power systems.
4. To explore the emerging trends and technologies in power electronics and their applications in automation and robotics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Power Semiconductor Devices: Power Diodes, TRIACs, GTOs, Power BJTs, Power MOSFETs, and IGBTs. Thyristors, Static and Dynamic Characteristics. Gate characteristics. Firing methods. Gate drive circuits. Thyristor protection circuits. Series and parallel combinations. Commutation types (no analysis).	9
2	AC-DC Converters: Single-phase Uncontrolled and Controlled Rectifiers. Three-phase Controlled Rectifiers. Inductive, and Capacitive Loads. Power Factor Improvement in Rectifiers. DC-DC Converters: Step up and step down Chopper. Buck, Boost, Buck-Boost Converters. Isolated and non-isolated converters	9
3	Inverters: Single-phase and three-phase inverters. Voltage source inverters – series, parallel, and bridge inverters – current source inverters. 1800 and 1200 modes. PWM inverters.	9

	AC-AC Converters: Single-phase and three-phase AC-AC Converters. Cycloconverters (single phase)	
4	Applications of Power Electronics: D.C Motor control - phase control, Single phase SCR drive, Three phase SCR drive, Chopper controlled DC drives. A.C. Motor control: Controlled – slip system and slip power recovery system, stepper motor drive. UPS, Battery charging techniques. Recent Developments in Power Electronics for Automation and Robotics.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the operating principles of power semiconductor devices and their characteristics.	**K2**
CO2	Design rectifier and chopper circuits for various applications.	**K3**
CO3	Develop inverters and AC-AC converters for various applications.	**K3**
CO4	Apply power electronics concepts in the control of electric drives and power systems. Also, explore the recent trends.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	2	2	1	1							2
CO2	3	3	3	2	2							3
CO3	3	3	3	3	3	1						3
CO4	3	3	3	3	3	2						3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Power Electronics: Circuits, Devices and Applications	Muhammad H. Rashi	Pearson	4th edition, 2013
2	Power Electronics: Converters, Applications, and Design	Ned Mohan, Tore M. Undeland, and William P. Robbins	Wiley	4th edition, 2020

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Modern Power Electronics and AC Drives	Bimal K. Bose	Pearson	1st edition, 2001
2	Fundamentals of Power Electronics	Robert W. Erickson and Dragan Maksimovic	Springer	3rd edition, 2020

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc24_ee88/preview?user_email=sreedeep.ra@adishankara.ac.in
2	https://onlinecourses.nptel.ac.in/noc24_ee88/preview?user_email=sreedeep.ra@adishankara.ac.in
3	https://onlinecourses.nptel.ac.in/noc24_ee88/preview?user_email=sreedeep.ra@adishankara.ac.in
4	https://onlinecourses.nptel.ac.in/noc24_ee88/preview?user_email=sreedeep.ra@adishankara.ac.in

SEMESTER S6

ENERGY SYSTEMS

Course Code	PERAT637	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Introduces various types of renewable energy sources
2. Various means of generating and storing energy and the importance of renewable energy

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Energy Scenario:** Indian Energy Scenario, World Energy Scenario, Indian Energy Sector Reforms, Energy and Environment, Energy Security, Energy conservation act **Energy Efficient Systems:** Reducing pollution and improving efficiency in buildings, Green Building Standards, Types of lamps and their efficiencies	9
2	**Renewable Energy Resources:** Solar Thermal System-Working Principle-Block diagram, Solar Photovoltaic System- Working Principle-Block diagram, Solar cell efficiency calculation, Wind Energy Systems- Working Principle-Block diagram, wind power equation, Energy from Waves and tides- Working Principle-Block diagram, Ocean Thermal Energy System-Working Principle-Block diagram, Energy from Biomass	9
3	**Energy Storage:** Importance of Energy Storage- Means of Storing Energy-Principle of operation and performance comparison. Compressed air storage, Fly wheel Energy Storage, Battery Storage-Battery: Specification, Charging/Discharging rate, Primary and secondary cells-Dry cell, lead acid, lithium ion, Lithium air, Nickel Cadmium, Nickel Metal Hydride Fuel Cell: Working Principle, efficiency	9

4	**Energy Standards** – International Energy Standards-ISO50001, Bureau of Energy Efficiency, star rating **Energy Management:** Significance and general principles of Energy Management, Energy audit-types and procedure, Energy audit report, Instruments for energy auditing **Energy Economics:** Traditional Types of Rates - Single-Part Rates - Two-Part Rates - ThreePart Rates – Numerical problems	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Illustrate Indian and global energy scenario	**K2**
CO2	Elaborate different conventional and non-conventional energy generation schemes and the economics of generation	**K2**
CO3	Analyse principle of operation and performance comparison of various energy storage schemes	**K4**
CO4	Identify major Global and Indian standards for Energy Management	**K2**
CO5	Appraise various aspects of energy economics	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3	3										2
CO3	3	3										2
CO4	3	3										2
CO5	3	3										2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Energy Storage for Power Systems	A.G.Ter-Gazarian	The Institution of Engineering and Technology (IET) Publication, UK	Second Edition
2	Guide to Energy Management	Barney L. Capehart, Wayne C. Turner and William J. Kennedy	The Fairmont Press Inc., 2012	Seventh Edition
3	Electric Power Systems Planning	S. Pabla	Mac Millan India Ltd	First Edition

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Renewable Energy Sources and Emerging Technology	K.C. Kothari, D.P.Ranjan, Rakeshsingal	PHI	2nd Revised edition
2	Energy Resources: Conventional & Non-Conventional	M.V.R. Koteswara Rao	BS Publications/BSP Books (2017)	First Edition
3	Efficient Lighting Applications & Case Studies	Albert Thumann, Scott Dunning	The Fairmont Press, Inc.	First Edition
4	Energy Economics-Concepts, Issues, Markets and Governance	Subhes C. Bhattacharyya	Springer	First Edition

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.youtube.com/watch?v=hMtZl4Op5VQ
2	https://archive.nptel.ac.in/courses/121/106/121106014/
3	https://onlinecourses.nptel.ac.in/noc21_mm34/preview
4	https://onlinecourses.nptel.ac.in/noc20_hs68/preview

SEMESTER S6

SPECIAL ELECTRICAL MACHINES AND DRIVES

Course Code	PERAT638	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand various drives and their control systems used in EV technology.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Stepper Motors and Drives: Types -Variable reluctance, permanent magnet and hybrid motors - Constructional features - principle of operation - modes of excitation. Torque production in variable reluctance stepping motor: Static torque characteristics - position error due to low torque; Performance parameters - resolution - single step response and accuracy; Pull-in and pull-out characteristics – Resonance issues. Unipolar and bipolar drive schemes – Bifilar drives – Open loop position control - starting/stopping rate – Velocity profiling -Closed loop control of stepper motors (Assignment)	9
2	Permanent Magnet Synchronous Motor (PMSM) Drives: Permanent magnet materials and characteristics, PMSM – Principle of operation, SPM and IPM machines, Torque equation, Phasor diagram, Machine modeling, Switching schemes, Motor control - Field oriented control, Flux weakening, Sensorless control, Power controllers. Brushless DC Motor (BLDC) Drives: BLDC motors- constructional details – principle of operation; Speed-Torque characteristics -Torque Pulsation, Machine modeling. Switching schemes for BLDC operation, Motor control - Power controllers	9

	– full wave and half wave converters – regeneration – Hall sensor based control, Phase advance angle control, Sensorless control	
3	Switched Reluctance Motors (SRM) and Drives: SRM – Principle of operation, Inductance profile, torque equation, Motoring and regeneration. Low speed and high speed operation – Characteristics, Energy conversion loop - Energy effectiveness. Power controllers, Control schemes – Six switch converter – Split DC supply converter – R dump, C dump converters. Soft switching converter topologies, Torque ripple minimization	**9**
4	Advanced Magnetless Motor Drives: Synchronous reluctance motors, Permanent magnet synchronous reluctance motors –- Constructional details, Principle of operation, Different topologies, Vernier reluctance motors, Applications. Field oriented control principles, Direct torque and flux control. Vernier permanent magnet motors: Constructional details, Principle of operation, Modelling. Inverters for Vernier PM motors, Vernier PM Motor control, Applications	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the knowledge of stepper motor drives for industrial applications	**K3**
CO2	Apply the knowledge of switched reluctance motor drives for industrial applications	**K3**
CO3	Apply the knowledge of PMSM and BLDC motor drives for industrial applications	**K3**
CO4	Apply the knowledge of magnetless motor drives for industrial applications	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	2				2						2
CO2	3	2				2						2
CO3	3	2				2						2
CO4	3	2				2						2
CO5	3	2				2						2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Electric Vehicle Machines and Drives – Design, Analysis and Application	K T Chau,	John Wiley & Sons Singapore Pte. Ltd.,	2015
2	Modern Electric, Hybrid Electric and Fuel Cell Vehicles - Fundamentals, Theory and Design	Mehrdad Ehsani, YiminGao, Sebastien E. Gay & Ali Emadi,	CRC Press	2005
3	Reluctance Electric Machines-Design and Control	Ion Boldea, Lucian Tutelea	CRC Press	2019

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Electric Motor Drives – Modeling, Analysis and Control	R Krishnan	Pearson Education Ltd.	2015
2	Switched Reluctance Motor Drives–Modeling, Simulation, Analysis, Design and Applications	R. Krishnan	CRC Press	2001
3	Permanent Magnet Synchronous and brushless dc drives	R Krishnan	CRC Press	2010
4	Brushless Permanent Magnet Motor Design	D. C. Hanselman	Magna Physics Pub	2006
5	Advanced Electric Drives	Ned Mohan	John Wiley & Sons Inc.	2014
6	Stepping motors - a guide to theory and practice	Paul Acarnley	IET UK	4th Edition 2002.

SEMESTER S6

ADVANCED CONTROL SYSTEMS

Course Code	PERAT635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	PCRAT501	Course Type	Theory

Course Objectives:

1. This course aims to provide a strong foundation on advanced control methods for modelling, time domain analysis, and stability analysis of linear and nonlinear systems. The course also includes the design of feedback controllers and observers

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	State Space Representation of Systems Introduction to state space and state model concepts- State equation of linear continuous time systems, matrix representation- features- Examples of electrical circuits and dc servomotors. Phase variable forms of state representation- Diagonal Canonical forms- Similarity transformations to diagonal canonical form	9
2	State Space Analysis State transition matrix- Properties of state transition matrix- Computation of state transition matrix using Laplace transform and Cayley Hamilton method. Derivation of transfer functions from state equations. Solution of time invariant systems: Solution of time response of autonomous systems and forced systems. State space analysis of Discrete Time control systems: Phase variable form and Diagonal canonical form representations- Pulse transfer function from state matrix- Computation of State Transition Matrix (problems from 2nd order systems only)	9

3	State Feedback Controller Design Controllability & observability: Kalman's, Gilbert's and PBH tests. - Duality principle State feedback controller design: State feed-back design via pole placement technique State observers for LTI systems- types- Design of full order observer	**9**
4	Nonlinear Systems and stability Types and characteristics of nonlinear systems- Jump resonance, Limit cycles and Frequency entrainment Phase plots: Concepts- Singular points – Classification of singular points. Definition of stability- asymptotic stability and instability. Construction of phase trajectories using Isocline method for linear and nonlinear systems. Lyapunov stability analysis: Lyapunov function- Lyapunov methods to stability of nonlinear systems- Lyapunov methods to LTI continuous time systems	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	*Internal Ex*	*Evaluate*	*Analyse*	*Total*
5	**15**	**10**	**10**	**40**

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Students should be able to solve control system problems and solve using Matlab or any other programming language.

Evaluation Methods:

1. Simulations using software tools: (10 marks)

2. Course Project applying the principles of nonlinear control system analysis:(10 marks)

Project phases: Proposal, Implementation, Testing, Final Report, Presentations and Viva Voce:

Faculty may direct students for the selection of topics for projects from real world examples.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Develop the state variable representation of physical systems	**K3**
CO2	Analyse the performance of linear and nonlinear systems using state variable approach	**K4**
CO3	Design state feedback controller for a given system	**K3**
CO4	Explain the characteristics of nonlinear systems	**K2**
CO5	Apply Lyapunov method for the stability analysis of physical systems	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3	3	2									2
CO3	3	3	3									2
CO4	3	3										2
CO5	3	3	2									2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Control System Engineering,	Nagarath I. J. and Gopal M.,	New Age Publishers,	5/e, 2007
2	Modern Control Engineering,	Ogata K.,	Prentice Hall of India,	5/e, 2010
3	Modern Control System Theory,	Gopal M,	New Age Publishers,	2/e, 1984
4	Analysis and Synthesis of Sampled Data Systems,	Kuo B.C,	Prentice Hall Publications,	2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Nonlinear Systems,	Khalil H. K	Prentice Hall	3/e, 2002
2	Nonlinear Automatic Control,	Gibson J.E.	Mc Graw Hill	1963
3	Control Systems Principles and Design	Gopal M.,	Tata McGraw Hill,	4/e, 2012
4	Applied Nonlinear Control,	Slotine J. E and Weiping Li,	Prentice-Hall,	1991
5	Discrete Time Control Systems,	Ogata K.,	Pearson Education, Asia	2/e, 2015

SEMESTER S6

MOBILE ROBOTICS

Course Code	**PBRAT604**	**CIE Marks**	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	**ESE Marks**	40
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To give students the ability to apply mathematical modelling to kinematics, localization and path planning of mobile robots
2. To give students the ability to apply various strategies to system level control of mobile robots

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Locomotion:** Introduction, Key issues for locomotion, **Legged Mobile Robots** - Leg configurations and stability, **Wheeled Mobile Robots** - Wheeled locomotion: the design space, Wheel design, Wheel geometry, Stability, Maneuverability, Controllability, **Aerial robots** – advantages of UAVs over conventional manned aircraft, Types of UAVs - Fixed-wing UAV or airplane – characteristics, Rotary-wing UAV - Multi-rotor, Tricopter, Quadcopter or quadrotor, Hexacopter, Y3/Y6, Octocopter **Mobile Robot Kinematics:** Introduction, Kinematic Models and Constraints - Representing robot position, Forward kinematic models, Wheel kinematic constraints for Fixed standard wheel, Steered standard wheel and Castor wheel only, Robot kinematic constraints Example of Robot kinematic models and constraints: Differential-drive robot only Mobile Robot Maneuverability - Degree of mobility, Degree of steerability,	11

	Robot maneuverability, Mobile Robot Workspace - Degrees of freedom, Holonomic robots	
2	**Mobile Robots Fundamental problems**: Path Planning, Localization, Sensing, Mapping, Simultaneous Localization and Mapping **Mobile Robot Localization:** Introduction, The Challenge of Localization - Noise and Aliasing, Localization-Based Navigation versus Programmed Solutions, Belief Representation - Single-hypothesis belief, Multiple-hypothesis belief, Differences between Markov and Kalman Localization (detailed explanation not required) **Representation and Reasoning about Space:** Representing Space - Spatial Decomposition, Geometric Representations, Topological Representations	**11**
3	**Path Planning for Mobile Robots:** Free-space path versus Semi-free-space path, Non-homotopic paths versus Homotopic paths **Constructing a discrete search space** - Visibility graph, Tangent graph, Retraction Methods **Searching a Discrete State Space** - Graph search - Depth-first Search, Breadth-first Search, Dijkstra's Algorithm, Best first search – A and A* Algorithm **Searching a Continuous State Space** – Potential Fields, Vector field histogram, Bug 1, 2 and Tangent Bug algorithms **Probabilistic path planning** – RPP, PRM, RRT	**11**
4	**System Control:** Horizontal Decomposition, Vertical Decomposition, Hybrid Control Architectures, Middleware, Alternative Control Formalisms, The Human Robot Interface **Off-Board Communication:** Tethered, Untethered – types of untethered communication	**11**

Suggestion on Project Topics

Students may be given projects on modelling to kinematics, localization and path planning of mobile robots

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks **(8x2 =16 marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 2 sub divisions. • Each question carries 6 marks. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Derive kinematic constraints and models for mobile robots	**K2**
CO2	Describe the techniques used in mobile robot localization	**K2**
CO3	Apply various algorithms to path planning for mobile robots	**K3**
CO4	Apply various techniques for mobile robotic system control	**K3**
CO5	Implement a mobile robot that incorporates functionalities suitable for an application	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3		3							3
CO2	3	3	3		3							3
CO3	3	3	3		3							3
CO4	3	3	3		3							3
CO5	3	3	3	3	3				3	3	3	3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Autonomous Mobile Robots	R Siegwart, IR Nourbakhsh, D Scaramuzza	MIT Press	Second Edition 2012
2	Computational Principles of Mobile Robotics	Gregory Dudek, Michael Jenkin	Cambridge University Press	2nd edition, 2010
3	Chapman & Hall_CRC Artificial Intelligence and Robotics Series - A First Course in Aerial Robots and Drones-	Yasmina Bestaoui Sebbane	CRC Press	1st edition, 2022

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Mobile Robot Control	Spyros G. Tzafestas	Elsevier Science Publishing Co Inc	1st edition, 2013
2	Autonomous Mobile Robots and Multi-Robot Systems - Motion-Planning, Communication, and Swarming	Kagan, Shvalb, Ben-Gal	Wiley	1st edition, 2019

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/112/106/112106298/
2	https://archive.nptel.ac.in/courses/112/106/112106298/
3	https://archive.nptel.ac.in/courses/112/106/112106298/
4	https://archive.nptel.ac.in/courses/112/107/112107289/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		**30**

1. **Project Planning and Proposal (5 Marks)**
    - Clarity and feasibility of the project plan
    - Research and background understanding
    - Defined objectives and methodology

2. **Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**
    - Individual contribution to the presentation
    - Effectiveness in answering questions and handling feedback

3. **Involvement in the Project Work and Team Work (3 Marks)**
    - Active participation and individual contribution
    - Teamwork and collaboration

4. **Execution and Implementation (10 Marks)**
    - Adherence to the project timeline and milestones
    - Application of theoretical knowledge and problem-solving
    - Final Result

5. **Final Presentation (5 Marks)**
    - Quality and clarity of the overall presentation
    - Individual contribution to the presentation
    - Effectiveness in answering questions
6. **Project Quality, Innovation, and Creativity (3 Marks)**
    - Overall quality and technical excellence of the project
    - Innovation and originality in the project
    - Creativity in solutions and approaches

SEMESTER S6

DESIGN THINKING AND PRODUCT DEVELOPMENT

(Common to Group A & Group B)

Course Code	**GXEST605**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	2:0:0:0	**ESE Marks**	60
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To guide students through the iterative stages of design thinking, including empathizing with users, defining problems, ideating solutions and developing Proof of Concepts (PoC) and technical feasibility studies.
2. To promote the development of critical thinking skills by engaging students in integrative inquiry, where they ask meaningful questions that connect classroom knowledge with real-world applications.
3. To equip students with the ability to involve in product design considering the sustainability, inclusivity, diversity and equity aspects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Fundamentals of design thinking and product development:** Overview of stages of product development lifecycle; Design thinking -Definition-Design thinking for product innovation; Bringing social impact in ideation-Identifying societal needs-understanding multi-faceted issues-community engagement and empathetic design- technological innovation meeting societal needs; Understanding and Bridging the divide using Human Centered Design (HCD); Designing for inclusivity in product development-embracing user diversity - Long term impact - sustainability encompassing environmental,economic and social dimensions; Technology Readiness Level in the Innovation Life-cycle; Performing a self-check on innovative	**6**

	ideas - Originality of idea- understanding innovation landscape - patentability - understanding the economic landscape - Unique Selling Proposition (USP) - Repeatability and Manufacturability - Sustainability - Leveraging business models for comprehensive analysis	
2	**Empathize:** Design thinking phases; Role of empathy in design thinking; Methods of empathize phase - Ask 5 Why/ 5 W+H questions; Empathy maps - Things to be done prior to empathy mapping - Activities during and after the session; Understanding empathy tools - Customer Journey Map - Personas. **Define**: Methods of Define Phase: Storytelling, Critical items diagrams, Define success.	**6**
3	**Ideation** : Stages of ideation; Techniques and tools - Divergent thinking tools - Convergent thinking tools - Idea capturing tools; Cross-industry inspiration; Role of research in ideation - Market research - consumer research - leveraging research for informed ideation; Technological trends - navigating the technological landscape - Integrating emerging technologies; Feasibility studies - technical, economic, market, operational, legal, and ethical feasibility; Ideation session- techniques and tips. **Proof of Concept (PoC) :** Setting objectives; Risk assessment; Technology scouting; Document and process management; Change management; Knowledge Capture; Validating PoC; Story telling in PoC presentation	**6**
4	**Design:** Navigating from PoC to detailed design; Developing Specification Requirement Document (SRD)/Software Requirement Specification (SRS); Design for manufacturability; Industrial standards and readability of code; Design to cost; Pre-compliance; Optimized code; Design Failure Mode and Effects Analysis (DFMEA); Forecasting future design changes. **Prototyping:** Alpha prototypes; Beta prototypes; Transition from design to prototype; Goals and expectations for Alpha and Beta prototypes; Effective strategies for maintaining timeline in prototyping; Testing and refining Alpha prototypes; Transitioning to Beta prototypes. **Pilot build:** Definition and purpose of a pilot build; setting objectives; Identification and selection of manufacturing partner for pilot build; Testing procedures in pilot build; Scaling from pilot build to full-scale production / implementation.	**6**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignments	Internal Examination	Reflective Journal and Portfolio	Total
5	**20**	**10**	**5**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Empathize to capture the user needs and define the objectives with due consideration of various aspects including inclusivity, diversity and equity	**K5**
CO2	Ideate using divergent and convergent thinking to arrive at innovative ideas keeping in mind the sustainability, inclusivity, diversity and equity aspects.	**K6**
CO3	Engage in Human Centric Design of innovative products meeting the specifications	**K5**
CO4	Develop Proof of Concepts (PoC), prototypes & pilot build of products and test their performance with respect to the Specification Requirement Document.	**K4**
CO5	Reflect on professional and personal growth through the learnings in the course, identifying areas for further development	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		2	3	3	3	2	2		3
CO2	3	2	3		2	3	3	3	2	2		3
CO3	3	2	3		2	3	3	2	2	2		3
CO4	3	2	2		3	3	3	2	2	2		3
CO5	3					3	3	2	2	2		3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Product Sense: Engineering your ideas into reality	Dr. K R Suresh Nair	NotionPress.com	2024
2	Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation	Tim Brown	HarperCollins Publishers Ltd.	2009
3	Design Thinking for Strategic Innovation	Idris Mootee	John Wiley & Sons Inc.	2013

Sample Assignments:

1. Evaluate and prepare a report on how the aspects including inclusivity, diversity and equity are taken into consideration during the empathize and define phases of the Miniproject course.
2. Evaluate and prepare a report on how the aspects including sustainability, inclusivity, diversity and equity are taken into consideration during the ideate phase of the Miniproject course.
3. Evaluate and prepare a report on how User-Centric Design (UCD) is used in the design and development of PoC of the product being developed in the Miniproject course.
4. Prepare a plan for the prototype building of the product being developed in the Miniproject course.
5. Report on the activities during the empathize phase including the maps & other materials created during the sessions.
6. Report on the activities during the define phase including the maps & other materials created during the sessions.
7. Report of all the ideas created during the ideation phase of the Miniproject course through the tools including SCAMPER technique, SWOT analysis, Decision matrix analysis, six thinking hats exercise
8. Prepare a full scale production plan for the product being developed in the Miniproject course.
9. Create a Stanford Business Model Canvas related to the Miniproject.

An industrial visit of at least a day for experiential learning and submit a report on the learnings, for example industry standards and procedures.

SEMESTER S6

FUNDAMENTALS OF ROBOTICS

Course Code	**OERAT611**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Students will be able to apply mathematical modelling to kinematics, dynamics, control and trajectory planning of robotic manipulators
2. Students will be able to apply various concepts to design automated work cells that incorporate robotic manipulators

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Components and Structure of Robots:** Symbolic Representation, Degrees of Freedom and Workspace, Classification of Robots - Power Source, Application Area, Method of Control, Geometry, Common Kinematic Arrangements (Structure and Workspace) - Articulated Configuration (RRR), Spherical Configuration (RRP), SCARA Configuration (RRP), Cylindrical Configuration (RPP), Cartesian configuration (PPP), Parallel Manipulator, Spherical wrist **End Effectors:** Mechanical, Magnetic and Vacuum Grippers, Active and Passive Grippers **Kinematics of Robots:** Rotations - Fundamental and Composite Rotations, Homogeneous coordinates, Translations and Rotations, Composite homogeneous transformations, Screw transformations, Inverse of Homogeneous Transformation Matrices, Other representations of Rotations: Definitions of RPY, Euler Angles and Quaternions	9
2	**Forward Kinematics:** Definition, Kinematic Chains, Denavit Hartenberg Representation, Assigning the coordinate frames, Algorithm based on DH	9

	Convention for Forward Kinematics, Forward Kinematics Examples - up to 3-DoF (Higher degrees can be given as assignment) **Inverse Kinematics:** Definition, The General Inverse Kinematics Problem, General properties of solutions to the inverse kinematics problem, Theory of Kinematic Decoupling, Inverse Kinematics Example - up to 2-DoF (Higher degrees can be given as assignment) **Velocity Kinematics:** Derivation of the Jacobian – Angular Velocity, Linear Velocity, combining linear and angular Jacobians, Velocity Jacobian Examples given forward kinematic matrices - 2 link planar manipulator, Singularities – Definition and importance of identifying singularities	
3	**Dynamic Analysis:** Force-mass-acceleration and torque-inertia-angular-acceleration relationships, Lagrangian Mechanics – Overview, Dynamic Analysis Examples – 1 DOF spring cart system **Robot Control:** The control problem, Single axis PID control - Control Law, Block Diagram, Derivation of Closed Loop Transfer Function, advantages and disadvantages	**9**
4	**Trajectory Planning:** Definitions of Path planning and Trajectory Planning, Joint space versus Cartesian Space trajectory planning, Cubic Polynomial based trajectory planning, Point to Point and Continuous Path Planning **Robot selection considerations for an application**: Number of axes, work volume, capacity and speed, stroke and reach, Repeatability Precision and Accuracy, Operating environment	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe various robotic manipulator configurations and end effectors	**K2**
CO2	Formulate forward and inverse kinematics for robotic manipulators	**K2**
CO3	Analyse robotic manipulator configurations using velocity kinematics and dynamic analysis	**K4**
CO4	Plan trajectories and control schemes for robotic manipulators	**K3**
CO5	Design robotic work cells for various applications	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3	2	1									2
CO3	3	2	1									2
CO4	3	2	1									2
CO5	3	2	1									2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robot Dynamics and Control	Mark W. Spong, Seth Hutchinson, and M. Vidyasagar	Wiley	2nd edition, 2020
2	Fundamentals of robotics – Analysis and control	Robert. J. Schilling	Prentice Hall of India	1996
3	Introduction to Robotics - Analysis, Control, Applications	Saeed Benjamin Niku	John Wiley & Sons	2nd edition, 2010
4	Industrial Robotics – Technology, Programming and Applications	Mikell P. Groover, Mitchel Weiss, Roger N. Nagel, Nicholas G. Odrey, Ashish Dutta	Tata McGraw Hill	2nd edition, 2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Robotics	S K Saha	McGraw-Hill	2008
2	Robotics and Control	R K Mittal and I J Nagrath	Tata McGraw Hill	2003
3	Introduction to Robotics – Mechanics and Control	John J. Craig	Pearson	4^{th} edition, 2017
4	Robotics-Fundamental concepts and analysis	Ashitava Ghosal	Oxford University Press	2006

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/112/105/112105249/
2	https://archive.nptel.ac.in/courses/107/106/107106090/
3	https://archive.nptel.ac.in/courses/112/107/112107289/

SEMESTER S6

BASICS OF ROBOT SENSING

Course Code	**OERAT612**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Understand and discuss the fundamental elementary concepts of Robotics
2. Provide insight into different types of robots
3. Understand basic laws and phenomena on which sensors operate
4. Understand working principle of different actuators used in robotics

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Robotics and Automation:- laws of robot, brief history of robotics, robotic system components, safety measures in robotics Types of robots: Manipulator, Legged robot, wheeled robot, aerial robots, Industrial robots, humanoids, Cobots, Autonomous robots and Swarm robots, Robotics Applications.	9
2	Transducers and sensors:- Static and Dynamic Characteristics, Classification of sensors, Robotic Sensors Proximity sensor- Eddy current proximity sensor, Inductive Proximity sensor, Capacitive Proximity sensor, Force Sensor, Piezoelectric Sensor, Tactile sensor- Touch Sensor/Contact Sensor. Temperature Sensors: Thermistors, Thermocouples	9
3	Motion sensor: Encoder sensors, LVDT, Accelerometer, gyroscope-working principle only, PIR sensor, Range Sensors: RF beacons, Ultrasonic Ranging, Reflective beacons	9

	Optical sensors: Photoconductive cell, photovoltaic, Photo resistive, Photodiodes, Phototransistors, Laser Range Sensor (LIDAR) Special sensors: Acoustic Sensors, vision and imaging sensors, micro and nano sensors.	
4	Definition, Types, characteristics and selection of Actuator; linear, rotary, cylindrical, Logical and Continuous Actuators, Pneumatic actuator and Electro-Pneumatic actuator Mechanical actuation systems: Hydraulic actuator - Control valves, applications Electrical actuating systems: Electric motors as actuators Overview of electric motors. Solid-state switches, Solenoids, Piezoelectric Actuator	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the significance, social impact and future prospects of robotics and automation in various engineering applications	**K1,K2**
CO2	Demonstrate the working principle and characteristics of proximity, force and pressure sensors	**K1,K2**
CO3	Categorize and choose the suitable sensor to measure position, motion, and range of the obstacles	**K1,K2**
CO4	Describe the working principle of different actuators used in robotics	**K1,K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1			1		1				1
CO2	2	1	1			1		1				1
CO3	2	1	1			1		1				1
CO4	2	1	1			1	1	1				1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Sensors and actuators: Engineering system instrumentation	DeSilva, Clarence W	CRC Press	2018
2	Instrumentation: Devices and Systems	Rangan & Mani	McGraw Hill	
3	Process Control Instrumentation Technology	Curtis D. Johnson	Prentice Hall India	
4	Industrial Robots - Technology, Programming and Applications	Mikell P. Groover et. al	McGraw Hill, Special Edition	2012
5	Robotics Technology and flexible automation	S.R. Deb	Tata McGraw-Hill Education	2009

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Sensor, Actuators and their Interfaces: A Multidisciplinary Introductions. (1st eds)	Ida.N	SciTech, Edison	
2	Fundamentals of robotics – Analysis and control	Robert. J. Schilling	Prentice Hall of India	1996

SEMESTER S6

INDUSTRIAL AUTOMATION

Course Code	OERAT613	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the concepts of automation methodologies and trends in manufacturing automation
2. Design different pneumatic and hydraulic circuits

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Automation methodologies: Concept of Mechanization and Automation, Types of Automation, Automated flow lines, Fundamentals of Transfer Lines. Trends in manufacturing – GT and Cellular Manufacturing, Flexible manufacturing systems – features of FMS, Computer integrated manufacturing – Need for AI and expert systems in CIM, Automated assembly system – flexible assembly automation	9
2	Elements of CNC systems: servomotor and servo system design trends, stepper motors and controls, Automated tool changers and pallet changers, Material Handling and Identification Technologies: Overview of Material Handling Systems, Principles and Design Consideration, Material Transport Systems, Storage Systems, Automated Storage and Retrieval Systems, Overview of Automatic part Identification Methods, Automatic Guided Vehicles. Sensor systems for automated inspection, Automatic gauging and size control systems, thickness measurement, Machine vision systems.	9
3	Sensors and actuators for automation, Classification of position, proximity and motion sensors, electromechanical switches, LVDT, RVDT. Electrical, Hydraulic, and pneumatic actuators and their comparison (Concept only), Examples - use of Electrical, Hydraulic, and Pneumatic actuators in	9

	industrial automation. Pneumatic/Hydraulic Automation: control valves – direction, pressure and flow, sequential control of single /multiple actuator systems, cascade and Karnaugh Veitch map methods. Electro pneumatic/electrohydraulic automation: Symbols: Basic electrical elements – relay, solenoid, timers, pneumatic – electrical converters, Design of circuits and hands on models on material handling systems.	
4	Automation Control: Sequence control and programmable controllers – logic control and sequencing elements, Programmable Logic Controller – Architecture, PLC ladder diagram, Relay logic, Analog and digital I/Os, Timers, Counters, Program control instructions. Case studies on PLC ladder programming. SCADA – Building blocks of SCADA systems, RTU and HMI subsystem, Architecture of Distributed Control System (DCS), Interfacing PLC to SCADA/DCS using communication link (RS232, RS485), Protocols (Modbus ASCII/RTU) and OPC.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of automation methodologies and trends in manufacturing automation.	**K2**
CO2	Explain the design aspects of modern CNC machines, different automated inspection methods and operation of different types of material handling devices.	**K2**
CO3	Design different pneumatic and hydraulic circuits based on their applications.	**K3**
CO4	Develop the basic concepts of PLC programming.	**K3**
CO5	Illustrate SCADA systems and Distributed Control systems for industrial automation.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	2	-	-	-	-	-	-	3
CO2	3	2	2	1	2	1	-	-	-	-	-	2
CO3	3	3	2	2	2	1	-	-	-	-	-	2
CO4	3	3	2	2	2	1	-	-	-	-	-	2
CO5	3	3	2	2	2	-	-	-	-	-	-	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Automation, Production Systems and Computer Integrated Manufacturing	Mikell P Groover	Pearson	4^{th} Edition, 2014
2	Programmable Logic Controllers	Frank D. Petruzella	McGraw Hill	4^{th} Edition, 2011

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Computer Control of Manufacturing Systems	Yoram Koren	Tata McGraw-Hill	1st Edition, 2005
2	Standard Handbook of Industrial Automation	Douglas M Considine & Glenn D Considine	Springer US	1986
3	Pneumatic Control for Industrial Automation	Peter Rohner& Gordon Smith	John Wiley and Sons	1987
4	Pneumatics, Concepts, Design and Applications	Jagadeesha T	University Press	2015

SEMESTER: S6

ROBOTICS AND MACHINE VISION LAB

Course Code	**PCRAL607**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Lab

Course Objectives:

1. Grasp the fundamentals of image processing and computer vision

2. Develop robot control logic

3. Explore object detection and recognition techniques

Expt. No.	Experiments
1	Line Following Robot
2	Object Detection and Picking
3	Colour Sorting Robot
4	Obstacle Avoidance Robot
5	Hand Gesture Recognition
6	Visual Servoing
7	Simultaneous Localization and Mapping (SLAM)
8	Traffic Sign Recognition
9	Facial Expression Recognition
10	Object Counting and Tracking

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Apply image processing and computer vision principles	**K3**
CO2	Develop robot control logic based on visual information	**K6**
CO3	Utilize object detection and recognition techniques	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	1			1		2		2	
CO2	3	2	3	1			1		2		2	
CO3	3	2	3	1			1		2		2	

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robot Vision	Berthold K P Horn	MIT Press	
2	Computer Vision and Sensor-Based Robots	George G Dodd & Lothar Rossol	Springer-Verlag New York Inc.	2011

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robotics, Vision and Control: Fundamental Algorithms in MATLAB	Peter Corke	Springer	2012

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.classcentral.com/course/youtube-robotics-by-prof-d-k-pratihar-47367

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**
   - Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
   - Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.
2. **Conduct of Experiments (7 Marks)**
   - Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.

- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

- Accuracy of Results: Precision and correctness of the obtained results.

- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

ROBOTICS AND ARTIFICIAL INTELLIGENCE

SEMESTER S7

ROBOT NAVIGATION

Course Code	**PERAT741**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Robot navigation deals with the ability of the robot to determine its own position in its frame of reference and then to plan a path towards a goal location
2. Students will learn the various aspects and techniques associated with robot navigation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Building blocks of navigation:** Definitions - perception, localization, cognition, motion control **Perception:** Sensors for Perception - Schematic representation of the sensing unit, Obstacle Sensor (Bumper), the odometry sensor, Heading sensors - Gyroscopes, IMUs, Distance sensors - ToF, Phase shift, triangulation, ultrasonic rangefinders, Ground based beacons, GPS, Motion field and optical flow, Color tracking sensors, Feature Extraction - Feature Definition, Target Environment, Environment representation	**9**
2	**Localization**: General schematic for mobile robot localization, Challenge of Localization - Sensor noise and Sensor aliasing, Effector noise, Error model for odometric position estimation Localization-Based Navigation versus Programmed Solutions, Belief Representation, Map Representation - Continuous Representation,	**9**

	Decomposition Strategies, current challenges in map representation, Probabilistic Map-Based Localization - Markov localization, Kalman filter localization, Application to mobile robots: Kalman filter localization - schematic, Robot position prediction, Observation, Measurement prediction, Matching, Estimation, Landmark-based navigation, Globally unique localization, Positioning beacon systems, Route-based localization, Autonomous Map Building - stochastic map technique, Challenges - Cyclic environments, Dynamic environments	
3	**Motion in Potential Field and Navigation Function** Problem Statement, configuration space, free configuration space, Gradient Descent Method of Optimization, Gradient Descent without and with Constraints, Minkowski Sum, Potential Field, Navigation Function - in Static Deterministic Environment, in Static Uncertain Environment, Navigation Function and Potential Fields in Dynamic Environment - Estimation, Prediction, Optimization	**9**
4	**Satellite Navigation** Introduction to Satellite Navigation, Trilateration, Position Calculation - Multipath Signals, GNSS Accuracy Analysis, Dilution of Precision, Coordinate Systems, UTM Projection, Local Cartesian Coordinates, Velocity Calculation, Urban Navigation - Urban Canyon Navigation, Map Matching, Dead Reckoning – Inertial Sensors, Incorporating GNSS Data with INS - Modified Particle Filter, Estimating Velocity by Combining GNSS and INS, A-GPS, DGPS Systems, RTK Navigation, GNSS Threats	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe sensors for perception in navigation	**K2**
CO2	Compare various methods for robot localization	**K2**
CO3	Explain the techniques of motion in Potential Field and Navigation Function	**K2**
CO4	Apply information from satellites for navigation	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									2
CO2	3	2	3									2
CO3	3	2	3									2
CO4	3	2	3									2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Autonomous Mobile Robots	Roland Siegwart and Illah R. Nourbakhsh	MIT Press	
2	Autonomous Mobile Robots and Multi-Robot Systems - Motion-Planning, Communication, and Swarming	Kagan, Shvalb, Ben-Gal	Wiley	
3	Probabilistic Robots	Thrun, Burgard	MIT Press	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Planning Algorithms	Steven M. LaValle	Cambridge University Press	
2	Robot Motion Planning	Jean- Claude Latombe	Springer Science+Business Media	

SEMESTER S7

NATURAL LANGAUGE PROCESSING

Course Code	PERAT742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To introduce ARM microcontrollers
2. To introduce the design of embedded electronic systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	NLP Tasks and Applications, Language-Building Blocks, Challenges of NLP, Regular Expressions, Text Normalization, Edit Distance, N-gram Language Models, Approaches to NLP-- Heuristics-Based NLP, Machine Learning-based NLP.	9
2	Text Representation—Preprocessing, Sentence segmentation, Word tokenization, Stemming and lemmatization, Vector Space Models--Basic Vectorization Approaches--One-Hot Encoding, Bag of Words, Bag of N-Grams TF-IDF; Distributed Representations-- Word Embeddings, Doc2Vec	9
3	Text Classification--Text classification applications – Pipeline for building text classification systems, Naïve Bayes for Sentiment Classification – Naïve Bayes Classifier Training – Optimizing for Sentiment Analysis, Logistic Regression, Support Vector Machine for Text Classification	9
4	Information Extraction(IE)—IE Applications – The General Pipeline for IE - Named Entity Recognition(NER), Ambiguity in Named Entity	9

	Recognition – NER as Sequence Labeling – Evaluation of NER. Information Retrieval – Term weighting and document scoring – Inverted Index – Evaluation of Information Retrieval Systems	

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamental concepts of NLP	**K1**
CO2	Explain the different representation methods on text data	**K2**
CO3	Make use of NLP techniques in Text Classification	**K1**
CO4	Demonstrate the NLP techniques in Information Extraction and Information Retrieval	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										2
CO2	3	2	2									2
CO3	3	2		2								2
CO4	3	2	2	2								2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Natural Language Understanding	James Allen	Pearson	Second Edition 2014
2	Statistical Natural Language Processing	Christopher Manning and Hinrich Schutze	MIT Press	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc22_cs93/preview
2	https://onlinecourses.nptel.ac.in/noc22_cs93/preview

SEMESTER S7

BEHAVIORAL ROBOTICS

Course Code	**PERAT743**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Students will be able to understand the concepts of robots that are able to exhibit complex-appearing behaviours despite little internal variable state to model its immediate environment.
2. Students will be able to understand corrective actions of such robots via sensory-motor links

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Behavior-based techniques on robots, Features of behavior-based robots, Behavioural architectures, robot design Linkages, Types, Transmission elements, Flexible connectors, pulley-and- Belt drives, variable speed transmission, Comparison with classical AI	9
2	Introduction to embodied cognitive science and behaviour-based robotics, reactive behaviour-based architectures, perception, deliberative systems, hybrid systems, sub-sumption architecture. representational issues for behavioral systems, adaptive behavior, social behavior, fringe robotics	9
3	Robot work cell design and control-Sequence control, Operator interface, Safety monitoring devices in Robot-Mobile robot working principle, Robot applications- Material handling, Machine loading and unloading, assembly, Inspection, Welding, Spray painting and undersea robot. Fabrication of micro/ Nano grippers	9

4	Robot Languages-Classifications, Structures- VAL- language commands, motion control, hand control, program control, Robot welding application using VAL program- RAPID- language basic commands- Motion Instructions-. VAL-II programming-basic commands, Applications-Simple pick and place applications-. AML Language-General description, elements and functions, mProgram control statements.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe behaviour-based techniques on robots	**K1**
CO2	Understand methods and models in embodied cognitive science and artificial intelligence	**K2**
CO3	Describe models and architectures with respect to their conceptual clarity, supported by empirical data, robotic programming	**K2**
CO4	Apply embodied cognitive science and robot work cell design	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1										3
CO2	2	1										3
CO3	2	1										3
CO4	2	1	2									3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Behaviour-Based Robotics"	Arkin, C. Ronald	MIT Press, Cambridge: MA	1998
2	Introduction to AI Robotics	Murphy, R.,	Prentice Hall of India	1996
3	Introduction to Robotics - Analysis, Control, Applications	Saeed Benjamin Niku	John Wiley & Sons	2nd edition, 2010
4	Industrial Robotics – Technology, Programming and Applications	Mikell P. Groover, Mitchel Weiss, Roger N. Nagel, Nicholas G. Odrey, Ashish Dutta	Tata McGraw Hill	2nd edition, 2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Robotics	S K Saha	McGraw-Hill	2008
2	Robotics and Control	R K Mittal and I J Nagrath	Tata McGraw Hill	2003
3	Introduction to Robotics – Mechanics and Control	John J. Craig	Pearson	4th edition, 2017
4	Robotics-Fundamental concepts and analysis	Ashitava Ghosal	Oxford University Press	2006

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/112/105/112105249/
2	https://archive.nptel.ac.in/courses/107/106/107106090/
3	https://archive.nptel.ac.in/courses/112/107/112107289/
4	https://archive.nptel.ac.in/courses/112/105/112105249/

SEMESTER S7

FUNDAMENTALS OF HUMANOID ROBOTS

Course Code	PERAT744	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. The goal of this course is to introduce the students to the field of humanoid robots, to let them learn the basic aspects, to know the evolution to the current technology, and to inspire them to contribute to the field

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to humanoid robots: Autonomous robots and humanoids, Evolution of humanoid robots; Kinematics of Humanoid robots: Creating model, Forward and inverse kinematics	9
2	Action: Legged locomotion and balance, Biped 2D walking; stability; Arm movement, Gaze and auditory orientation control, Neurobotics, Facial expression, Hands and manipulation, Sound and speech generation, Mechanisms	9
3	Perception: Motion capture/Learning from demonstration: Directly recording human motions, Kinesthetic teaching, Immersive teleoperation, Human activity recognition using vision, touch, sound, Tactile Sensing. Audition, Speech recognition, Face perception. Thinking: Models of emotion and motivation	9
4	Performance: Perception of motion: over view of ZMP, 2-D Analysis, Ground reaction forces; Dynamics: Humanoid robot motion and Ground reaction forces, calculation of centre of mass; Anthropomorphic characters	9

	Case studies: Technical details of the following Humanoid robots as samples: Honda Robots (ASIMO), Sony Dog and Biped (mpg1, mpg2), Hiroshi face robot, Mitra,	

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Differentiate different types of Humanoids based on their underlying technology	
CO2	Analyse various actions of humanoid robots	
CO3	Rate the performances of the robots	
CO4	Analyse in details various types of existing Humanoids through case studies	

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2								2
CO2	3	2		2								2
CO3	3	2		2								2
CO4	3	2		3								2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Towards A Better Pose Understanding for Humanoid Robots	Motaz Abdul Aziz Al-Hami	Temple University Libraries	2016
2	Humanoid Robots Modeling and Control	Dragomir N. Nenchev, Atsushi Konno, Teppei Tsujita	Publisher:Elsevier Science	2018
3	Humanoid Robots	Edited by Ben Choi	INTECH Open Access publisher	
4	Humanoid Robots	Mary Lindeen	Lerner Publishing Group	
5	Neural systems for robotics	Omid Omidvar, Patrick van der Smagt	Academic Press	1997

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Humanoid Robots	Mary Lindeen	Lerner Publishing Group	
2	Neural systems for robotics	Omid Omidvar, Patrick van der Smagt	Academic Press	1997

SEMESTER S7

MICROROBOTICS

Course Code	PERAT746	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understanding Core Concepts and able to explain the science and technology behind micro-robots
2. Explain the design methodology, Microfabrication and Technology
3. Explain design principles of actuators
4. Explain the design selection criteria for micro machining applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Micro Robot: Science and Technology behind micro-robot – definition of microrobot. Fundamental concepts of linear elasticity, stress, strain, torsion, yield criteria. Fundamental concepts of kinematics – kinetics, dynamics, linear and angular momentum, equation of motion, multi body systems	9
2	Physics for Micro Robots: Scaling effect – scaling laws – scaling effect on surface and volume, electrical properties, fluids, physical forces. Concept of torsion – dummy suffix notation – matrix notation – transformations. Flexures: Definition of flexure – design methodology – bending mode, torsion mode. Elemental flexure – cantilever beam. Range motion, Notch Hinge, Cross pivot – cross strip flexure–monolithic cross pivot. Flexure systems – Linear guidance systems.	9

3	Actuators for Micro Robot: Introduction to Actuators – Design principles of actuators – Amplification of motion – Bimorph Assembly – Motion Amplification in plane – electrostatic actuator – Thermal-based Actuator – Shape memory alloys – piezoelectric actuators – piezoelectric motors – magnetic shape memory alloys.	**9**
4	Sensors for Micro Robots: Sensors in micro-robots, sensors terminology, sensing technology for displacements – Types of sensors –tracking methods - Implementation, application and future prospects of Micro Robots:	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe micro system technology and micro robots	**K2**
CO2	Explain the physics of micro robots and describe flexures used in micro robots	**K2**
CO3	Describe various actuators used un micro robots	**K2**
CO4	Discuss about the various sensors used in micro robots	**K2**
CO5	Apply appropriate design criteria suitable for micromachining and micro assembly	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	0	0	0	1	1	1	1	2	1	0	2
CO2	2	1	2	0	1	1	1	1	1	0	0	1
CO3	2	0	0	0	1	1	1	1	2	1	0	2
CO4	2	1	2	0	1	1	1	1	1	0	0	1
CO5	2	0	0	0	1	1	1	1	2	1	0	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Microrobotics Methods and Applications	Yves Bellouard	CRC Press, Massachusetts	2019

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Micro robotics for Micromanipulation"	Nicolas Chaillet, Stephane Regnier	Wiley	2013
2	Sergej Fatikow, Ulrich Rembold,	Microsystem Technology and Micro robotics	Springer	2013
3	Vikas Choudhry, Krzystof	MEMS: Fundamental Technology and Applications	CRC Press	2013
4	NadimMaluf and Kirt Williams	An Introduction to Microelectromechanical systems Engineering	Artech House, MA	2002

SEMESTER S7

REINFORCEMENT LEARNING FOR ROBOTICS

Course Code	**PERAT747**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Students will be able to formulate Reinforcement Learning problems for various applications
2. Apply dynamic programming and Monte Carlo methods to Reinforcement Learning

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Reinforcement Learning (RL):** Introduction, Example applications of RL, Elements of Reinforcement Learning **Finite Markov Decision Processes:** The Agent–Environment Interface, Examples - Pick-and-Place Robot, Recycling Robot; Goals and Rewards, Returns and Episodes, Example – Pole Balancing, Unified Notation for Episodic and Continuing Tasks, Policies and Value Functions, Example – Grid world, Optimal Policies and Optimal Value Functions, Bellman optimality equation, Example - Bellman Optimality Equations for the Recycling Robot	**9**
2	**Dynamic Programming:** Introduction, Policy Evaluation (Prediction), Iterative Policy Evaluation, Example – 4x4 grid world, Policy Improvement, Policy Iteration, Value Iteration **Monte Carlo Methods:** Monte Carlo Prediction, Monte Carlo Estimation of Action Values, Monte Carlo Control, Monte Carlo Control without Exploring Starts, Off-policy	**9**

	Prediction via Importance Sampling, Incremental Implementation, Off-policy Monte Carlo Control	
3	**Temporal-Difference Learning:** TD Prediction, Advantages of TD Prediction Methods, Optimality of TD(0), Sarsa: On-policy TD Control, Q-learning: Off-policy TD Control, Expected Sarsa **n-step Bootstrapping:** n-step TD Prediction, n-step Sarsa, n-step Off-policy Learning, Off-policy Learning Without ImportanceSampling: The n-step Tree Backup Algorithm	**9**
4	**On-policy Prediction with Approximation:** Value-function Approximation, The Prediction Objective, Stochastic-gradient and Semi-gradient Methods, Linear Methods **Eligibility Traces:** The λ-return, TD(λ), n-step Truncated λ -return Methods, Sarsa(λ) **Policy Gradient Methods:** Policy Approximation and its Advantages, The Policy Gradient Theorem, REINFORCE: Monte Carlo Policy Gradient, REINFORCE with Baseline, Actor–Critic Methods	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Formulate Reinforcement Learning problems for various applications	**K3**
CO2	Apply dynamic programming and Monte Carlo methods to Reinforcement Learning	**K3**
CO3	Describe temporal difference Reinforcement Learning algorithms	**K2**
CO4	Explain On-policy Prediction and Policy gradient methods with approximation	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	3	-	-	-	-	-	-	3
CO2	3	3	3	-	3	-	-	-	-	-	-	3
CO3	3	3	3	-	3	-	-	-	-	-	-	3
CO4	3	3	3	-	3	-	-	-	-	-	-	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Reinforcement Learning: An Introduction	Sutton, Barto, Bach	MIT Press	2nd edition, 2018
2	Reinforcement Learning in Robotics: A Survey	Kober, Peters	Springer Tracts in Advanced Robotics	2014

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	A Survey on Policy Search for Robotics	Deisenroth, Neumann, Peters	Now Publishers Inc	2013
2	Algorithms for Reinforcement Learning	Csaba Szepesv´ari	Morgan & Claypool Publishers	2009

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/106/106/106106143/
2	https://archive.nptel.ac.in/courses/106/106/106106143/
3	https://archive.nptel.ac.in/courses/106/106/106106143/
4	https://archive.nptel.ac.in/courses/106/106/106106143/

SEMESTER S7

AERIAL ROBOTICS

Course Code	**PERAT745**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs.30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Explain the principle of UAV systems different Aerodynamics, Stability and Control
2. Demonstrate the hardware anatomy of quadcopter Power Train and Propeller functionality
3. Classify the use of various Testing and Maintenance of Quadcopter, Crash analysis and Voltage testing.
4. Model the Control of Quadcopter Performance Analysis and Challenge.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to UAV Systems-Introduction, Classes of UAV Systems, Basic Aerodynamics, Stability and Control, Propulsion, Payloads, Launch and Recovery	**9**
2	Hardware Anatomy of Quadcopter-Power Train – Propellers, Motors- Total Lift - Electronic Speed Controllers – Flight Battery – Radio transmitter and receiver – Flight Controller – Sensor, GPS, Compass, Camera Assembling for Quad copter – Connectors, Mounting of Propellers and Powering up	**9**
3	Testing and Maintenance of Quadcopter-Key Flight Safety Rules - Preflight Checklist and Flight Log Information – Flight Instructions - Repair and Maintenance: Crash analysis, Common issues, Voltage testing	**9**
4	Design ,Control of Quadcopter and Case Studies-Flight Performance Analysis – Dynamics and Design – Design Challenge – Guidance,	**9**

	Navigation, and Control of Drones, Applications of drones in agriculture, smart cities, delivery drones, health care, defense surveillance	

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	***Internal Ex***	***Evaluate***	***Analyse***	***Total***
5	**15**	**10**	**10**	**40**

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Students may be given microprojects in the field of UAV systems, Design and control of Quadcopter and Case Studies.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the principle of UAV systems different Aerodynamics, Stability and Control,Propulsion	**K2**
CO2	Design the hardware anatomy of quadcopter Power Train and Propeller functionality	**K3**
CO3	Evaluate the use of various Testing and Maintenance of Quadcopter Crash analysis and Voltage testing.	**K5**
CO4	Design the Control of Quadcopter Performance Analysis and Challenge.	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2										2
CO2	2	2			2							2
CO3	2	2			1							2
CO4	3	3	2	2	2				2	1		3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Unmanned Aircraft Systems: UAVs Design, Development and Deployment	Wiley Publications	Reg Austin	2010
2	Build Your Own Quadcopter - Power Up Your Designs with the Parallax Elev-8	Donald Norris	McGraw-Hill Education	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Building a Quadcopter with Arduino	VasilisTzivaras	Packt Publishing	2016
2	Drones and Unmanned Aerial Systems	Zavrsnik	Springer	2015
3	Drone Technology- Future Trends and Practical Applications	Sachi Nandan Mohanty	Scrivener Publishing	2023
4	Building your own Drones	John Baichwal	Que Publishing	2016

Video Links (NPTEL, SWAYAM…)	
Module No.	Link ID
1	https://youtu.be/S-XiFIRVkgQ
2	https://youtu.be/mv81oj5PBwM
3	https://youtu.be/9c769xiEXn0?list=PLFW6lRTa1g83B1HdU2mece6QLeBrtspL7
4	https://youtu.be/9c769xiEXn0?list=PLFW6lRTa1g83B1HdU2mece6QLeBrtspL7

SEMESTER S7

DATA STRUCTURES & ALGORITHMS

Course Code	PERAT751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	2:1:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide an insight of data structures and their applications
2. To develop an understanding of the basic design considerations of algorithms
3. To apply these algorithms in various computational scenarios

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Data Structures, Arrays and Stacks:** Basic definitions, Classifications of data structures- ADT and CDT, Linear and Non-linear, Static and Dynamic. Operations on data structures, Asymptotic notations, Notion of recursive algorithms, Recurrence relations. Representation of Arrays, Applications of arrays, Sparse matrix and its representation. Stack definitions and concepts, Operations on stacks, Applications of stacks, Polish Expression, Reverse Polish Expression and their Compilation, Recursion.	9
2	**Queues, Linked lists, Trees and Graphs:** Representation of Queues, Operations on queues, Circular queue, Priority queue, Double-ended queue, Applications of queues. Notion of linked lists, Singly linked list, doubly linked list, Circular linked list, Operations on linked list. Concepts and types of Trees, Tree traversal algorithms, Search trees. Definition of Graphs, Directed and undirected graph, Adjacency matrix and Adjacency list representation.	9
3	**Indexing Structure, Graph Algorithms and Searching & Sorting**	9

	Algorithms: Concepts and implementations of B-Tree, B+ tree, Hashing, Dictionary. Graph algorithms- Depth-first search, Breadth-first search, Shortest path problem-Dijkstra's algorithm. Searching and sorting algorithms- Linear search, Binary search, Hash tables, Internal and external sorting algorithms, Sorting without comparison.	
4	**Algorithm Design and Analysis:** Greedy algorithm, Divide and conquer, Dynamic programming, Backtracking, Branch and bound, Randomized algorithms. Asymptotic notations, Recurrences, NP- complete problems.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse various data structures and their uses	**K4**
CO2	Apply appropriate data structures like arrays, linked lists, stacks and queues to solve real world problems efficiently.	**K3**
CO3	Represent and manipulate data using nonlinear data structures like trees and graphs to design algorithms for various applications.	**K3**
CO4	Comprehend and implement various techniques for searching, sorting and hashing	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	-	-	-	-	-	-	-	3
CO2	2	1	-	-	-	-	-	-	-	-	-	3
CO3	3	2	-	-	-	-	-	-	-	-	-	3
CO4	2	1	-	-	-	-	-	-	-	-	-	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Classic Data Structures	Samanta D.	Prentice Hall India	2/e, 2009
2	Data Structures: A Pseudocode Approach with C	Richard F. Gilberg, Behrouz A. Forouzan	Cengage Learning	2/e, 2005
3	Data Structures and Algorithms	Aho A. V., J. E. Hopcroft and J. D. Ullman	Pearson Publication	1983

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fundamentals of Data Structures in C	Horwitz E., S. Sahni and S. Anderson	University Press (India)	2008
2	Introduction to Data Structures with Applications	Tremblay J. P. and P. G. Sorenson	Tata McGraw Hill	1995
3	Advanced Data Structures	Peter Brass	Cambridge University Press	2008

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/106102064 https://www.youtube.com/watch?v=zWg7U0OEAoE
2	https://www.youtube.com/watch?v=PGWZUgzDMYI
3	https://www.youtube.com/watch?v=CIm6RzdoPCI
4	https://www.youtube.com/watch?v=EcT-Jt5WStw

SEMESTER S7

ADAPTIVE SIGNAL PROCESSING

Course Code	PERAT752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To have basic understanding of digital filter responses and characteristics
2. To design various types of digital filters for practical applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Estimation and Stochastic process-basics; Basic kinds of estimation, linear filter structures with finite and infinite memory, approaches for developing linear adaptive filters, applications. Discrete time Stochastic process-characterisation, mean ergodic theorem, correlation matrix-definition, properties, correlation matrix of sine wave plus noise, stochastic models- AR, MA, ARMA, Wold decomposition, complex gaussian processes, power spectral desnsity-properties and estimation.	9
2	**Wiener Filters:** Linear optimum filtering, principle of orthogonality, minimum mean square error, Wiener Hopf equation The steepest descent algorithm- basic ideas and stability consideration, application to wiener filtering, advantages and limitations	9
3	**Least mean square (LMS) algorithm**; Signal flow graph, convergence of LMS algorithm; optimality considerations, applications. normalized LMS algorithm, affine projection adaptive filters.	9
4	**RLS Algorithm and Kalman filtering**: Limitations of LMS algorithm, formulation of recursive least squares (RLS) based adaptive filters, matrix inversion lemma, exponentially weighted RLS algorithm, single-weight	9

	adaptive noise canceller. Kalman filtering-recursive minimum mean square estimation, statement of kalman filtering problem, Innovations process, filtering.	

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic principles of Adaptive filters	**K2**
CO2	Understand the basic principles of Wiener filters and steepest descent algorithm	**K2**
CO3	Apply the Least Mean Squared (LMS) algorithm for certain applications	**K3**
CO4	Analyze the Recursive Least Squares (RLS) algorithm and kalman filtering for adaptive filters	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	-	-	-	-	-	-	-	-	2
CO2	3	3	3	-	-	-	-	-	-	-	-	2
CO3	3	3	3	-	-	-	-	-	-	-	-	2
CO4	3	3	3	-	-	-	-	-	-	-	-	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Adaptive Filter Theory	Haykins	Prentice-Hall	4th , 2001
2	Fundamentals of Adaptive Filtering	Ali H. Sayed,,	John WIley	2003

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Statistical Digital Signal Processing And Modeling	Monson H. Hayes	Wiley India	1st , 2008
2	Adaptive Filters: Theory and Applications	B. Farhang-Boroujen	, John Wiley and Sons,	2013
3	Digital Signal Processing, Principles algorithms and applications	Proakis J. G. and Manolakis D. G	Pearson Education	2007

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	NPTEL videos on Adaptive Signal Processing, by ProfM. Chakraborty, IIT Kharakpur (https://nptel.ac.in/courses/117105075 etc.)
2	
3	
4	

SEMESTER S7

MODERN COMMUNICATION SYSTEMS

Course Code	PERAT753	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To explain OFDM, OFDMA and SC-FDMA techniques used in cellular communication
2. To explain communication standards and IoT architecture and various connectivity technologies used in IoT Systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Cellular Communication System Need for Multi carrier system, Basics of Orthogonal Frequency Division Multiplexing (OFDM), Multiple access for OFDM systems, Orthogonal Frequency Division Multiple Access (OFDMA), Single carrier Frequency Division Multiple Access (SC-FDMA). Cellular concept, path loss and shadowing, Doppler shift, Multipath effect, Significance of diversity in wireless communication systems.	9
2	Short Range Communication System Introduction to current wireless technologies, background and current scenario, future wireless network requirements, IEEE 802.11 (Wi-Fi) standards and applications (IEEE 802.11a/b/g/n/ac/ax,ay), Space time wireless standards, 3GPP-LTE standard, Millimeter wave characteristics, Channel performance at 60 GHz, Development of millimeter wave standards, Indoor and outdoor applications for millimeter wave communications. 6G Networks – Use Cases and Technologies.	9
3	IoT System	9

	Introduction of IoT, characteristics, physical and logical design of IoT, IoT Enabling Technologies – Wireless Sensor networks, Cloud computing. Introduction to IoT, Evolution of IoT, IoT Networking Components. IoT Connectivity Technologies – Zigbee, Wireless HART, RFID, NFC, LoRa, WiFi, Bluetooth. IoT Communication Technologies – Infrastructure Protocols – IPv6, 6LoWPAN, Data Protocols – MQTT, MQTT-SN, CoAP.	
4	Intelligent Transport System Introduction to Intelligent Vehicular Communication – Evolution, Vehicular Networks and ITS, Vehicular Communication Standards/ Technologies – DSRC, IEEE 802.11p WAVE, IEEE 1609, IEEE 02.15.7 - Visible Light Communication (VLC), 4G/5G-Device to Device (D2D), 6G Cellular Networks and Connected Autonomous Vehicles, Operational Scenario – Collision Avoidance.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain OFDM, OFDMA and SC-FDMA techniques used in cellular communication	**K2**
CO2	Explain different communication standards	**K3**
CO3	Explain the IoT architecture and various connectivity technologies used in IoT Systems	**K2**
CO4	Understand the various communication standards for connected autonomous vehicles	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	2	-	-	-	-	-	-	-
CO2	3	3	3	-	-	3	-	-	-	-	-	1
CO3	3	3	3	-	-	3	-	-	-	-	-	-
CO4	3	3	3	-	-	3	-	-	-	-	-	-

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Principles of Modern Wireless Communication Systems	Aditya K. Jagannatham	Tata McGraw Hill	2016
2	Wireless Communications	T.L. Singal	Tata McGraw Hill	Second Edition, 2011

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Millimeter Wave Communication systems	K. C. Huang, Z. Wang	John Wiley &Sons.	
2	Introduction to IoT	Sudip Misra, Anandarup Mukherjee & Arijit Roy	Cambridge University Press.	2021
3	Current Technologies in Vehicular Communication	George J. Dimitrakopoulos	Springer International Publishing	2017

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc24_ee151
2	
3	
4	

SEMESTER S7

PATTERN RECOGNITION

Course Code	PERAT 754	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To introduce the fundamental algorithms for pattern recognition
2. To instigate the various classification techniques
3. To introduce deep learning and convolution neural networks

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Pattern Recognition: Classification of Pattern recognition systems, Design of Pattern recognition system, Various applications; Learning paradigms, supervised and unsupervised learning. Statistical Pattern Recognition: Review of probability theory, Gaussian distribution, Bayes decision theory and Classifiers, Optimal solutions for minimum error and minimum risk criteria, Normal density and discriminant functions, Decision surfaces.	9
2	Parameter estimation: Maximum likelihood and Bayesian Estimation; Hidden Markov models; Nonparametric techniques: Parzen-window method, K-Nearest Neighbour method; Decision trees: Concept of construction, splitting of nodes, choosing of attributes, overfitting, pruning.	9
3	Linear discriminants, binary class, multiple classes, cost functions; Linear Discriminant based algorithm: Perceptron, Support Vector Machines, Fisher's linear discriminant;	

	Nonlinear classifiers, the XOR problem, multilayer perceptrons, Back Propagation algorithm, Artificial neural networks; Classifier Ensembles: Bagging, Boosting / AdaBoost.	
4	Introduction to deep learning networks, deep feedforward networks, ReLU, bias-variance tradeoff, regularization, dropout, vanishing/exploding gradients, weight initialization for deep networks, basics of convolutional neural networks, layers of convolutional neural networks.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basics of statistical pattern recognition.	**K2**
CO2	Apply linear algebra and statistical methods in parameter and non-parameter estimation.	**K3**
CO3	Apply statistical methods in linear and non-linear classification.	**K3**
CO4	Understand the basics of deep learning networks, convolutional neural networks.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	**CO1**	3	-	-	-	-	-	-	-	-	-	-
CO2	**CO2**	3	3	3	-	3	-	-	-	-	-	-
CO3	**CO3**	3	3	3	-	3	-	-	-	-	-	-
CO4	**CO4**	3	3	-	-	-	-	-	-	-	-	-

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Pattern Recognition and Machine Learning	Christopher M. Bishop	Springer	2011
2	Pattern Classification and scene analysis	R O Duda, P.E. Hart and D.G. Stork,	John Wiley	Second Edition,2000

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Pattern Recognition Principles	Tou and Gonzales	Wesley Publication Company, London,	1974
2	Pattern Recognition	S.Theodoridis and K. Koutroumbas	Academic Press	Fourth Edition,2009
3	Pattern Recognition : Statistical, Structural and Neural Approaches	Robert J. Schalkoff	John Wiley & Sons Inc., New York	2007
4	The Elements of Statistical Learning: Data Mining, Inference, and Prediction	Trevor Hastie Robert Tibshirani Jerome Friedman	Springer	Second Edition

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	NPTEL :: Electronics & Communication Engineering - Pattern Recognition and Application
2	NPTEL :: Electronics & Communication Engineering - Pattern Recognition and Application
3	NPTEL :: Electronics & Communication Engineering - Pattern Recognition and Application
4	Deep Learning - Course (nptel.ac.in)

SEMESTER S4

SECURE COMMUNICATION

Course Code	PERAT756	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand and discuss the fundamental concepts of encryption
2. Provide insight into different types of encryption standards
3. Understand basic concepts of Cryptography

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction and Classic Encryption Techniques:-OSI security architecture, Security attacks – Passive attacks, Active attacks, Security services- Authentication, Access Control, Data Confidentiality, Data integrity, Nonrepudiation, Availability service. Model for network security. Symmetric cipher model, Cryptography, Substitution techniques- Hill Cipher, Transposition Techniques. Finite Fields: -Groups, Rings and Fields, Modular arithmetic, Euclidian algorithm, Finite Fields of the form GF(p), Polynomial arithmetic	9
2	Block Ciphers: - Data Encryption Standard, Block Cipher Principles – Stream Ciphers and Block Ciphers, Feistel Cipher, Feistel Decryption algorithm, The Data encryption standard, DES Decryption, The AES Cipher, substitute bytes transformation, Shift row transformation, Mix Column transformation.	9
3	Public Key Cryptography: -RSA and Key Management, Principles of public key cryptosystems-Public key cryptosystems, Application for Public key cryptosystem requirements, Fermat's theorem, Euler's Totient	9

	Function, Euler's theorem, RSA algorithm, Key management, Distribution of public keys, Publicly available directory, Public key authority, public key certificates, Distribution of secret keys using public key cryptography	
4	Message Authentication and Hash Function: - Authentication requirements, Authentication functions- Message Encryption, Public Key Encryption, Message Authentication Code, Hash function	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain network security services and mechanisms and the types of attacks they are designed for and apply the concepts of modular arithmetic, Euclidean algorithm, polynomial arithmetic.	**K3**
CO2	Illustrate the principles of modern symmetric ciphers like Data Encryption Standard and Advanced Encryption Standard.	**K3**
CO3	Outline the concepts of public key cryptography, RSA algorithm, key distribution, and management for public key systems.	**K2**
CO4	Explain the requirements for authentication and the types of functions used to produce an authenticator	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3	3										2
CO3	3	3										2
CO4	3	3										2
CO5	3	3										2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Cryptography and Network security: principles and practice	William Stallings	Prentice Hall of India	4^{th} Edition, 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cryptography and Network security	Behrouz A. Forouzan	Tata McGraw-Hill	2008
2	Abstract Algebra	David S. Dummit & Richard M Foote	Wiley India Pvt. Ltd	2nd Edition, 2008.
3	Cryptography, Theory and Practice	Douglas A. Stinson,	Chapman & Hall CRC Press Company	2nd Edition, 2005.
4	Elliptic Curves: Theory and Cryptography	Lawrence C. Washington	Chapman & Hall, CRCPress Company, Washington	2008
5	A course in Number theory and Cryptography	N. Koeblitz		2008
6	Elementary Number Theory with Applications	Thomas Koshy	Academic Press	2nd Edition, 2007
7	Cryptography and network security	Tyagi and Yadav	Dhanpat Rai & Co	2012

SEMESTER S7

REAL TIME OPERATING SYSTEM

Course Code	PERAT757	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Understand the fundamental concepts and characteristics of real-time systems.
2. Analyze and implement real-time scheduling algorithms and techniques.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Real-Time Systems** Overview of Real-Time Systems: Definition and types of real-time systems, Hard vs. soft real-time systems.Basic Concepts: Real-time tasks and their characteristics, Task scheduling, Timing constraints and requirements. RTOS Architectures: Monolithic kernels vs. microkernels. RTOS examples: commercial vs Open RTOS and their comparison, examples.Inter-Process Communication (IPC): Shared memory, Message passing. RTOS Environment Setup: Installation and setup of an RTOS on a microcontroller (e.g., ARM Cortex-M), Task Creation and Management: Writing simple tasks, Task states and transitions, Scheduling and Context Switching:Implementing basic scheduling algorithms,Demonstrating context switching with example tasks	9
2	**Real-Time Scheduling and Synchronization**	

	Real-Time Scheduling Algorithms: Fixed-priority scheduling (Rate-Monotonic, Deadline-Monotonic), Dynamic priority scheduling (Earliest Deadline First), Priority based preemption, Round Robin, Task Synchronization: Mutual exclusion, Priority inversion and inheritance Inter-Task Communication: Semaphores, Mutexes, Event flags Implementing Scheduling Algorithms: Practical implementation of scheduling, **Synchronization Mechanisms:** Practical implementation of semaphores and mutexes in task synchronization, Demonstrating priority inversion and its mitigation: **Real-Time Task Communication:** Implementing inter-task communication using queues and mailboxes	**9**
3	Real-Time System Design and Analysis System Design Principles: Modular design, Time-triggered vs. event-triggered systems, Worst-Case Execution Time (WCET) Analysis: Techniques for WCET estimation, Timing analysis, Reliability and Fault Tolerance: Redundancy, Error detection and recovery. Designing a Real-Time System: Case study: Designing a real-time control system, WCET Analysis Tools: Using tools for WCET analysis and timing verification, Implementing Fault Tolerance: Practical implementation of redundancy and error recovery mechanisms	**9**
4	Real-Time Operating System Services and Applications Real-Time Operating System Services: Memory management, I/O management. Real-Time Middleware: Middleware services for real-time systems, Case Studies and Applications: Automotive systems, Aerospace and defense, Medical devices Memory Management in RTOS: Implementing dynamic memory allocation, Real-Time Middleware Implementation: Developing middleware components for a real-time application Case Study Implementation: Implementing a real-time system for a specific application (e.g., real-time data acquisition)	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the fundamental concepts and characteristics of real-time systems.	**K1, K2**
CO2	Analyze and implement real-time scheduling algorithms and techniques.	**K4**
CO3	Conduct worst-case execution time (WCET) analysis for real-time tasks.	**K3, K4**
CO4	Utilize RTOS services and middleware for developing real-time applications	**K3,K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	**3**	**3**	**2**									**3**
CO2	**3**	**3**	**2**									**3**
CO3	**3**	**3**	**2**									**3**
CO4	**3**	**3**	**2**									**3**

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Real-Time Operating Systems Book 1: The Theory	Jim Cooling	CreateSpace Independent Publishing Platform	1st 2018
2	Real-Time Systems: Theory and Practice	Rajib Mall	Pearson Education	2007
3	Real-Time Systems: Design Principles for Distributed Embedded Applications	Hermann Kopetz	Springer	2nd 2011
4	Embedded Systems: Real-Time Operating Systems for Arm Cortex-M Microcontrollers	Jonathan W. Valvano	CreateSpace Independent Publishing Platform	3rd, 2017

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Real-Time Systems	C. M. Krishna, Kang G. Shin,	McGraw-Hill	2010
2	Real-Time Systems	Jane W. S. Liu	Pearson Education	2009
3	Real-Time Systems Design and Analysis	Philip A. Laplante, Seppo J. Ovaska,	Wiley	2012
4	Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C	Yifeng Zhu	E-Man Press LLC	3rd , 2017

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://elearn.nptel.ac.in/shop/iit-workshops/completed/lab-workshop-on-embedded-rtos/?v=c86ee0d9d7ed https://onlinecourses.nptel.ac.in/noc21_cs98/preview
2	https://elearn.nptel.ac.in/shop/iit-workshops/completed/lab-workshop-on-embedded-rtos/?v=c86ee0d9d7ed
3	https://elearn.nptel.ac.in/shop/nptel/real-time-operating-system/?v=c86ee0d9d7ed https://onlinecourses.nptel.ac.in/noc21_cs98/preview
4	https://elearn.nptel.ac.in/shop/iit-workshops/completed/lab-workshop-on-embedded-rtos/?v=c86ee0d9d7ed

SEMESTER S7

DIGITAL FILTER DESIGN

Course Code	PERAT758	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To have a basic understanding of digital filter responses and characteristics
2. To design various types of digital filters for practical applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Digital Filtering**-Review of Z-Transform; Z-domain transfer function, concept of poles and zeroes, location of poles and zeroes and their effect on filter responses, FIR and IIR type of digital filters, ideal and practical responses of Low pass, high pass, band pass and band reject filters, Analog filter responses-Butterworth, Chebyschev and elliptic responses, applications and advantages of digital filtering.	9
2	**FIR Filter Design**-linear phase filters, Location of zeroes, Design using Fourier series, window method-rectangular, hanning, hamming windows-comparison of window frequency responses; LP, HP, BP and BR type designs. FIR Filter banks- quadrature mirror sub band filters-tree structured design	9
3	**IIR Filter Design**- Design from analog filters-Design of Butterworth analog filter transfer function, specifications of analog and digital frequency responses, conversion to digital responses-impulse invariant and bi-linear transformations, frequency transformation in analog domain, design of LP and HP filters, design of comb filter by pole-zero placement.	9
4	**Filter Structures**: FIR filter structures-direct, linear phase and lattice structures.	

	IIR filter structures-direct, cascade, parallel, lattice structures. Coefficient quantization effects in FIR and IIR filters-basic concepts (No analysis required)	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze transfer functions of digital filters	**K3**
CO2	Design of FIR type digital filters	**K2**
CO3	Design of IIR type digital filters	**K4**
CO4	Realize structures of digital filters	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	-	-	-	-	-	-	-	-	2
CO2	3	3	3	-	-	-	-	-	-	-	-	2
CO3	3	3	3	-	-	-	-	-	-	-	-	2
CO4	3	3	3	-	-	-	-	-	-	-	-	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Digital filters, analysis design and applications	Andreas Antoniou	Mc Graw Hill	$2^{nd,}$ 2000
2	Discrete-Time Signal Processing	Alan V Oppenheim, Ronald W. Schafer	Pearson	3^{rd}, 2010

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Digital Signal Processing, Signals, systems and Filters	Andreas Antoniou	Pearson	1^{st} , 2005
2	Digital Signal Processing: A Computer - Based Approach	Sanjith K Mitra	Mc Graw Hill	4^{th} , 2013
3	Digital Filter Design	T. W. Parks and C. S. Burrus	John Wiley & Sons	1^{st} , 1987
4	Signals & Systems-Continuous and Discrete,	Rodger E. Ziemer	Pearson	4^{th} 2013
5	Digital Signal Processing	Proakis J. G. and Manolakis D. G	Pearson Education	2007
6	Digital Signal Processing-a practical approach	Ifeachor E.C. and Jervis B. W	Pearson education	2009
7	Schaums outline: Digital Signal Processing	Monson H Hayes	McGraw Hill Professional	1999
8	Digital Signal Processing	Salivahanan S	McGraw Hill	2019

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	NPTEL videos on Digital Signal Processing, by Prof. T. K. Basu, IIT Kharakpur (https://archive.nptel.ac.in/courses/108/105/108105055/ etc.)
2	
3	
4	

SEMESTER S7

FPGA BASED SYSTEM DESIGN

Course Code	**PERAT755**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs.30 Min.
Prerequisites (if any)	Digital Electronics	**Course Type**	Theory

Course Objectives:

1. Apply hardware description languages (HDLs) to design digital circuits and verify the functionality using test benches
2. Understand FPGA architecture, logic blocks, and programmable interconnect principles.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction:** Digital system design options and tradeoffs, Design methodology and technology overview. **High Level System Architecture and Specification:** Behavioral modelling and simulation, Hardware description languages (emphasis on Verilog), combinational and sequential design, state machine design, test benches.	9
2	**Programmable logic Devices:** ROM, PLA, PAL, CPLD, FPGA Features, Limitations, Architectures and Programming. Implementation of MSI circuits using Programmable logic Devices.	9
3	FPGA architecture: FPGA Architectural options, granularity of function and wiring resources, coarse V/s fine grained, vendor specific issues (emphasis on Xilinx and Altera), Logic block architecture: FPGA logic cells, timing models, power dissipation I/O block architecture: Input and Output cell characteristics, clock input, Timing, Power dissipation.	9
4	**Placement and Routing:** Programmable interconnect - Partitioning and Placement, Routing resources, delays; Applications -Embedded system	9

	design using FPGAs, DSP using FPGAs. **Commercial FPGAs:** Xilinx, Altera, Actel (Different series description only).	

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	***Internal Exam***	***Evaluate***	***Analyse***	***Total***
5	**15**	**10**	**10**	**40**

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Evaluation Methods:

1. Simulation Assignments Using FPGA Design and Development Tools: (10 marks)

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Each assignment can focus on developing Verilog modules of digital circuits, Simulate the design and implement the design on FPGA boards.

2. Course Project:

Comprehensive project involving design and implementation of digital circuits.

(10 marks)

Project phases: Proposal, Design, Implementation, Testing, Final Report.

Presentations and Viva Voce:

Students present their projects explaining design choices, methodologies, and results.

Viva voce to assess understanding and ability to answer related questions.

Example topics for Experiments/ Miniproject

1. Design and Verilog modeling of simple Combinational Logic circuits.
2. Design and Verilog modeling of various data path elements – Adders, Multipliers etc.
3. Design and Verilog modeling of simple sequential circuits like Counters and Shift registers
4. Implementation of combinational/ sequential circuits on FPGA evaluation Boards.
5. Implementation of digital clock on FPGA
6. Implementation of basic calculator on FPGA

Criteria for Evaluation: Experiments (10 marks)

1. Understanding of Concepts (3 marks)
   - Demonstrates a clear understanding of the theoretical concepts related to the experiment.
2. Implementation and Accuracy (3 marks)
   - Correctly implements the design using appropriate tools.
   - The design functions as expected without errors.
3. Analysis and Problem-Solving (2 marks)
   - Effectively analyse the design to identify and resolve issues.
   - Demonstrates problem-solving skills in addressing any encountered challenges.
4. Documentation and Reporting (1 mark)
   - Provides clear and concise documentation of the steps and processes followed.
   - The report includes diagrams, code snippets, and simulation results.
5. Presentation and Communication (1 mark)
   - Clearly presents the experiment and its results.
   - Able to answer questions and explain the design choices.

Criteria for Evaluation: Course Project (10 marks)

1. Project Proposal and Planning (2 marks)
   - Submits a well-defined project proposal outlining objectives, methodology, and expected outcomes.
   - Demonstrates thorough planning and a clear timeline for the project.

2. Design and Implementation (3 marks)
    - Implements the project design accurately using appropriate tools and techniques.
    - The design is functional and meets the project objectives.
3. Innovation and Creativity (2 marks)
    - Introduces innovative ideas or unique approaches in the design and implementation.
    - Demonstrates creativity in solving problems or optimizing designs.
4. Analysis and Testing (2 marks)
    - Effectively analyzes the project design to identify and address any issues.
    - Conducts thorough testing to verify the functionality and performance of the design.
5. Final Report and Presentation (1 mark)
    - Submits a comprehensive final report detailing the project, including objectives, design, methodology, analysis, and results.
    - Clearly presents the project and its outcomes, and effectively communicates the key points.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply hardware description languages (HDLs) to design digital circuits and verify the functionality using test benches	K3
CO2	Design simple digital systems with programmable logic devices	K3
CO3	Understand FPGA architecture, logic blocks, and programmable interconnect principles.	K2
CO4	Understand the design considerations of FPGA	K2
CO5	Analyse combinational and sequential circuits using FPGA	K4

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	**3**	**3**	**3**	**1**	**3**							**3**
CO2												**3**
CO3												**3**
CO4												**2**
CO5												**3**

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	FPGA-Based System Design	Wayne Wolf	Prentice Hall	2004
2	Field-Programmable Gate Arrays	Stephen D. Brown, Robert J. Francis, Jonathan Rose, Zvonko G. Vranesic	Springer	1992

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Modern VLSI Design: System-on-Chip Design	Wayne Wolf	Prentice Hall	2002
2	Verilog HDL	Samir Palnitkar	Prentice Hall India	2003
3	The Design Warrior's Guide to FPGAs	Clive Maxfield	Elsevier	2000.

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.youtube.com/playlist?list=PLbMVogVj5nJSY-1XxFHgwgtj2F7mB7NuV
2	https://www.youtube.com/playlist?list=PLbMVogVj5nJSY-1XxFHgwgtj2F7mB7NuV
3	https://www.youtube.com/playlist?list=PLbMVogVj5nJSY-1XxFHgwgtj2F7mB7NuV
4	https://www.youtube.com/playlist?list=PLbMVogVj5nJSY-1XxFHgwgtj2F7mB7NuV

SEMESTER S7

ARTIFICIAL INTELLIGENCE FOR ROBOTICS

Course Code	**OERAT 721**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Understand the Fundamentals of Robotics and AI:
2. Utilize supervised and unsupervised learning techniques for robotic perception and action.
3. Develop and evaluate novel AI algorithms for specific robotic applications through projects and case studies.
4. Analyze recent advancements and research trends in AI and robotics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Artificial intelligence : Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview, Intelligent Agents-Agents and Environments, Good behaviour- The concept of rationality, nature of Environments, structure of Agents, Expert systems - basic overview, Application areas of AI - vision and speech processing, robotics.	**9**
2	Learning - Forms of learning: Supervised Learning Algorithms- Decision Trees,Logistic Regression,Support vector Machines, Unsupervised Learning Algorithms: K-means clustering k-Nearest Neighbors, Principle Component Analysis, Reinforcement based learning	**9**
3	Deep Feedforward Networks: Perceptrons – Single layer & Multi-layer, Learning XOR ,Convolutional Neural Networks -basic outline and functions of each layers only, Sequence Modeling-Need for sequence	**9**

	models, basic RNN architecture and types	
4	Robotics - Robotic perception, Localization and mapping, Machine learning in robot Perception, Application domains.Mobile Robot Navigation-Path planning ,Path Planning and obstacle avoidance-A* algorithm	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze the role of AI in solving problems in different domains and its evolution.	**K3**
CO2	Explain the different learning techniques used in Machine learning	**K2**
CO3	Identify the need for multilayer neural network for solving complex tasks	**K2**
CO4	Identify suitable neural network architectures for different types of data and tasks.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2								2
CO2	3	3	3	2	2							2
CO3	3	3	3	2	2							2
CO4	3	3	3	2	2							2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Deep Learning	Ian Goodfellow, Yoshua Bengio, Aaron Courville	MIT Press	2016
2	Artificial Intelligence - A Modern Approach	Stuart J. Russell and Peter Norvig,	Pearson	Third Edition, 2016
3	Pattern Recognition and Machine Learning,	Bishop, C. ,M.,	Springer	2006.
4	A survey of deep learning techniques for autonomous driving.	Grigorescu, Sorin, et al		Journal of Field Robotics 37.3 (2020): 362-386.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to AI Robotics,	Robin R. Murphy	The MIT Press	Second Edition: 2019
2	Artificial Intelligence and Machine Learning,	Chandra S.S.V, Anand Hareendran S	PHI	2021
3	Models, Learning and Inference	Simon J. D. Prince	Cambridge University Press	2012

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc22_cs56/preview
2	https://archive.nptel.ac.in/courses/106/106/106106139/
3	https://archive.nptel.ac.in/courses/106/106/106106184/
4	https://archive.nptel.ac.in/courses/112/106/112106298/

SEMESTER S7

INTRODUCTION TO ROBOT OPERATING SYSTEM

Course Code	OERAT722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Apply ROS fundamentals and topics
2. Use ROS services and actions
3. Describe robots and simulators

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**ROS Fundamentals, Topics:** The ROS Graph, roscore, catkin, Workspaces, and ROS Packages, rosrun, Names, Namespaces, and Remapping, roslaunch, tf: Coordinate Transforms - Poses, Positions, and Orientations, tf, Publishing to a Topic, Subscribing to a Topic, Latched Topics, Defining Message Types, Mixing Publishers and Subscribers	9
2	**Services and Actions:** Defining a Service, Implementing a Service, Using a Service, Defining an Action, Implementing a Basic Action Server, Using an Action, Implementing a More Sophisticated Action Server, Using the More Sophisticated Action	9
3	**Robots and Simulators:** Subsystems - Actuation: Mobile Platform, Actuation: Manipulator Arm, Sensors, Computation Complete Robots – PR2, Fetch, Robonaut 2, TurtleBot, Simulators – Stage, Gazebo. Other Simulators	9

4	**Building Maps of the World and Navigating About the World:** Maps in ROS, Recording Data with rosbag, Building Maps, Starting a Map Server and Looking at a Map, Localizing the Robot in a Map, Using the ROS Navigation Stack, Navigating in Code	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Apply ROS fundamentals and topics	**K3**
CO2	Use ROS services and actions	**K3**
CO3	Describe robots and simulators	**K2**
CO4	Apply ROS to build maps of the World and navigate	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	-	-	2	-	-	-	-	-	-	1
CO2	3	1	-	-	2	-	-	-	-	-	-	1
CO3	3	1	-	-	2	-	-	-	-	-	-	1
CO4	3	1	-	-	2	-	-	-	-	-	-	1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Programming Robots with ROS: A Practical Introduction to the Robot Operating System	Morgan Quigley, Brian Gerkey, and William D. Smart	O'Reilly	2015
2	A Gentle Introduction to ROS	Jason M. O'Kane		2016

SEMESTER S7

ADVANCED TOPICS IN ROBOTICS

Course Code	**OERAT723**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. This course is intended to provide an insight into the advanced applications and branches of robotics

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to mobile robots and mobile manipulators. Principle of locomotion and types of locomotion. Types of mobile robots: ground robots (wheeled and legged robots), aerial robots, underwater robots and water surface robots. degree of freedom and manoeuvrability, generalized wheel model, different wheel configurations, holonomic and non-holonomic robot. Sensors for mobile robots. Popular algorithms for mobilising robots.	**9**
2	**Field and Service Robots:** History of service robotics–Present status and future trends–Need for service robots-applications-examples and Specifications of service and field Robots. Non-conventional Industrial robots. Ariel robots- Collision avoidance-Robots for agriculture, mining, exploration, underwater, civilian and military applications, nuclear applications, Space applications.	**9**

3	**Medical Robots** Types of medical robots - Navigation - Motion Replication - Imaging - Rehabilitation and Prosthetics - State of art of robotics in the field of healthcare. Minimally invasive surgery and robotic integration - surgical robotic sub systems- –Types of assistive robots.	**9**
4	**Soft Robotics:** Introduction to soft robotics: definitions and applications, Morphological computation and Bioinspired design- Elastomers -Thermoplastics and textiles-cable driven soft robots -Fluidic actuation -Dielectric elastomers - Shape memory alloys- Soft resistive, capacitive, and inductive sensing	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Familiarize mobile robotics.	**K2**
CO2	Understand the concepts of field and service robots	**K2**
CO3	Understand the role of robots in the field of medicine	**K2**
CO4	Understand the concepts of soft robotics	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										1
CO2	3	3										1
CO3	3	3										1
CO4	3	3										1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robotics: Fundamental Concepts and Analysis	JGhosal,A	Oxford University Press,	2nd reprint, 2008
2	Medical Robotics	Achim Schweikard, Floris Ernst	Springer	2015
3	Introduction to Autonomous Mobile Robots	Roland Siegwart, Illah Reza Nourbakhsh, Davide Scaramuzza	Bradford Company Scituate, USA,	2004

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robot Analysis and Control	H. Asada, J. J. E. Slotine	John Wiley & sons,	1991.
2	Elements of Robotics	N. Ben- Ari	Springer	2017

SEMESTER S7

BASICS OF MOBILE ROBOTICS

Course Code	OERAT724	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course introduces the student with the computational fundamentals behind the design and control of autonomous robots
2. This course takes the students through all the important concepts that is required to build an autonomous robot.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Locomotion, manipulation and their representation:** Static and dynamic stability – Degrees of freedom – Coordinate systems and Frames of Reference – Matrix notation – Mapping from one frame to another – Concatenation of Transformations **Kinematics:** Forward Kinematics – The Denavit-Hartenberg notation - Inverse Kinematics –- Differential Kinematics – Inverse Differential Kinematics	9
2	**Forces:** Statics – Kineto- Static Duality – Manipulability Ellipsoid in Velocity Space, Force Space, Manipulability considerations **Grasping:** The theory of grasping – Simple Grasping Mechanisms – 1-DoF scissor like gripper, Parallel Jaw, 4-bar linkage parallel gripper, Multi-fingered hands	9

	Actuators: Electric Motors - Hydraulic and pneumatic actuators – Safety Considerations **Sensors:** Terminology – Sensors that measure the robot's joint configuration – sensors that measure ego-motion, measuring force – Sensors to measure distance – Sensors to measure global pose	
3	**Vision:** Images as two dimensional signals – Translation of Signal to information - Basic image operations – Extracting structure from Vision – Computer vision and Machine learning **Feature Extraction:** Feature detection as an information-reduction problem – Features – Line recognition – Scale-invariant feature transforms – Feature detection and Machine learning	**9**
4	**Artificial Neural Network:** The simple perception – Activation function - Simple perceptron to multilayer neural network – Single outputs to higher dimensional data – Objective Functions and optimization **Task Execution:** Reactive Control – Finite State Machines – Hierarchical Finite State Machine- Behaviour Trees – Mission Planning **Mapping:** Map representations – Iterative Closest point for sparse mapping – Octomap: dense mapping of voxels – RGB-D mapping: dense mapping of surfaces	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the challenges involved in robotic locomotion and the kinematics of robots	**K2**
CO2	Identify the appropriate sensors and actuators for a particular robotic application	**K2**
CO3	Describe the principles behind extracting information using visual techniques	**K2**
CO4	Explain the design of control loops that determine robotic behaviour and mapping techniques used by the robots to understand its surroundings	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2										2
CO2	2		2									2
CO3	3		2		2							2
CO4	3	2	3	2	2							2
CO5												2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Autonomous Robots: Mechanisms, Sensors, Actuators and Algorithms	Nikolaus Correll, Bradley Hayes, Christoffer Heckman and Alessandro Roncone	MIT Press	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Mobile Robot Control	Spyros G. Tzafestas	Elsevier, USA	2012
2	Introduction to Autonomous Mobile Robots	R Siegwart, IR Nourbakhsh, D Scaramuzza	MIT Press	2011
3	Sensors for mobile robot	HR Everett	CRC Press	2012

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://open.umn.edu/opentextbooks/textbooks/316

SEMESTER 8

ROBOTICS AND ARTIFICIAL INTELLIGENCE

SEMESTER S8

MECHATRONIC SYSTEM DESIGN

Course Code	PERAT861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Focus on understanding the interdisciplinary nature of mechatronics and the synergy between various components to create efficient and reliable systems.
2. Enable students to develop and apply advanced modeling, simulation, and control techniques for mechatronic systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Overview of Mechatronics** Definition, scope, and applications of mechatronic systems. Interdisciplinary approach: integration of mechanical, electronic, and software components.	9
2	**Mechanical Components** Introduction to mechanical components: actuators, sensors, and mechanical structures. Selection criteria and compatibility considerations. **Electronic Components** Overview of electronic components: microcontrollers, sensors, and signal conditioning circuits. Interfacing techniques and circuit design for integration	9
3	**Modeling Techniques** Introduction to modeling methodologies: mathematical, physical, and empirical models. Modeling mechanical, electrical, and control subsystems. **Simulation Tools**	9

	Hands-on experience with simulation software (e.g., MATLAB/Simulink, LabVIEW). Simulating mechatronic systems to analyze performance and behavior.	
4	Control Theory Basics Introduction to control systems: feedback, PID control, and system stability. Design considerations for mechatronic control systems. Implementation and Testing Practical implementation of control algorithms on hardware platforms. Testing and validation of mechatronic systems using simulation and real-world experiments.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the interdisciplinary nature of mechatronics by examining how mechanical, electronic, and software components are integrated to create efficient and reliable systems.	**K2**
CO2	apply advanced modeling and simulation techniques to predict and analyze the behavior of mechatronic systems, enabling them to make informed design decisions.	**K3**
CO3	develop integrated mechatronic systems by designing and integrating mechanical, electronic, and software components, demonstrating proficiency in system-level design and integration	**K3**
CO4	evaluate and assess control system designs for mechatronic applications, demonstrating competence in designing, implementing, and evaluating control strategies to achieve desired system performance	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1			1							1
CO2	3	1			3							1
CO3	3	1		3	3							1
CO4	3	1		3	3							1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Mechatronics: Principles and Applications"	Godfrey C. Onwubolu	CRC Press	2015
2	Mechatronics: A Foundation Course"	Clarence W. de Silva	CRC Press	2020
3	Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering"	W. Bolton	Pearson	2018

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Mechatronics: Integrated Mechanical Electronic Systems"	R. K. Rajput	S. Chand & Company Ltd.	2016
2	Mechatronics: Principles,Technologies and Applications"	J. K. Choudhury	CRC Press	2017

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.youtube.com/watch?v=0RATDMszxoE
2	https://www.youtube.com/watch?v=1_Jfxh5EouI
3	https://www.youtube.com/watch?v=jsAqyIGr-5o
4	https://www.youtube.com/watch?v=wkfEZmsQqiA

SEMESTER S8

CO-OPERATIVE ROBOTICS

Course Code	PERAT862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. The objective of this course is to give students a basic idea of cooperative robotics and design considerations of various algorithms.
2. Familiarise the various control problems, consensus algorithm and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to robotics: Components, Odometry, Kinematics, Control, Finite State Machines as Robot Controllers, State Transitions based on Robot–Robot Interactions, Early Micro-Macro Problems, Macroscopic Perspective, Expected Macroscopic Dynamics and Feedback	8
2	Discrete Consensus Achievement in Artificial Systems: Consensus Achievement, The Best-of-n Problem, Overview of Current Design Approaches, Bottom–Up Design Approaches, Top–Down Design Approaches. Consensus Algorithms in Cooperative Control: Fundamental Consensus Algorithms, Convergence Analysis of Consensus Algorithms, Synthesis and Extensions of Consensus Algorithms, Design of Coordination Strategies via Consensus Algorithms	10
3	Consensus-based Design Methodologies for Distributed Multivehicle Cooperative Control: Coupling in Cooperative Control Problems: Objective Coupling, Local Coupling, Full Coupling, Dynamic Coupling. Approach to Distributed Cooperative Control Problems with an Optimization Objective: Cooperation Constraints and Objectives, Coordination Variables	9

	and Coordination Functions, Centralized Cooperation Scheme, Consensus Building Approach to Distributed Cooperative Control Problems Without an Optimization Objective: Coordination Variable Constituted by a Group-level Reference State, Coordination Variable Constituted by Vehicle States	
4	Applications to Multivehicle Cooperative Control:Rendezvous and Axial Alignment with Multiple Wheeled Mobile Robots, Distributed Formation Control of Multiple Wheeled Mobile Robots with a Virtual Leader, Deep Space Spacecraft Formation Flying, Cooperative Fire Monitoring with Multiple UAVs, Cooperative Surveillance with Multiple UAVs	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Familiarise with various robotic components and controllers	**K2**
CO2	Describe various design approaches	**K2**
CO3	Understand various consensus algorithms and their applicability	**K2**
CO4	Analyse the use of coupling in cooperative control problems	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1										3
CO2	2	1										3
CO3	2	1										3
CO4	3	2	2									3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Swarm robotics: A formal approach	Hamann, Heiko	Springer	2018
2	Distributed consensus in multi vehicle cooperative control	Ren, Wei, and Randal W. Beard	Springer London	2008
3	Achieving consensus in robot swarms	Valentini, Gabriele	Studies in computational Intelligence, Springer	2017

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Path planning of cooperative mobile robots using discrete event models	Mahulea, Cristian, Marius Kloetzer, and Ramón González	John Wiley & Sons	2020
2	Cooperative Robots and Sensor	Koubâa, Anis, and	Springer	2014
3	Distributed autonomous robotic systems 2	Asama, Hajime, Toshio Fukuda, Tamio Arai, and Isao Endo	Springer Science & Business Media	2013

SEMESTER S8

PROBABILISTIC ROBOTICS

Course Code	**PERAT863**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Teach students to design and implement probabilistic algorithms to enhance the autonomy and reliability of robotic systems.
2. Cover advanced topics such as sensor fusion, decision-making under uncertainty, and learning from data to prepare students for cutting-edge research and industry challenges in probabilistic robotics

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Probabilistic Robotics, Importance and applications of probabilistic methods in robotics. Historical context and development of probabilistic robotics. Fundamentals of Probability and Statistics Basic probability theory, random variables, and distributions. Statistical inference, Bayesian reasoning, and their significance in robotics. Probabilistic Models Types of probabilistic models used in robotics. Real-world examples and case studies demonstrating these models. Assignments and Assessments: Problem sets on probability and statistics. Readings and essays on the significance of probabilistic robotics	11
2	Bayesian Filters Understanding Bayes' rule and its application in filtering. Implementing Bayes filters for robotic localization. Kalman Filters	10

	Linear Gaussian models and the Kalman filter. Practical applications and limitations in robotics. Particle Filters Non-parametric filters and the particle filter algorithm. Handling non-linearities and non-Gaussian noise. Assignments and Assessments: Programming tasks to implement Bayesian, Kalman, and particle filters. Simulation projects for robotic localization	
3	Sensor Fusion Techniques Principles and methodologies for sensor fusion. Practical examples and case studies of sensor fusion in robotics. Decision Making Under Uncertainty Markov Decision Processes (MDPs) and Partially Observable MDPs (POMDPs). Algorithms for decision making in uncertain environments. Practical Applications Case studies and practical implementations. Hands-on projects with real or simulated robotic systems. Assignments and Assessments: Projects focused on sensor fusion and decision making. Case study analyses and practical demonstrations.	8
4	Learning Algorithms in Robotics Overview of supervised, unsupervised, and reinforcement learning. Implementing learning algorithms in robotic systems. Advanced Probabilistic Methods Exploration of advanced filtering techniques. Multi-robot systems and cooperative localization. Future Trends and Research Directions Emerging trends and future directions in probabilistic robotics. Assignments and Assessments: Final project integrating learned probabilistic algorithms. Review and presentation of contemporary research papers	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand and Explain the foundational principles of probability and statistics as they apply to robotic systems	**K2**
CO2	Design and Implement probabilistic algorithms, including Bayesian filters, Kalman filters, and particle filters, to enhance robotic localization and mapping capabilities.	**K3**
CO3	Analyze various sensor fusion techniques and decision-making algorithms to improve the autonomy and reliability of robotic systems under uncertain conditions.	**K4**
CO4	Develop and Synthesize advanced probabilistic methods and learning algorithms to address complex challenges in robotic research and industry applications.	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1			1							1
CO2	3	1			3							1
CO3	3	1		3	3							1
CO4	3	1		3	3							1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Probabilistic Robotics"	Sebastian Thrun, Wolfram Burgard, Dieter Fox	MIT Press	2005
2	"Introduction to Autonomous Mobile Robots"	Roland Siegwart, Illah R. Nourbakhsh, Davide Scaramuzza	MIT Press	2011
3	"Bayesian Reasoning and Machine Learning"	David Barber	Cambridge University Press	2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Artificial Intelligence: A Modern Approach"	Stuart Russell, Peter Norvig	Pearson	2021
2	"Machine Learning: A Probabilistic Perspective"	Kevin P. Murphy	MIT Press	2012

SEMESTER S8

CLOUD ROBOTICS

Course Code	PERAT864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Provide an overview of cloud robotics
2. Understand the concept of programmable robots for automated tasks, principles of cloud computing technologies and robotics-related software paradigm such as the Robotics Operating System (ROS)
3. Provide knowledge on how cloud computing techniques can automate manufacturing tasks using algorithms designed based on machine learning and big data analytics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Challenges in cloud robotics: Technology for cloud computing. Constraints and features of programmable robots. Development tools and software. Robot operating systems and software services. Current trends and advancements in cloud robotics computing.	9
2	Programmable robots: Microprocessors such as Arduino/Raspberry Pi with consumer robotic appliances. Robotics design. Navigation planning. Simultaneous Localisation and Mapping algorithm. Robot operating systems and its architecture	9
3	Cloud computing and networking: Cloud computing systems. Client-server programming and programming API in the cloud. Software programming and API services to integrate ROS and the Cloud.	9
4	Cloud robotics and automation architectures: robotics and human automation by cloud computing, current trends of networked robotics using Internet and the Cloud. System components and architectures of the Internet of Things. Optimisation and Data analytics algorithms	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Identify the basic problems, limitations, strengths and current trends of programmable robotics	**K2**
CO2	Explain the current cloud computing technologies and computing mechanisms for robotics such as ROS	**K2**
CO3	Analyse and critique the performance of robotics algorithms and data analytics algorithms for cloud robotics	**K4**
CO4	Develop the attitude to use software programming and cloud computing solutions to create cloud robotics prototype	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										3
CO2	3	3			2							3
CO3	3	3	3		3							3
CO4	3	3	3		3							3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Cloud Robotics – Distributed Robotics using Cloud Computing	Joao Pedro, Carvalho Rosa		Coimbra, 2016
2	A Survey of Research on Cloud Robotics and Automation	Ben Kehoe, Sachin Patil, Pieter Abbeel, Ken Goldberg	IEEE Transactions on Automation Science and Engineering (T-ASE)	Special Issue on Cloud Robotics and Automation. Vol. 12, no. 2, 2015
3	Programming Robots with ROS: A Practical Introduction to the Robot Operating System	Morgan Quigley, Brian Gerkey and William Smart	O'Reilly Media	1st Edition, 2015.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	A Gentle Introduction to ROS	Jason M. O'Kane,	CreateSpace Independent Publisher;	1st Edition, 2013.
2	Cloud Computing: Principles and Paradigms	Rajkumar Buyya, James Broberg, Andrzej M. Goscinski	Wiley	2011
3	Enterprise Cloud Computing - Technology, Architecture, Applications,	Gautam Shroff,	Cambridge University Press,	2010

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc21_cs14/preview
2	https://www.youtube.com/watch?v=Bcz3a-klKGk
3	https://onlinecourses.nptel.ac.in/noc21_cs14/preview
4	https://onlinecourses.nptel.ac.in/noc21_cs14/preview

SEMESTER S8

UNDER WATER ROBOTICS

Course Code	PERAT866	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To describe the principle of locomotion and different types of mobile robots.
2. To impart knowledge of kinematics of underwater vehicle.
3. To develop skills in motion path planning, control and stability of underwater robots.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Mobile Robot: Introduction to mobile robots - principle of locomotion - types of mobile robots: ground robot, aerial robot, underwater robot and water-surface robot - principles of underwater vehicle construction	9
2	Kinematics of underwater vehicle: Equations for moving frame - rigid motion in a plane - representation of a rotated frame - holonomic and non-holonomic systems - kinematic modeling with respect to global coordinates.	9
3	Sensors for robot navigation: Types of sensors - magnetic and optical position sensor - gyroscope - accelerometer - magnetic compass inclinometer - tactile and proximity sensor - ultrasound range finder - laser scanner, infrared range finder - visual and motion sensing systems.	9
4	Motion path planning, control and stability: path planning algorithms - collision-free path planning - sensor-based obstacle avoidance - motion control methods: kinematic control, dynamic control, controllability and stability about a point and trajectory	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the principle of locomotion and described different types of mobile robots	**K2**
CO2	Summarize the degree of freedom to manoeuvrability of various robots	**K2**
CO3	Identify the use of various sensors deployed in autonomous robots	**K2**
CO4	Develope motion path planning and its control for an autonomous robot	**K3**
CO5	Explain the robust feedback control methods	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2								
CO2	3	2		2	2							
CO3	2	3	2	2								2
CO4	3	3	2	2	2							2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Autonomous Underwater Vehicles	Sabiha Wadoo, Pushkin Kachroo,	CRC Press	2011
2	Visual Perception and Control of Underwater Robots	Yu Junzhi	CRC Press	2018
3	Underwater Robots	Gianluca Antonelli	Springer Cham	2018
4	Underwater Robots	Dan Faust	Rosen Publishing	2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Under water Robotics Science Design & Fabrication	Steven W. Moore	Marine Advanced Technology Education	2010
2	Introduction to Autonomous Robots,	Nikolaus Correll	The MIT Press	2022
3	Navigation, Control and Sensing, Surface	Gerald Cook, Feitian Zhang,	Wiley-IEEE Press	2020

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/f5GMjGRsKi4?list=PLyqSpQzTE6M-YPOEUVrgd_2UnIRvWUcRr
2	https://youtu.be/U0gGoPKgays
3	https://youtu.be/lRm9GiGoZKg?list=PLLy_2iUCG87AjAXKbNMiKJZ2T9vvGpMB0
4	https://youtu.be/A7wHSr6GRnc

SEMESTER S8

GENERATIVE AI

Course Code	**PERAT867**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To learn the fundamentals of Neural Networks and their various types.
2. To explore Generative AI models like GANs, VAEs, and Transformers.
3. To analyze the limitations of traditional RNNs and LSTMs.
4. To discuss current trends and future directions in Generative AI research.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Foundations of AI and Neural Networks:** History and evolution of AI/ML, Deep learning revolution, Transfer learning, History of Neural Natural Language Processing, Structure of Artificial Neural Networks, Steps in Training an Artificial Neural Network, Parameters and Hyperparameters, Backpropagation	9
2	**Advanced Neural Network Architectures:** Introduction to advanced architectures, Introduction to Generative AI Models: Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Transformers, Attention Mechanism in detail Long Short-Term Memory Networks (LSTMs).	9
3	**Data Pre-processing:** Probability and Statistics, Data Pre-processing Techniques, Model Training Techniques	9
4	**Generative AI Applications:** Applications in Various Fields : Art and Creativity, Image and Video Generation, Text Generation, Music Composition, Healthcare	9

	Finance.Real-world use cases and challenges in deploying generative AI models	

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the evolution of AI and the significance of Deep Learning.	**K2**
CO2	Apply various Neural Network architectures for tasks like image recognition and sequence modelling.	**K3**
CO3	Analyze data pre-processing and training techniques for neural networks.	**K4**
CO4	Design practical solutions using advanced neural networks for diverse applications.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1		3	1	2		2	1		2	1	
CO2		2							1	1	2	
CO3	1		1	1	1		1	1	2	3	1	
CO4		1		2		1	1		3			

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Generative AI for everyone: Understanding the essentials and applications of this breakthrough technology".	Altaf Rehmani	Altaf Rehmani	February 12, 2024
2	Introduction to Generative AI	Numa Dhamani	Manning	March 5, 2024

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Generative Adversarial Networks Cookbook: Over 100 recipes to build generative models using Python, TensorFlow, and Keras	Josh Kalin	Packt Publishing	31 December 2018
2	Generative AI in Software Development: Beyond the Limitations of Traditional Coding	Jesse Sprinter	IBS	2024

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://elearn.nptel.ac.in/shop/iit-workshops/completed/leveraging-generative-ai-for-teaching-programming-course s/?v=c86ee0d9d7ed
2	https://elearn.nptel.ac.in/shop/iit-workshops/completed/introduction-to-language-models/?v=c86ee0d9d7ed

SEMESTER S8

SOFT ROBOTICS

Course Code	PERAT868	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the fundamental principles and characteristics of soft robotics.
2. To explore the design and fabrication methods used in creating soft robotic systems.
3. To study the control strategies and material properties that enable soft robotics functionality.
4. To investigate the applications and future trends of soft robotics in various domains

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Soft Robotics: Definition and Importance-Comparison with Traditional Rigid Robotics- Fundamental Principles of Soft Robotics: Compliance, Adaptability, and Safety Materials for Soft Robotics: Polymers, Gels, and Flexible Materials Historical Development and Milestones in Soft Robotics Current Trends and Future Directions in Soft Robotics	9
2	Design and Fabrication of Soft Robots :Design Principles for Soft Robots- Bio inspiration and Functional Requirements- Fabrication Techniques: Molding, Casting, 3D Printing, and Pneumatic Actuation- Materials Used in Soft Robotics: Silicone, Elastomers, and Hydrogels- Mechanical Properties and Characterization of Soft Materials- Case Studies: Soft Grippers, Soft Actuators, and Soft Sensors	9
3	Control and Actuation in Soft Robotics: Control Strategies for Soft Robotics: Kinematics and Dynamics- Actuation Mechanisms: Pneumatic, Hydraulic, and Electro active Polymers- Integration of Sensors and Feedback Systems- Modeling and Simulation of Soft Robotic Systems- Case Studies: Soft Robotic Manipulators and Crawlers	9

4	Applications and Future Trends in Soft Robotics: Applications of Soft Robotics-Industrial Automation, Healthcare, and Consumer Products- Soft Robotics in Medical Applications- Soft Endoscopy, Wearable Robots, and Rehabilitation -Research Trends in Soft Robotics- Bio hybrid Systems and Soft-Body Locomotion- Future Directions: Soft Robotics in Space Exploration and Extreme Environments- Challenges and Limitations: Durability, Control Precision, and Integration-Ethical and Societal Implications of Soft Robotics	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts and significance of soft robotics.	**K2**
CO2	Design and fabricate simple soft robotic systems.	**K3**
CO3	Apply appropriate control strategies for the movement and operation of soft robots.	**K3**
CO4	Evaluate the potential and challenges of soft robotics in real-world applications.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3	3	3	2		3						3
CO3	3		1			1						3
CO4	3											2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Soft Robotics: Transferring Theory to Application"	Cecilia Laschi and Matteo Cianchetti	Springer	2016
2	"Fundamentals of Soft Matter Science"	Linda S. Hirst	CRC Press	2020
3	Soft Robotics: Modeling, Simulation, and Control of Deformable Robots" Ivan Petrov, Springer, 2021.	Ivan Petrov	Springer	2021
4	Soft Robotics: Technologies and Systems for Soft Machines and Robots".	Mohammad D. Hossain	Springer	2021

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Soft Robotics: Materials, Design, and Applications	Dong Sun and Shuozhi Xu	CRC Press	2020
2	Bio inspired Soft Robotics: Soft Matter in Engineering and Life Sciences	Eduard Arzt, De Gruyter,	CRC Press	2016
3	Fundamentals of Soft Robotics: Theory and Applications	Qingsong Xu,	Springer	2022
4	Soft Robotics: Challenges and Applications	Xianmin Zhang, Yujiang Shi, and Kwang Kim,	Springer	2021
5	Wearable Robotics: Challenges and Trends	Jose L. Pons	Springer	2019

SEMESTER S8

VISUAL AND AUGMENTED REALITY FOR ROBOTICS

Course Code	**PERAT865**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	

Course Objectives:

1. Understand the taxonomy, technology, and features of Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR)
2. Apply AR Tools and Techniques
3. Apply VR Hardware and Interaction

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction - Taxonomy, Technology and Features of Augmented Reality, Difference between AR, VR and MR, Challenges with AR, AR Systems and Functionality, Augmented Reality Methods, Visualization Techniques for Augmented Reality	**9**
2	Introduction to AR Tools - Camera Parameters and Calibration, Marker-Based Augmented Reality – AR Toolkit, Case Study: AR for Human Robot Communication - AR and Cobots.	**9**
3	Basic Features and Architecture - VR Input Hardware: Tracking Systems, Motion Capture Systems, Data Gloves, VR Output Hardware: Visual Displays	**9**
4	3D Manipulation Tasks, Manipulation Techniques and Input Devices, Interaction Techniques for 3D Manipulation, Applications of VR and AR: Medical Applications - Educational Applications - Public Safety and Military Applications	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Students may be given microprojects on the application of AR and VR tools

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the taxonomy, technology, and features of Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR)	**K2**
CO2	Apply AR Tools and Techniques	**K3**
CO3	Apply VR Hardware and Interaction	**K3**
CO4	Analyse AR and VR Applications	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	1	1	1	-	1	1	-	-	-	-	1
CO2	2	1	2	1	2	1	1	1	1	1	1	2
CO3	2	1	2	1	2	1	1	1	1	1	1	2
CO4	2	2	2	2	2	2	2	2	2	2	1	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Virtual & Augmented Reality for Dummies	Paul Mealy	John Wiley and Sons	2018
2	Learning Virtual Reality: Developing Immersive Experiences and Applications for Desktop, Web, and Mobile	Tony Parisi	O' Reilly	2015

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Understanding Augmented Reality: Concepts and Applications	Alan B. Craig	Morgan Kaufmann	2013
2	Developing Virtual Reality Applications: Foundations of Effective Design	Alan Craig, William Sherman and Jeffrey Will	Morgan Kaufmann	2009
3	Practical Augmented Reality: A Guide to the Technologies, Applications, and Human Factors for AR and VR (Usability)	Steve Aukstakalnis	Addison-Wesley Publisher	1st Edition, 2016.
4	Virtual Reality Technology	Burdea, G. C. and P. Coffet	Wiley-IEEE Press	2nd Edition, 2017.

SEMESTER S8

INTRODUCTION TO SIGNAL PROCESSING FOR ROBOTICS

Course Code	**OERAT831**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To understand and analyse various signal processing techniques for Robotics
2. To analyse continuous and discrete time systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Elementary signals, Continuous time and Discrete time signals and systems, Signal classifications, operations, linearity, stability and causality of systems, LTI systems-definition and examples Convolution, Correlation and Orthogonality of signals	9
2	Frequency domain representation-Continuous time Fourier series, Continuous time Fourier transform, Laplace transform, Inverse Laplace transform, Differential equation representation of continuous time systems,Transfer function, Frequency response	9
3	Sampling, Aliasing, Z transform, ROC, Inverse Z transform, Difference equation representation of discrete time systems, Frequency domain representation of discrete time signals-Discrete time Fourier transform (DTFT), Analysis of discrete time LTI systems using the above transforms	9
4	Discrete Fourier Transform (DFT), computational complexity, circular convolution, long convolution, Fast Fourier Transform (FFT)-Decimation in Time algorithm.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Perform basic operations on continuous time discrete time signals and systems	**K2**
CO2	Analyze continuous time systems in the frequency domain by finding the frequency response	**K4**
CO3	Analyze discrete time systems using tools like Z-transform and DTFT	**K4**
CO4	Efficiently compute the DFT using fast algorithms(FFT)	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2									2
CO2	3	3	3									2
CO3	3	3	3									2
CO4	3	3	3									2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Discrete-Time Signal Processing	Alan V Oppenheim, Ronald W. Schafer	Pearson	3rd, 2010
2	Signals & Systems	Simon Haykin	PHI	2nd , 2003

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Signals & Systems	Alan V. Oppenheim and Alan Willsky,	PHI	2nd, 2009
2	Signals & Systems	Anand Kumar,	PHI	3rd, 2013
3	Signals & Systems	P Ramakrishna Rao, Shankar Prakriya	Mc Graw Hill	2013
4	Signals & Systems-Continuous and Discrete,	Rodger E. Ziemer	Pearson	4th 2013
5	Digital Signal Processing	Proakis J. G. and Manolakis D. G	Pearson Education	2007
6	Digital Signal Processing-a practical approach	Ifeachor E.C. and Jervis B. W	Pearson education	2009
7	Schaums outline: Digital Signal Processing	Monson H Hayes	McGraw HillProfessional	1999
8	Digital Signal Processing	Salivahanan S	McGraw Hill	2019

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	NPTEL videos on Signals and Systems, by Prof. Aditya K Jagannatham, IIT Kanpur (https://archive.nptel.ac.in/courses/108/104/108104100/ etc.)
2	NPTEL videos on DSP, by Prof. S. C. Dutta Roy, IIT Delhi (https://nptel.ac.in/courses/117102060 etc.)
3	NPTEL videos on Real TIme DSP, by Prof. Rathna G. N., IISc, Bangalore (https://archive.nptel.ac.in/courses/108/108/108108185/ etc.)

SEMESTER S8

IMAGE PROCESSING AND ROBOT VISION

Course Code	**OERAT832**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Understand the basic principles and techniques in digital image processing.
2. Provide insight into various methods of intensity transformations and spatial filtering techniques.
3. Provide students with a comprehensive understanding of morphological operations and image segmentation techniques.
4. Equip students with theoretical knowledge and practical skills in depth estimation and multi-camera views

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamental Steps in Digital Image Processing: Elements of Visual Perception, Image Sensing and Acquisition, A Simple Image Formation Model, Image Sampling and Quantization, Basic Relationship between pixels, Distance Measures	**9**
2	Intensity Transformation Functions- Image Negatives, Log Transformations, Power Law Transformations, Piecewise-Linear Transformation Functions, contrast stretching, intensity level slicing, bit plane slicing, Fundamentals of spatial Filtering-mechanics of spatial filtering, vector representation of linear filtering, generating spatial filter masks, smoothing linear filters and nonlinear filters	**9**
3	Reflection of a set and translation of a set, erosion and dilation, duality, Opening and closing, Image Segmentation, Classification, Region Approach to Image Segmentation, Clustering Techniques-Hierarchical, K- Means, Thresholding- global thresholding, adaptive thresholding	**9**

4	Perspective, Binocular Stereopsis: Camera and Epipolar Geometry, Homography, Rectification, DLT, RANSAC, 3-D reconstruction framework, Background Subtraction and Modelling	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic principles, techniques, and algorithms essential for digital image processing	**K2**
CO2	Apply intensity transformation and spatial filtering techniques to enhance image quality	**K3**
CO3	Demonstrate various mathematical morphology and image segmentation techniques	**K2**
CO4	Explain perspective geometry, binocular stereopsis, camera calibration, 3D reconstruction, motion analysis, and optical flow computation	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	2	2	1			1	1		1
CO2	3	2	2	2	1	1			1	1	1	
CO3	3	2		2	2					1	2	1
CO4	2	2	2		1				2		1	1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Digital Image Processing	Rafael C Gonzalez, Richard E Woods	Prentice Hall	3rd edition 2008
2	Digital Image Processing	S Jayaraman, S Esakkirajan, T Veerakumar	McGraw Hill	14th edition, 2015
3	Computer and Machine Vision-Theory Algorithm and Practicalities	E.R.Davies	Academic Press	2012
4	Computer Vision: Algorithms and Applications	Richard Szeliski	Springer	2011

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Computer Vision: A Modern Approach	David Forsyth and Jean Ponce	Pearson	2002

SEMESTER S8

COMMUNICATION AND COGNITIVE SYSTEMS

Course Code	**OERAT 833**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. This course let the students to study the fundamentals of wireless mobile communication, software defined radio and cognitive radio techniques and their essential functionalities

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Digital Modulation: QPSK, M-Ary QAM; OFDM: OFDM symbol, cyclic prefix, Transmit symbol; Fundamentals of Multiple access methods: TDMA, FDMA, CDMA, OFDMA; Wireless propagation mechanism, free space propagation model, pathloss and shadowing, Introduction to fading: Frequency selective and FNS fading; Power Delay Profile; Coherence bandwidth and Coherence time; Diversity techniques: Diversity combining methods (combined SNR equation of each); Diversity Gain, Diversity Order	9
2	Cellular Communications: Cellular concept, Interference and system capacity: Cell structure, Cell splitting, Sectoring, Repeaters, and Microcells; Cellular System Design Fundamentals: Frequency Reuse, channel assignment strategies, handoff Strategies; GSM system Wireless Mobile Communication Generations: Specifications, modulation and MA methods	9
3	Introduction to MIMO system: System model, concept of Diversity-Multiplexing trade-off, parallel decomposition, beamforming; Multi-User MIMO: Structure, beamforming, Downlink, Uplink; Massive MIMO: Spatial multiplexing, spatial diversity, beamforming; Milli-meter wave	9

	communication; Concepts of Cooperative Communication system: Basic system model, Amplify and Forward, Decode and Forward; Basics of Vehicular Communication (V2V, V2G)	
4	Introduction to Cognitive Radio(CR): Fundamentals of Software-Defined Radio(SDR), architecture, Difference between SDR and CR, Cognitive Resource Manager Framework (CRM). CRN Architecture: Architectures for Spectrum Sensing, Network Optimization through Utilities, Policy Support as a Part of the Architecture, Spectrum Brokering Services. Radio frequency spectrum regulations; Spectrum access methods: SINR of Overlay, Underlay and Interweave methods	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Compare different digital modulation schemes, MA methods, and Generations of mobile communication	**K2**
CO2	Apply the knowledge of wireless channel models to learn the parameters of a new channel	**K3**
CO3	Apply the knowledge of Cellular mobile parameters in cell design	**K3**
CO4	Describe MIMO systems, beamforming, Cooperative communication and vehicular communication	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2								2
CO2	3	2		2								2
CO3	3	2		2								2
CO4	3	2		2								2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Wireless Communications	Andrea Goldsmith	Cambridge University Press	2015

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Cognitive Radio Communications and Networks	Alexander M. Wyglinski, Maziar Nekovee, Thomas Hou	Academic Press Elsevier	2010
2	Cognitive Radio, Software Defined Radio, and Adaptive Wireless Systems	Huseyin Arslan (Ed.),	Springer	2007